NEET 2023

OFFICIAL QUESTION PAPERS WITH DETAILED SOLUTION

All questions with detailed solutions of Physics, Chemistry and Biology

CAREERS360

Physics

Q. 1 The errors in the measurement which arise due to unpredictable fluctuations in temperature and voltage supply are

Option 1:
Instrumental errors

Option 2:
Personal errors

Option 3:
Least count errors

Option 4:
Random errors

Correct Answer:
Random errors

Solution:
The cause is not known so, It is a Random error.

Hence, the answer is the option (4).

Q. 2 A metal wire has mass $(0.4 \pm 0.002) g$, radius $(0.3 \pm 0.001) \mathrm{mm}$ and length $(5 \pm 0.02) \mathrm{cm}$. The maximum possible percentage error in the measurement of density will nearly be:

Option 1:
1.2%

Option 2:
1.3%

Option 3:
1.6%

Option 4:
1.4%

Correct Answer:
1.6%

Solution:

$\rho=\frac{\text{Volume}}{\text{Vos}}=\frac{\mathrm{m}}{\pi r^2 \ell}$

$\frac{\Delta \mathrm{m}}{\mathrm{m}} \times 100=\frac{0.002}{0.4} \times 100=0.5 \%$

$\frac{\Delta \mathrm{r}}{\mathrm{r}} \times 100=\frac{0.001}{0.3} \times 100=\frac{1}{3} \%$

$\frac{\Delta \ell}{\ell} \times 100=\frac{0.02}{5} \times 100=0.4 \%$

$\frac{\Delta \rho}{\rho} \times 100=\left(\frac{\Delta \mathrm{m}}{\mathrm{m}} \times 100\right)+2 \frac{\Delta \mathrm{r}}{\mathrm{r}} \times 100+\frac{\Delta \ell}{\ell} \times 100$

$=1.56 \%$

$\simeq 1.6 \%$

Hence, the answer is the option (3).

Q. 3 A bullet is fired from a gun at the speed of $280 \mathrm{~m} \mathrm{~s}^{-1}$ in the direction 30° above the horizontal. The maximum height attained by the bullet is $\left(g=9.8 \mathrm{~m} \mathrm{~s}^{-2}, \sin 30^{\circ}=0.5\right)$:

Option 1:
$2800 \mathrm{~m}$

Option 2:
$2000 \mathrm{~m}$

Option 3:
$1000 \mathrm{~m}$

Option 4:
$3000 \mathrm{~m}$

Correct Answer:
$1000 \mathrm{~m}$

Solution:

$\mathrm{H}=\frac{\mathrm{u}^2 \sin ^2 \theta}{2 \mathrm{~g}}$

$\mathrm{u}=280 \mathrm{~m} / \mathrm{s}, \sin 30^{\circ}=0.5$

$\mathrm{g}=9.8 \mathrm{~m} / \mathrm{s}^2$

$\mathrm{H}=1000 \mathrm{~m}$

Hence, the answer is the option (3).

Q. 4 The angular acceleration of a body, moving along the circumference of a circle, is :

Option 1:
along the radius, away from centre

Option 2:
along the radius towards the centre

Option 3:
along the tangent to its position

Option 4:
along the axis of rotation

Correct Answer:
along the axis of rotation

Solution:
Angular acceleration is an axial vector.

Hence, the answer is the option (4).

Q. 5 A vehicle travels half the distance with speed v and the remaining with speed 2v. Its average speed is :

Option 1:
$\frac{\mathrm{v}}{3}$

Option 2:
$\frac{2\mathrm{v}}{3}$

Option 3:
$\frac{4\mathrm{v}}{3}$

Option 4:
$\frac{3\mathrm{v}}{4}$

Correct Answer:
$\frac{4\mathrm{v}}{3}$

Solution:

$$\mathrm{v}_{\mathrm{avg}}=\frac{2 v_1 v_2}{v_1+v_2} \quad [\text{when distance is same}]$$
$$v_{\mathrm{avg}}=\frac{2 v(2 v)}{3 v}=\frac{4 v}{3}$$

Hence, the answer is the option (3).

Q. 6 A horizontal bridge is built across a river. A student standing on the bridge throws a small ball vertically upwards with a velocity $4\,\mathrm{ms}^{-1}$. The ball strikes the water surface after 4s. The height of bridge above water surface is $\left(\text{ Take } g=10\ \mathrm{ms}^{-2}\right)$:

Option 1:
$56\,\mathrm{m}$

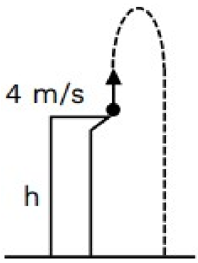
Option 2:
$60\,\mathrm{m}$

Option 3:
$64\,\mathrm{m}$

Option 4:
$68\,\mathrm{m}$

Correct Answer:
$64\,\mathrm{m}$

Solution:



$$-\mathrm{h}=\mathrm{ut}-\frac{1}{2}\mathrm{gt}^2 \quad \Rightarrow \quad -\mathrm{h}=4\times 4-\frac{1}{2}\times 10\times 16 \quad \Rightarrow \quad h=64\ \mathrm{m}$$

Q. 7 A bullet from a gun is fired on a rectangular wooden block with velocity u. When bullet travels 24 cm through the block along its length horizontally, velocity of bullet becomes $\frac{u}{3}$ .Then it further penetrates into the block in the same direction before coming to rest exactly at the other end of the block. The total length of the block is

Option 1:
$27\ \mathrm{cm}$

Option 2:
$24\ \mathrm{cm}$

Option 3:
$28\ \mathrm{cm}$

Option 4:
30 cm

Correct Answer:
27 cm

Solution:

$$v^2 = u^2 - 2as$$
$$\left(\frac{u}{3}\right)^2 = u^2 - 2a(24)....(1)$$
$$Now\ 0 = \left(\frac{u}{3}\right)^2 - 2a(x)....(2)$$
$$x = 3 \text{ cm}$$
$$\text{total length } = 24 + x$$
$$= 27 \text{ cm}$$

Hence, the answer is the option (1).

Q. 8 A football player is moving southward and suddenly turns eastward with the same speed to avoid an opponent. The force that acts on the player while turning is :

Option 1:
along eastward

Option 2:
along northward

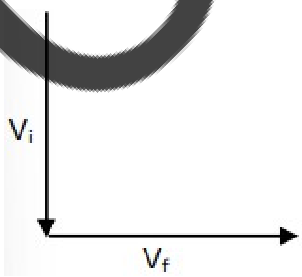
Option 3:
along north-east

Option 4:
along south-west

Correct Answer:
along north-east

Solution:

$$\text{Force } = \frac{\Delta P}{\Delta t}$$



direction of force = direction of change in velocity.

$$\Delta\vec{V} = \vec{V}_f - \vec{V}_i$$
$$= v\hat{i} + v\hat{j}$$
$$North - East$$

Q. 9 Calculate the maximum acceleration of a moving car so that a body lying on the floor of the car remains stationary. The coefficient of static friction between the body and the floor is $0.15(g = 10\ \mathrm{ms}^{-2})$.

Option 1:
$1.2\ \mathrm{ms}^{-2}$

Option 2:
$150\ \mathrm{ms}^{-2}$

Option 3:
$1.5\ \mathrm{ms}^{-2}$

Option 4:
$50\ \mathrm{ms}^{-2}$

Correct Answer:
$1.5\ \mathrm{ms}^{-2}$

Solution:
$a_{\max} = \mu g$
$= 0.15 \times 10$
$a_{\max} = 1.5\ \mathrm{m/s}^2$

Hence, the answer is the option (3).

Q. 10 The potential energy of a long spring when stretched by 2 cm is U. If the spring stretched by 8 cm, potential energy stored in it will be :

Option 1:
$2U$

Option 2:
$4U$

Option 3:
$8U$

Option 4:
$16\mathrm{U}$

Correct Answer:

16U

Solution:

$U=\frac{1}{2}kx^2$

$U=\frac{1}{2}k(2)^2$

$U'=\frac{1}{2}k(2\times 4)^2$

$U'=\frac{1}{2}k(2)^2\times 16$

$U'=16U$

Hence, the answer is the option (4).

Q. 11 The ratio of radius of gyration of a solid sphere of mass M and radius R about its own axis to the radius of gyration of the thin hollow sphere of same mass and radius about its axis is :

Option 1:

$3:5$

Option 2:

$5:3$

Option 3:

$2:5$

Option 4:

$5:2$

Correct Answer:

$3:5$

Solution:

$\frac{\mathrm{k}_1}{\mathrm{k}_2}=\frac{\sqrt{2/5}\mathrm{R}}{\sqrt{2/3}\mathrm{R}}$

$\frac{\mathrm{k}_1}{\mathrm{k}_2}=\sqrt{3}:\sqrt{5}$

Hence, the answer is the option (1).

Q. 12 A satellite is orbiting just above the surface of the earth with period T. If d is the density of the earth and G is the universal constant of gravitation, the quantity $\frac{3\pi}{\mathrm{Gd}}$ represents:

Option 1:
T

Option 2:
T^2

Option 3:
T^3

Option 4:
$\sqrt{T}$

Correct Answer:
T^2

Solution:
$$\mathrm{T}^2=\frac{4\pi^2}{\mathrm{GM}}\left(\mathrm{R}^3\right)$$
$$\mathrm{T}^2=\frac{4\pi^2\times\mathrm{R}^3}{\mathrm{Gd}\left(\frac{4}{3}\pi\mathrm{R}^3\right)}$$
$$\mathrm{T}^2=\frac{3\pi}{\mathrm{Gd}}$$

Hence, the answer is the option (2).

Q. 13 Two bodies of mass m and 9m are placed at a distance R. The gravitational potential on the line joining the bodies where the gravitational field equals zero, will be (G = gravitational constant) :

Option 1:
$-\frac{8\mathrm{Gm}}{\mathrm{R}}$

Option 2:
$-\frac{12\mathrm{Gm}}{\mathrm{R}}$

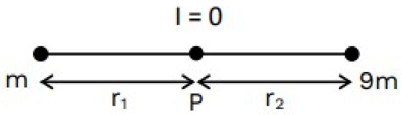
Option 3:
$-\frac{16\mathrm{Gm}}{\mathrm{R}}$

Option 4:
$-\frac{20\mathrm{Gm}}{\mathrm{R}}$

Correct Answer:
$-\frac{16\mathrm{Gm}}{\mathrm{R}}$

Solution:

Let point P is the point where the gravitational field equals zero



$$r_1 = \left(\frac{\sqrt{m}}{\sqrt{9m}+\sqrt{m}}\right) R = \frac{R}{4}$$

$$r_2 = \frac{3R}{4}$$

$$V = -\frac{Gm}{\left(\frac{R}{4}\right)} - \frac{G(9m)}{\left(\frac{3R}{4}\right)}$$

$$V = -\frac{16\mathrm{Gm}}{R}$$

Q. 14 Let a wire be suspended from the ceiling (rigid support) and stretched by a weight W attached at its free end. The longitudinal stress at any point of cross-sectional area A of the wire is :

Option 1:
2W/A

Option 2:
W/A

Option 3:
W/2A

Option 4:
Zero

Correct Answer:
W/A

Solution:

$$\text{Longitudinal stress} = \frac{\mathrm{F}_{\perp}}{\mathrm{A}} = \frac{\mathrm{Mg}}{\mathrm{A}} \Rightarrow \frac{\mathrm{W}}{\mathrm{A}}$$

Hence, the answer is the option (2).

Q. 15 The venturi-meter works on :

Option 1:
Huygen's principle

Option 2:
Bernoulli's principle

Option 3:
The principle of parallel axes

Option 4:
The principle of perpendicular axes

Correct Answer:
Bernoulli's principle

Solution:
Venturi meter-

- It is a device is used for measuring the rate of flow of liquid through pipes.
- This device is based on the application **of Bernoulli's theorem.**

Hence, the answer is the option (2).

Q. 16 The amount of energy required to form a soap bubble of radius 2 cm from a soap solution is nearly : $(\text{surface tension of soap solution } = 0.03 \text{ N m}^{-1})$

Option 1:
30.16×10^{-4} J

Option 2:
5.06×10^{-4} J

Option 3:
3.01×10^{-4} J

Option 4:
50.1×10^{-4} J

Correct Answer:
3.01×10^{-4} J

Solution:

$$Energy Required = 8\pi R^2 \text{ T}$$
$$= 8 \times \frac{22}{7} \times 4 \times 10^{-4} \times 3 \times 10^{-2} = 3.01 \times 10^{-4} \text{ J}$$

Hence, the answer is the option (3).

Q. 17 A Carnot engine has an efficiency of 50% when its source is at a temperature 327°C. The temperature of the sink is :

Option 1:
$27°\mathrm{C}$

Option 2:
$15°\mathrm{C}$

Option 3:
$100°\mathrm{C}$

Option 4:
$200°\mathrm{C}$

Correct Answer:
$27°\mathrm{C}$

Solution:

$$\eta = 1 - \frac{T_2}{T_1}$$
$$\frac{1}{2} = 1 - \frac{T_2}{600}$$
$$T_2 = 300\ \mathrm{K}$$

$So, \mathrm{T}_2 = 27°\mathrm{C}$

Q. 18 The temperature of a gas is -50° C. To what temperature the gas should be heated so that the rms speed is increased by 3 times?

Option 1:
$669\,°\,C$

Option 2:
$3295\,°\,C$

Option 3:
$3097 K$

Option 4:
$223\,K$

Correct Answer:
$3295\,°\,C$

Solution:

$v_{rms} \propto \sqrt{T}$

let initial speed is v As speed is increased by 3 times so final speed becomes 4v

$$\frac{V_2}{V_1} = \sqrt{\frac{T_2}{T_1}}$$
$$V_2 = 4\ V_1$$
$$T_2 = 3568\ K$$

$or\ 3295°C$

Hence, the answer is the option (2).

Q. 19 The ratio of frequencies of fundamental harmonic produced by an open pipe to that of closed pipe having the same length is :

Option 1:
$1:2$

Option 2:
$2:1$

Option 3:
$1:3$

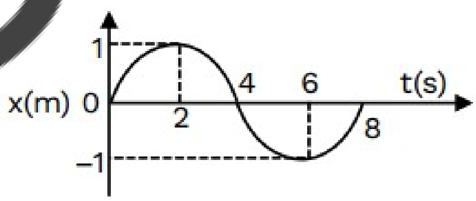
Option 4:
$3:1$

Correct Answer:
$2:1$

Solution:
$$\frac{n_{oop}}{n_{cop}} = \frac{\frac{v}{2\ell}}{\frac{v}{4\ell}} = \frac{2}{1}$$

Hence, the answer is the option (2).

Q. 20 The x-t graph of a particle performing simple harmonic motion is shown in the figure. The acceleration of the particle at t = 2 s is



Option 1:
$\frac{\pi^2}{8}\ \mathrm{ms}^{-2}$

Option 2:
$-\frac{\pi^2}{8}\ \mathrm{ms}^{-2}$

Option 3:
$\frac{\pi^2}{16}\ \mathrm{ms}^{-2}$

Option 4:
$-\frac{\pi^2}{16}\ \mathrm{ms}^{-2}$

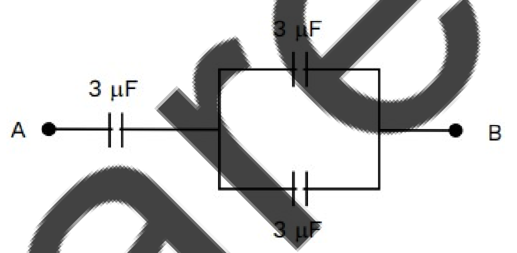
Correct Answer:
$-\frac{\pi^2}{16}\ \mathrm{ms}^{-2}$

Solution:
$a=-\omega^2 x$

$a=-\left(\frac{2\pi}{T}\right)^2\times 1$

$a=-\frac{\pi^2}{16}m/s^2$

Q. 21 The equivalent capacitance of the system shown in the following circuit is :



Option 1:
$2\,\mu\mathrm{F}$

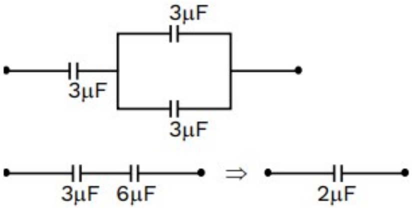
Option 2:
$3\,\mu\mathrm{F}$

Option 3:
$6\,\mu\mathrm{F}$

Option 4:
$9\,\mu\mathrm{F}$

Correct Answer:
$2\,\mu\text{F}$

Solution:



Q. 22 If $\oint_s \vec{E} \cdot \overrightarrow{ds} = 0$ over a surface, then:

Option 1:
the number of flux lines entering the surface must be equal to the number of flux lines leaving it.

Option 2:
the magnitude of electric field on the surface is constant.

Option 3:
all the charges must necessarily be inside the surface.

Option 4:
the electric field inside the surface is necessarily uniform

Correct Answer:
the number of flux lines entering the surface must be equal to the number of flux lines leaving it.

Solution:
$$\oint \overrightarrow{\mathrm{E}} \cdot \mathrm{d}\overrightarrow{\mathrm{S}} = \phi_{\text{in}} - \phi_{\text{out}} = 0 \Rightarrow \phi_{\text{in}} = \phi_{\text{out}}$$

Hence, the answer is the option (1).

Q. 23 An electric dipole is placed at an angle of 30° with an electric field of intensity $2 \times 10^5 \mathrm{NC}^{-1}$. It experiences a torque equal to 4 Nm. Calculate the magnitude of charge on the dipole, if the dipole length is 2 cm.

Option 1:
$8\,\mathrm{mC}$

Option 2:
$6\,\mathrm{mC}$

Option 3:
$4\,\mathrm{mC}$

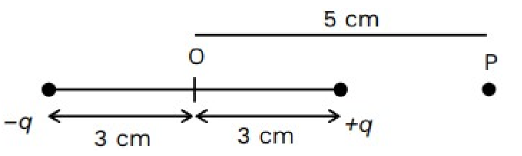
Option 4:
$2\,\mathrm{mC}$

Correct Answer:
$2\,\mathrm{mC}$

Solution:
$$\because \mathrm{q}=\frac{\tau}{\mathrm{dE}\sin\theta}=\frac{4}{0.02\times 2\times 10^5\times \sin 30^{\circ}}=2\times 10^{-3}\mathrm{C}=2\mathrm{mc}$$

Hence, the answer is the option (4).

Q. 24 An electric dipole is placed as shown in the figure.



The electric potential $(\text{in } 10^2\ \mathrm{V}\ )$ at point P due to the dipole is

$\left(\epsilon_0=\ \text{permittivity of free space and}\ \frac{1}{4\pi\epsilon_0}=K\right)$

Option 1:
$\left(\frac{3}{8}\right)\mathrm{qK}$

Option 2:
$\left(\frac{5}{8}\right)\mathrm{qK}$

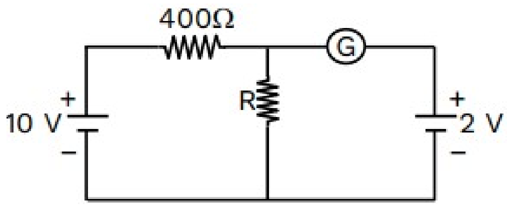
Option 3:
$\left(\frac{8}{5}\right)qK$

Option 4:
$\left(\frac{8}{3}\right)qK$

Correct Answer:
$\left(\frac{3}{8}\right)\mathrm{qK}$

Solution:
$$V=\frac{k(-q)}{8\times 10^{-2}}+\frac{k(q)}{2\times 10^{-2}}=\left(\frac{3}{8}\right)qk\times 10^2$$

Q. 25 If the galvanometer G does not show any deflection in the circuit shown, the value of R is given by :



Option 1:
$200\,\Omega$

Option 2:
$50\,\Omega$

Option 3:
$100\,\Omega$

Option 4:
$400\,\Omega$

Correct Answer:
$100\,\Omega$

Solution:
potential difference across r is given by

$$iR = V$$

$$\frac{10}{400+R} \times R = 2 \Rightarrow 5R = 400 + R$$
$$\Rightarrow 4R = 400$$
$$\Rightarrow R = 100\Omega$$

Q. 26 Resistance of a carbon resistor determined from colour codes is $(22000 \pm 5\%)\Omega$. The colour of third band must be:

Option 1:
Red

Option 2:
Green

Option 3:
Orange

Option 4:
Yellow

Correct Answer:
Orange

Solution:

$$\mathrm{R} = 22000 \pm 5\%$$

$$\mathrm{R} = 22 \times 10^{3} \pm 5\%$$

$$(\mathrm{I}, \mathrm{II} \times 10^{\mathrm{III}} \pm \mathrm{IV})$$

$$\mathrm{III} = \mathrm{Orange}$$

Q. 27 The resistance of platinum wire at $0^{\circ}\mathrm{C}$ is 2Ω and 6.8Ω at $80^{\circ}\mathrm{C}$. The temperature coefficient of resistance of the wire is:

Option 1:
$3 \times 10^{-4\circ}\mathrm{C}^{-1}$

Option 2:
$3 \times 10^{-3\circ}\mathrm{C}^{-1}$

Option 3:
$3 \times 10^{-2\circ}\mathrm{C}^{-1}$

Option 4:
$3 \times 10^{-1\circ}\mathrm{C}^{-1}$

Correct Answer:
$3 \times 10^{-2\circ}\mathrm{C}^{-1}$

Solution:

$$R_T = R_{\circ}(1 + \alpha T)$$

$$6.8 = 2[1 + \alpha(80)]$$

$$\alpha = \frac{3.4 - 1}{80} = \frac{2.4}{80} = 3 \times 10^{-2\circ}\mathrm{C}^{-1}$$

Hence, the answer is the option (3).

Q. 28 10 resistors, each of resistance R are connected in series to a battery of emf E and negligible internal resistance. Then those are connected in parallel to the same battery, the current is increased n times. The value of n is :

Option 1:
10

Option 2:
100

Option 3:
1

Option 4:
1000

Correct Answer:
100

Solution:

$$I_1 = \frac{E}{10R}$$

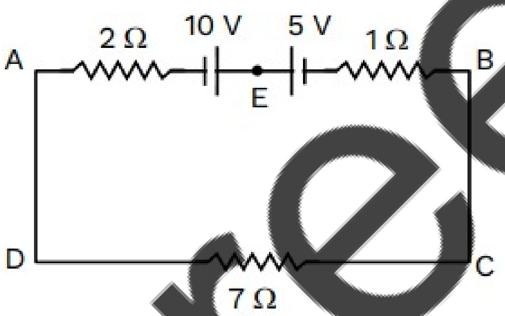
$$I_2 = \frac{E}{R/10} = \frac{10E}{R}$$

$$\Rightarrow I_2 = nI_1$$

$$\Rightarrow \frac{10E}{R} = n\left(\frac{E}{10R}\right) \Rightarrow n = 100$$

Hence, the answer is the option (2).

Q. 29 The magnitude and direction of the current in the following circuit are:



Option 1:
$0.2\ \mathrm{A}\ from\ B\ to\ A\ through\ E$

Option 2:
$0.5\ \mathrm{A}\ from\ A\ to\ B\ through\ E$

Option 3:
$\frac{5}{9}\ A\ from\ A\ to\ B\ through\ E$

Option 4:
$1.5\ \mathrm{A}\ from\ B\ to\ A\ through\ E$

Correct Answer:
$0.5 \text{ A} \, from \, A \, to \, B \, through \, E$

Solution:

$$I=\frac{\Delta V}{R_{\text{net}}}$$

$$\Rightarrow I=\frac{10-5}{2+7+1} \Rightarrow 0.5 \text{ A}$$

$from \, A \, to \, B \, through \, E$

Q. 30 The net magnetic flux through any closed surface is :

Option 1:
Zero

Option 2:
Positive

Option 3:
Infinity

Option 4:
Negative

Correct Answer:
Zero

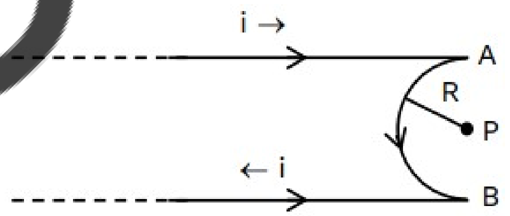
Solution:

$$\because \oint \vec{B} \cdot d\vec{s}=0$$

Because magnetic field lines make closed loops. So, for any closed surface. Net magnetic flux is always zero.

Hence, the answer is the option (1).

Q. 31 A very long conducting wire is bent in semi-circular shape from A to B as shown in figure. The magnetic field at point P for steady current configuration is given by:



Option 1:
$\frac{\mu_0 i}{4R}$ *pointed into the page*

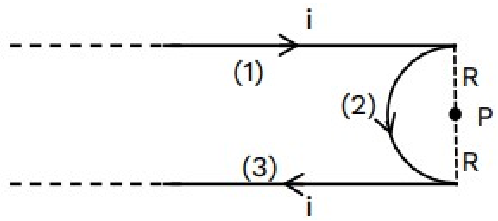
Option 2:
$\frac{\mu_0 i}{4R}$ *pointed away from the page*

Option 3:
$\frac{\mu_0 \mathrm{i}}{4\mathrm{R}}\left[1-\frac{2}{\pi}\right]$ *pointed away from page*

Option 4:
$\frac{\mu_0 \mathrm{i}}{4\mathrm{R}}\left[1-\frac{2}{\pi}\right]$ *pointed into the page*

Correct Answer:
$\frac{\mu_0 \mathrm{i}}{4\mathrm{R}}\left[1-\frac{2}{\pi}\right]$ *pointed away from page*

Solution:



$$\mathrm{B}_1=\mathrm{B}_2-\left(\begin{matrix}\mathrm{B}_1+\mathrm{B}_3\\ \otimes\end{matrix}\right)$$

$$B_p=\frac{\mu_0 i}{4R}-\left(\frac{\mu_0 i}{4\pi R}+\frac{\mu_0 i}{4\pi R}\right)$$

$$\mathrm{B}_{\mathrm{p}}=\frac{\mu_0 \mathrm{i}}{4\mathrm{R}}\left(1-\frac{2}{\pi}\right)\odot$$

Q. 32 A wire carrying a current I along the positive x-axis has length L. It is kept in a magnetic field $\tilde{\mathrm{B}}=(2\hat{\mathrm{i}}+3\hat{\mathrm{j}}-4\hat{\mathrm{k}})\mathrm{T}$. The magnitude of the magnetic force acting on the wire is:

Option 1:
$3\ \mathrm{IL}$

Option 2:
$\sqrt{5}\ \mathrm{IL}$

Option 3:
$5\ \mathrm{IL}$

Option 4:
$\sqrt{3}\ IL$

Correct Answer:
$5\ \mathrm{IL}$

Solution:
$$\vec{F}=I(\vec{L}\times\vec{B})$$
$$=I[(L\hat{i})\times(2\hat{i}+3\hat{j}-4\hat{k})]$$
$$\vec{F}=IL(3\hat{k}+4\hat{j})$$
$$|\vec{F}|=5IL$$

Hence, the answer is the option (3).

Q. 33 The magnetic energy stored in an inductor of inductance 4 μH carrying a current of 2 A is :

Option 1:
$4\ \mu\mathrm{J}$

Option 2:
$4\ \mathrm{mJ}$

Option 3:
$8\ \mathrm{mJ}$

Option 4:
$8\ \mu\mathrm{J}$

Correct Answer:
$8\ \mu\mathrm{J}$

Solution:
$$U=\frac{1}{2}Li^2$$
$$=\frac{1}{2}\times 4\times 10^{-6}\times 2^2$$
$$=8\times 10^{-6}\ \mathrm{J}$$
$$=8\mu\mathrm{J}$$

Hence, the answer is the option (4).

Q. 34 A 12 V, 60 W lamp is connected to the secondary of a step down transformer, whose primary is connected to ac mains of 220 V. Assuming the transformer to be ideal, what is the current in the primary winding ?

Option 1:
0.27 A

Option 2:
2.7 A

Option 3:
3.7 A

Option 4:
0.37 A

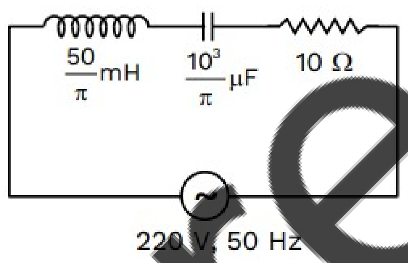
Correct Answer:
0.27 A

Solution:
For an ideal transformer,

$$P_{\text{in}} = P_{\text{out}}$$
$$\Rightarrow V_{\text{in}} I_{\text{in}} = V_{\text{out}} I_{\text{out}}$$
$$\Rightarrow 220 \times I_{\text{in}} = 60 \quad \Rightarrow I_{\text{in}} = 0.27 \text{ A}$$

Hence, the answer is the option (1).

Q. 35 The net impedance of circuit (as shown in figure) will be :



Option 1:
$10\sqrt{2}\Omega$

Option 2:
15Ω

Option 3:
$5\sqrt{5}\Omega$

Option 4:
25Ω

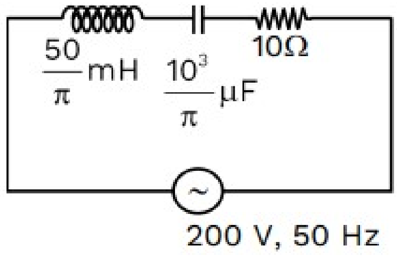
Correct Answer:
$5\sqrt{5}\Omega$

Solution:

$X_L = 2\pi \mathrm{fL}$

$= 2\pi \times 50 \times \dfrac{50}{\pi} \times 10^{-3}$

$X_L = 5\Omega$



$\mathrm{X_C} = \dfrac{1}{2\pi \mathrm{fC}} = \dfrac{10^6}{2\pi \times 50 \times \frac{10^3}{\pi}}$

$\mathrm{X_C} = 10\Omega$

$\mathrm{Z} = \sqrt{(10)^2 + (10 - 5)^2}$

$\mathrm{Z} = \sqrt{125} = 5\sqrt{5}\Omega$

Q. 36 In a series LCR circuit, the inductance L is 10 mH, capacitance C is 1μF and resistance R is $100\ \Omega$. The frequency at which resonance occurs is?

Option 1:
$15.9\ rad/\mathrm{s}$

Option 2:
$15.9\ \mathrm{kHz}$

Option 3:
$1.59\ \mathrm{rad/s}$

Option 4:
$1.59\ \mathrm{kHz}$

Correct Answer:
$1.59\ \mathrm{kHz}$

Solution:

$\mathrm{f_R} = \dfrac{1}{2\pi\sqrt{\mathrm{LC}}} = \dfrac{1}{2\pi\sqrt{10 \times 10^{-3} \times 10^{-6}}}$

$= \dfrac{10^4}{2\pi} \approx 1.59 \times 10^3\ \mathrm{Hz} = 1.59\mathrm{kHz}$

Hence, the answer is the option (4).

Q. 37 In a plane electromagnetic wave travelling in free space, the electric field component oscillates sinusoidally at a frequency of 2.0×10^{10} Hz and amplitude $48\, V m^{-1}$. Then the amplitude of oscillating magnetic field is :

$(\text{Speed of light in free space } = 3 \times 10^{8} \text{ m s}^{-1})$

Option 1:
1.6×10^{-9} T

Option 2:
1.6×10^{-8} T

Option 3:
1.6×10^{-7} T

Option 4:
1.6×10^{-6} T

Correct Answer:
1.6×10^{-7} T

Solution:

$$\because C = \frac{E_0}{B_0}$$

$$\Rightarrow \quad B_0 = \frac{E_0}{C} = \frac{48}{3 \times 10^{8}} = 16 \times 10^{-8} = 1.6 \times 10^{-7} \text{ T}$$

Hence, the answer is the option (3).

Q. 38 An ac source is connected to a capacitor C. Due to decrease in its operating frequency:

Option 1:
capacitive reactance decreases.

Option 2:
displacement current increases.

Option 3:
displacement current decreases.

Option 4:
capacitive reactance remains constant

Correct Answer:
displacement current decreases.

Solution:

$$Because\ I_c = I_d = \frac{v}{X_c}\left(\because x_c = \frac{1}{2\pi fc}\right)$$
$$Here\ if\ f \downarrow x_c \uparrow \Rightarrow I_c\ or\ I_d \downarrow (So\ displacement\ Decreases)$$

Hence, the answer is the option (3).

Q. 39 For Young's double slit experiment, two statements are given below:

Statement I : If screen is moved away from the plane of slits, angular separation of the fringes remains constant.

Statement II : If the monochromatic source is replaced by another monochromatic source of larger wavelength, the angular separation of fringes decreases.

In the light of the above statements; choose the correct answer from the options given below:

Option 1:
Both **Statement I** and **Statement II** are true.

Option 2:
Both **Statement I** and **Statement II** are false.

Option 3:
Statement I is true but **Statement II** is false.

Option 4:
Statement I is false but **Statement II** is true.

Correct Answer:
Statement I is true but **Statement II** is false.

Solution:

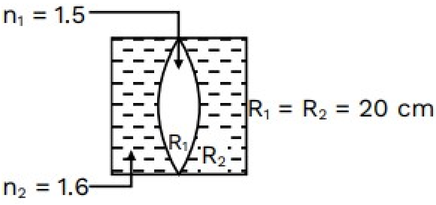
$\because \theta$ (angular fringe width) $= \frac{\lambda}{d}$

(i) In statement I : if $D \uparrow$ then $\theta \rightarrow$ same

(ii) In statement II : if $\lambda \uparrow$ then $\theta \uparrow$

Hence, the answer is the option (3).

Q. 40 In the figure shown here, what is the equivalent focal length of the combination of lenses (Assume that all layers are thin)?



Option 1:
$40 \mathrm{~cm}$

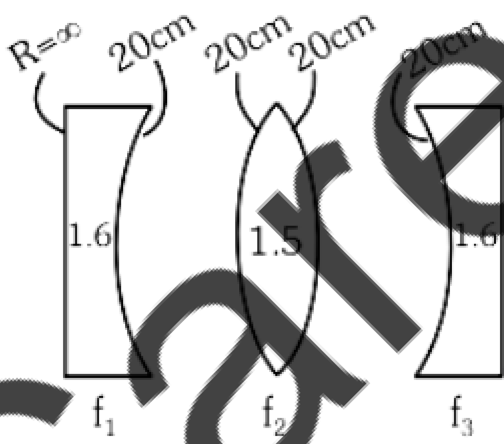
Option 2:
$-40 \mathrm{~cm}$

Option 3:
$-100 \mathrm{~cm}$

Option 4:
$-50 \mathrm{~cm}$

Correct Answer:
$-100 \mathrm{~cm}$

Solution:



$$\text{Use } \frac{1}{\mathrm{f}}=[\mu-1]\left[\frac{1}{\mathrm{R}_1}-\frac{1}{\mathrm{R}_2}\right]$$

$$\frac{1}{f_1}=[1.6-1]\left[\frac{1}{\infty}-\frac{1}{20}\right]=\frac{-3}{100}$$

$$\frac{1}{f_2}=[1.5-1]\left[\frac{1}{20}+\frac{1}{20}\right]=\frac{1}{20}$$

$$\frac{1}{f_3}=\frac{-3}{100}$$

$$\frac{1}{f_{eq}}=\frac{1}{f_1}+\frac{1}{f_2}+\frac{1}{f_3}$$
$$\frac{1}{f_{eq}}=-\frac{3}{100}+\frac{1}{20}-\frac{3}{100}=\frac{-1}{100}$$

$$f=-100 \text{ cm} \Rightarrow P=-1D$$

Q. 41 Two thin lenses are of the same focal lengths (f), but one is convex and the other one is concave. When they are placed in contact with each other, the equivalent focal length of the combination will be:

Option 1:
$Zero$

Option 2:
$f/4$

Option 3:
$f/2$

Option 4:
$Infinite$

Correct Answer:
$Infinite$

Solution:
$$\frac{1}{f_{\text{net}}}=\frac{1}{f}-\frac{1}{f} \quad \Rightarrow \quad \frac{1}{f_{\text{net}}}=0 \quad \Rightarrow f_{\text{net}}=\infty$$

Hence, the answer is the option (4).

Q. 42 Light travels a distance x in time t_1 in air and $10x$ in time t_2 in another denser medium. What is the critical angle for this medium?

Option 1:
$\sin^{-1}\left(\frac{t_2}{t_1}\right)$

Option 2:
$\sin^{-1}\left(\frac{10t_2}{t_1}\right)$

Option 3:
$\sin^{-1}\left(\frac{t_1}{10t_2}\right)$

Option 4:
$\sin^{-1}\left(\frac{10t_1}{t_2}\right)$

Correct Answer:
$\sin^{-1}\left(\frac{10t_1}{t_2}\right)$

Solution:

$$\sin\theta_c = \frac{v_2}{v_1} = \left(\frac{10x/t_2}{x/t_1}\right) = \frac{10t_1}{t_2}$$

$$\theta_c = \sin^{-1}\left(\frac{10t_1}{t_2}\right)$$

Hence, the answer is the option (4).

Q. 43 The work functions of Caesium (Cs), Potassium (K) and Sodium (Na) are 2.14 eV, 2.30 eV and 2.75 eV respectively. If incident electromagnetic radiation has an incident energy of 2.20 eV, which of these photosensitive surfaces may emit photoelectrons?

Option 1:
Cs only

Option 2:
Both Na and K

Option 3:
K only

Option 4:
Na only

Correct Answer:
Cs only

Solution:
For photoelectric effect

$$\left(\ E_{\text{photon incident}}\ \geq \phi\ (\text{work function})\ \right)$$

So the answer is Cs only.

Hence, the answer is the option (1).

Q. 44 The minimum wavelength of X-rays produced by an electron accelerated through a potential difference of V volts is proportional to:

Option 1:
$\sqrt{V}$

Option 2:
$\frac{1}{V}$

Option 3:
$\frac{1}{\sqrt{V}}$

Option 4:
V^2

Correct Answer:
$\frac{1}{V}$

Solution:
If the whole energy of e^- is converted into x-rays then the corresponding wavelength will be minimal.

$$\because \mathrm{eV}=\frac{\mathrm{hc}}{\lambda_{\min}}$$
$$\lambda_{\min}=\frac{\mathrm{hc}}{\mathrm{eV}} \Rightarrow \lambda_{\min} \propto \frac{1}{\mathrm{V}}$$

Hence, the answer is the option (2).

Q. 45 The half life of a radioactive substance is 20 minutes. In how much time, the activity of substance drops to $\left(\frac{1}{16}\right)^{\text{th}}$ of its initial value ?

Option 1:
20 minutes

Option 2:
40 minutes

Option 3:
60 minutes

Option 4:
80 minutes

Correct Answer:
80 minutes

Solution:
$$\left(\frac{1}{2}\right)^n = \frac{1}{16} \Rightarrow n = 4$$
$$\text{So, } \Delta T = 4 \times T_{\text{half}}$$
$$= 4 \times 20 \text{ min}$$
$$= 80 \text{ min}$$

Q. 46 The radius of inner most orbit of hydrogen atom is 5.3×10^{-11} m. What is the radius of third allowed orbit of hydrogen atom?

Option 1:
$0.53 \AA$

Option 2:
$1.06 \AA$

Option 3:
$1.59 \AA$

Option 4:
$4.77 \AA$

Correct Answer:
$4.77 \AA$

Solution:
$$r_n \propto n^2 \qquad (n = \text{ orbit No. })$$
$$\because r_1 = 5.3 \times 10^{-11} = 0.53 \AA$$
$$then r_3 = (3)^2 (0.53) = 4.77 \AA$$

Hence, the answer is the option (4).

Q. 47 In hydrogen spectrum, the shortest wavelength in the Balmer series is λ. The shortest wavelength in the Bracket series is :

Option 1:
2λ

Option 2:
4λ

Option 3:
9λ

Option 4:
16λ

Correct Answer:
4λ

Solution:
For Balmer series

$$\frac{1}{\lambda_{\min}}=\mathrm{R}\left(\frac{1}{(2)^2}-\frac{1}{\infty}\right)=\frac{\mathrm{R}}{4}$$
$$\Rightarrow \lambda_{\min}=\lambda=\frac{4}{\mathrm{R}} \quad \ldots(i)$$

For Bracket series

$$\frac{1}{\lambda'_{\min}}=\mathrm{R}\left(\frac{1}{(4)^2}-\frac{1}{\infty}\right)=\frac{\mathrm{R}}{16}$$
$$\Rightarrow \lambda'_{\min}=\frac{16}{\mathrm{R}} \quad \ldots(ii)$$
$$\Rightarrow \lambda'_{\min}=4\lambda$$

Hence, the answer is the option (2).

Q. 48 Given below are two statements:

Statement I : Photovoltaic devices can convert optical radiation into electricity.

Statement II : Zener diode is designed to operate under reverse bias in breakdown region.
In the light of the above statements, choose the most appropriate answer from the options given below :

Option 1:
Both **Statement I** and **Statement II** are correct ·

Option 2:
Both **Statement I** and **Statement II** are incorrect.

Option 3:
Statement I is correct but **Statement II** is incorrect.

Option 4:
Statement I is incorrect but **Statement II** is correct.

Correct Answer:
Both **Statement I** and **Statement II** are correct ·

Solution:
Both Statements I and Statement II are correct.

(I) Photo voltaic devices convert EMW into electricity

(II) Zener diode works as a voltage stabilizer in the reverse biased breakdown region.

Hence, the answer is the option (1).

Q. 49 A full wave rectifier circuit consists of two p-n junction diodes, a centre-tapped transformer, capacitor and a load resistance. Which of these components remove the ac ripple from the rectified output ?

Option 1:
A centre-tapped transformer

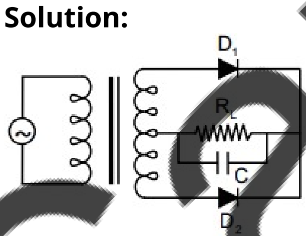
Option 2:
p-n junction diodes

Option 3:
Capacitor

Option 4:
Load resistance

Correct Answer:
Capacitor

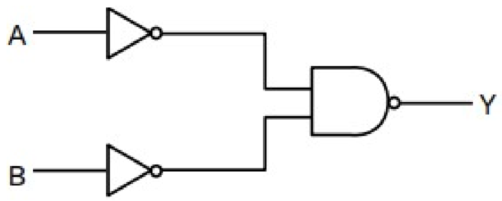
Solution:



Capacitor works here as a filter means remove ripples from rectified DC wave .

The role of the capacitor is to filters out the AC ripple and provides pure DC output.

Q. 50 For the following logic circuit, the truth table is:



Option 1:

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

Option 2:

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

Option 3:

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

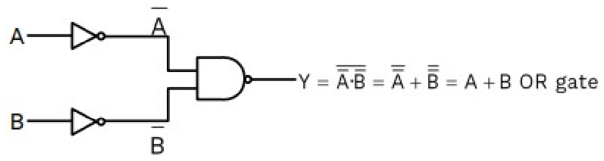
Option 4:

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

Correct Answer:

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

Solution:



relative table is

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

Chemistry

Q. 1 The right option for the mass of CO_2 produced by 20 g of 20% pure limestone is (Atomic mass of Ca =40)

$$CaCO_3 \overset{1200K}{\rightarrow} CaO + CO_2$$

Option 1:
1.12g

Option 2:
1.76g

Option 3:
2.64g

Option 4:
1,32g

Correct Answer:
1.76g

Solution:

$$\% \text{ purity of a reactant } \frac{(\text{ amount of pure reactant })}{\text{total amount of sample}} \times 100$$

$$20=\frac{\text{amount of pure reactant } \times 100}{20 \mathrm{~g}}$$

$$\therefore \text{ pure substance } = 4 \mathrm{~g} CaCO_3$$

$$CaCO_3 \longrightarrow CaCO + CO_2$$

$$100 \mathrm{~g}$$

$$\therefore 100 \text{ of } CaCO_3 \longrightarrow 44 \mathrm{~g} \text{ of } CO_2 \text{ is produced}$$

$$\therefore 4 \mathrm{~g} \text{ of } CaCO_3 \longrightarrow \frac{44}{100} \times 4 \mathrm{~g} \text{ of } CO_2 \text{ will be produced}$$

$$=1.76 \mathrm{~g}$$

Hence, the answer is the option (2).

Q. 2 Select the correct statement from the following:

A. Atoms of all elements are composed of two fundamental particles.

B. The mass of the electron is $9.10939 \times 10^{-31} kg$.

C. All the isotopes of given elements show the same chemical properties.

D. Protons and electrons are collectively known as nucleons

E. Dalton's atomic theory regarded the atom as an ultimate particle of matter.

Choose the correct answer from the option given below:

Option 1:
A, B and C only

Option 2:
C, D and E only

Option 3:
A and E only

Option 4:
B, C and E only

Correct Answer:
B, C and E only

Solution:

Statements (A) and (D) are incorrect.
Because → Atoms of all elements are composed of their fundamental particles (e, p, n)
(D) 'Protons' & 'Neutrons' are collectively known as nucleons.

Hence, the answer is the option (4).

Q. 3 Which of the following reaction will NOT given primary amine as the product?

Option 1:
$CH_3CONH_2 \overset{Br_2/KOH}{\rightarrow}$ product

Option 2:
$CH_3CN \xrightarrow[(ii) H_3O+]{(i) LiAlH_3}$ Product

Option 3:
$CH_3NC \xrightarrow[(ii) H_3O+]{(i) LiAlH_3}$ Product

Option 4:
$CH_3CONH_2 \xrightarrow[(ii) H_3O+]{(i) LiAlH_3}$ Product

Correct Answer:
$CH_3NC \xrightarrow[(ii) H_3O+]{(i) LiAlH_3}$ Product

Solution:

(1) $CH_3-\overset{\overset{O}{\|}}{C}-NH_2 \xrightarrow{Br_2.KOH} CH_3-NH_2 + 2NaBr + 2H_2O$
p-amine
[Hofmann bromamide degradation]

(2) $CH_3CN \xrightarrow[(ii)\ H_3O^{\oplus}]{(i)\ LiAlH_4} CH_3-CH_2-NH_2$
p-amine

(3) $CH_3NC \xrightarrow[(ii)\ H_3O^{\oplus}]{(i)\ LiAlH_4} CH_3-NH-CH_3$
Sec-amine

(4) $CH_3-\overset{\overset{O}{\|}}{C}-NH_2 \xrightarrow[(ii)\ H_3O^{\oplus}]{(i)\ LiAlH_4} CH_3-CH_2-NH_2$
p-amine

Q. 4 The relation between $n_m, (n_m = \text{the number of permissible values of magnetic quantum number})$ for a given value of azimuthal quantum number (l), is

Option 1:
$$l = \frac{n_m - 1}{2}$$

Option 2:
$$l = 2n_m + 1$$

Option 3:
$$n_m = 2l^2 + 1$$

Option 4:
$$n_m = l + 2$$

Correct Answer:
$$l = \frac{n_m - 1}{2}$$

Solution:
m (magnetic quantum No.)
Can have values -l to +l (including zero)
Total values of m in a given

$$\text{Subshell} = (2l + 1)$$
$$\downarrow$$
$$n_m = 2l + 1$$

$$l = \frac{n_m - 1}{2}$$

Q. 5 The element expected to form the largest ion to achieve the nearest noble gas configuration is:

Option 1:
O

Option 2:
F

Option 3:
N

Option 4:
Na

Correct Answer:
N

Solution:
$\mathrm{N}^{-3} > \mathrm{O}^{2-} > \mathrm{F}^{-} > \mathrm{Na}^{+}$
Hence, the answer is the option (3).

Q. 6 The correct order of energies of molecular orbitals of N_2 molecule is:

Option 1:
$\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < (\pi 2p_x = \pi 2p_y) < \sigma 2p_z < (\pi^* 2p_x = \pi^* 2p_y) < \sigma^* 2p_z$

Option 2:
$\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \sigma 2p_x < (\pi 2p_x = \pi 2p_y) < (\pi^* 2p_x = \pi^* 2p_y) < \sigma^* 2p_x$

Option 3:
$\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \sigma 2p_z < \sigma^* 2p_z < (\pi 2p_x = \pi 2p_y) < (\pi^* 2p_x = \pi^* 2p_y)$

Option 4:
$\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < (\pi 2p_x = \pi 2p_x) < (\pi^* 2p_x = \pi^* 2p_y) < \sigma 2p_z < \sigma^* 2p_z$

Correct Answer:
$\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < (\pi 2p_x = \pi 2p_y) < \sigma 2p_z < (\pi^* 2p_x = \pi^* 2p_y) < \sigma^* 2p_z$

Solution:
$\mathrm{N}_2 = 14 \text{ electron}$
$\sigma 1s^2 < \sigma^* 1s^2 < \sigma 2s^2 < \sigma^* 2s^2 < \pi 2P_x^2 = \pi 2p_y^2 < \sigma 2p_z^2 <$
$\pi \star 2P_x^0 = \pi \star 2p_y^0 < \sigma^* 2p_z^0$

Hence, the answer is the option (1).

Q. 7 Amongst the following, the total number of species NOT having eight electrons around central atom in its outer most shell, is $NH_3, \mathrm{AlCl}_3, \mathrm{BeCl}_2, \mathrm{CCl}_4, \mathrm{PCl}_5$:

Option 1:
3

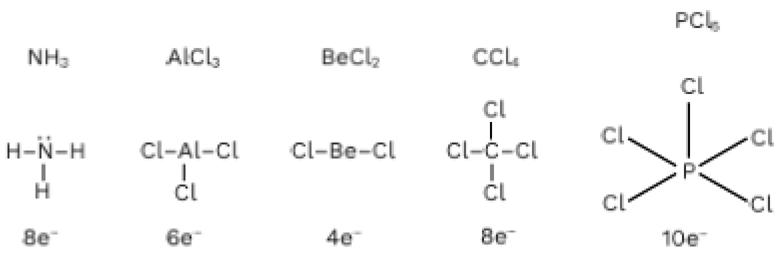
Option 2:
2

Option 3:
4

Option 4:
1

Correct Answer:
3

Solution:



So, $AlCl_3$, $BeCl_2$ and PCl_5 do not follow octet rule.

Q. 8 Intermolecular forces are forces of attraction and repulsion between interacting particles that will include:

A. Dipole - dipole forces
B. dipole - induced dipole forces.
C. hydrogen bonding
D. covalent bonding
E. dispersion forces.

Choose the most appropriate answer from the options given below:

Option 1:
B, C, D, E are correct

Option 2:
A, B, C, D are correct

Option 3:
A, B, C, E are correct

Option 4:
A, C, D, E are correct

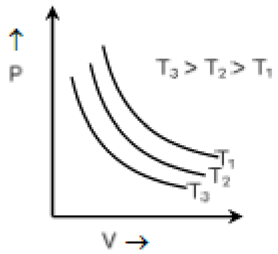
Correct Answer:
A, B, C, E are correct

Solution:
Covalent bonding is not Intermolecular forces, it is a bond.

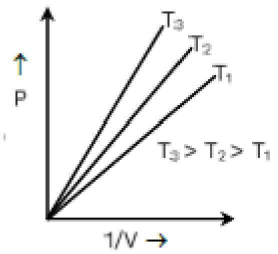
A, B, C, and E are correct

Q. 9 Which amongst the following options is correct graphical representation of Boyle's Law?

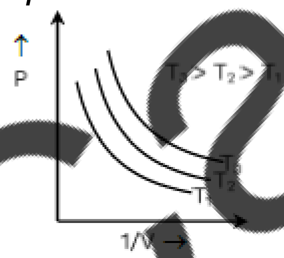
Option 1:



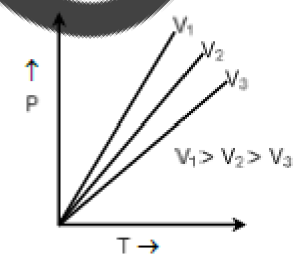
Option 2:



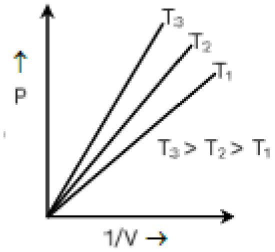
Option 3:



Option 4:



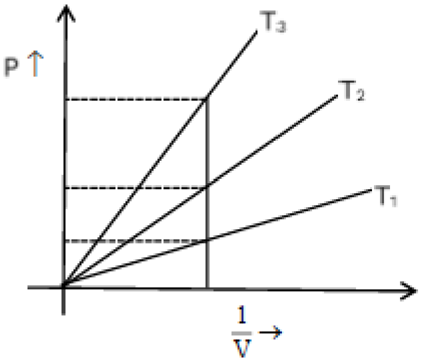
Correct Answer:



Solution:
Acoording to Boyle's Law,

$$P \propto \frac{1}{V}$$

At a constant T, P vs $\frac{1}{V}$ graph should be straight line.



At high P, temperature will be higher at given volume, so $T_3 > T_2 > T_1$

Q. 10 Which of the following options is the correct relation between change in enthalpy and change in internal energy?

Option 1:
$\Delta \mathrm{H} = \Delta U - \Delta n_g RT$

Option 2:
$\Delta \mathrm{H} = \Delta \mathrm{U} + \Delta \mathrm{n_g} RT$

Option 3:
$\Delta \mathrm{H} - \Delta \mathrm{U} = -\Delta n RT$

Option 4:
$\Delta \mathrm{H} + \Delta \mathrm{U} = \Delta \mathrm{nR}$

Correct Answer:
$\Delta \mathrm{H} = \Delta \mathrm{U} + \Delta \mathrm{n_g} RT$

Solution:
Relation between $\Delta \mathrm{H} \, \& \, \Delta \mathrm{U},$

$\Delta \mathrm{H} = \Delta \mathrm{U} + \Delta \mathrm{n_g} RT$

Hence, the answer is the option (2).

Q. 11 Amongst the given options which of the following molecules/ion acts as a Lewis acid?

Option 1:
NH_3

Option 2:
H_2O

Option 3:
BF_3

Option 4:
OH^-

Correct Answer:
BF_3

Solution:
Boron trihalides (BX_3) are electron deficient in nature, hence acting as Lewis acid.

Hence, the answer is the option (3).

Q. 12 The equilibrium concentrations of the species in the reaction $A + B \rightleftharpoons C + D$ are $2, 3, 10$ and $6 \mathrm{~mol~L^{-1}}$, respectively at $300 \mathrm{~K}. \Delta \mathrm{G}^{\circ}$ for the reaction is $(\mathrm{R} = 2 \mathrm{cal/mol~K})$

Option 1:
$1372.60 \mathrm{~cal}$

Option 2:
$-137.26 \mathrm{~cal}$

Option 3:
$-1381.80 \mathrm{~cal}$

Option 4:
$-13.73 \mathrm{~cal}$

Correct Answer:
$-1381.80 \mathrm{~cal}$

Solution:

$$A + B \rightleftharpoons C + D$$
$$2M \quad 3M \quad 10M \quad 6M$$
$$K_C = \frac{10 \times 6}{2 \times 3} = 10$$
$$\Delta G^\circ = -2.303RT \log K_C$$
$$= -2.303 \times 2 \times 300 \log 10$$
$$= -1381.80 \text{cal}$$

Hence, the answer is the option (3).

Q. 13 On balancing the given redox reaction,

$$aCr_2O_7^{2-} + bSO_3^{2-}(aq) + cH^+(aq) \rightarrow 2aCr^{3+}(aq) + bSO_4^{2-}(aq) + \frac{c}{2}H_2O(l)$$

Option 1:
$1, 3, 8$

Option 2:
$3, 8, 1$

Option 3:
$1, 8, 3$

Option 4:
$8, 1, 3$

Correct Answer:
$1, 3, 8$

Solution:

$Cr_2O_7^{2-} \longrightarrow 2Cr^{3+}$ (Reduction half cell)

$14H^+ + Cr_2O_7^{2-} \longrightarrow 2Cr^{3+} + 7H_2O$ (balance O & H)

$Cr_2O_7^{2-} + 14H^+ + 6e^- \longrightarrow 2Cr^{3+} + 7H_2O$ ... (1) (balance O & H)

$H_2O + SO_3^{2-} \longrightarrow SO_4^{2-} + 2H^+$ (balance O & H)

$H_2O + SO_3^{2-} \longrightarrow SO_4^{2-} + 2H^+ + 2e^-$ ... (2) (charge balance)

Multiply equation 2 by 3 and then adding to equation 1,

$$\underset{\substack{\downarrow \\ a}}{Cr_2O_7^{2-}} + 14H^+ + \underset{\substack{\downarrow \\ b}}{3SO_3^{2-}} + \underset{\substack{\downarrow \\ c}}{8H^+} \longrightarrow 2Cr^{3+} + 4H_2O + 6SO_4^{2-}$$

Q. 14 Which of the following statements are NOT correct?

A. Hydrogen is used to reduce heavy metal oxides to metals.

B. Heavy water is used to study reaction mechanism.

C. Hydrogen is used to make saturated fats from oils.

D. The H-H bond dissociation enthalpy is lowest as compared to a single bond between two atoms of any element.

E. Hydrogen reduces oxides of metals that are more active than iron.

Choose the most appropriate answer from the options given below.

Option 1:
B, C, D, E only

Option 2:
B, D only

Option 3:
D, E only

Option 4:
A, B, C only

Correct Answer:
D, E only

Solution:
The H-H bond dissociation enthalpy is NOT the lowest as compared to a single bond between two atoms of any element.

Hydrogen Cannot reduce oxides of metals that are more active than iron.

D and E are incorrect.

Q. 15 Which one of the following statements is correct?

Option 1:
The daily requirement of Mg and Ca in the human body is estimated to be 0.2-0.3 g

Option 2:
All enzymes that utilise ATP in phosphate transfer require Ca as the cofactor.

Option 3:
The bone in human body is an inert and unchanging substance.

Option 4:
Mg plays roles in neuromuscular function and interneuronal transmission.

Correct Answer:
The daily requirement of Mg and Ca in the human body is estimated to be 0.2-0.3 g

Solution:
The daily requirement of Mg and Ca in the human body is estimated to be 0.2-0.3 g

Correct option 1

Q. 16 Match List-I with List-II

List - I	List - II
A. Coke	I. Carbon atoms are sp^3 hybridised.
B. Diamond	II. Used as a dry lubricant
C. Fullerene	III. Use as a reducing agent
D. Graphite	IV. Cage like molecules

Choose the correct answer from the options given below:

Option 1:
A-II, B-IV, C-I, D-III

Option 2:
A-IV, B-I, C-II, D-III

Option 3:
A-III, B-I, C-IV, D-II

Option 4:
A-III, B-IV, C-I, D-II

Correct Answer:
A-III, B-I, C-IV, D-II

Solution:
Group - 14 (allotropes of carbon)
A. Coke : III. Use as a reducing agent.
B. Diamond : I. Carbon atoms are sp^3 hybridised.
C. Fullerene : IV. Cage like molecules
D. Graphite : II. Used as a dry lubricant

So, the correct match will be
A-III, B-I, C-IV, D-II

Correct option 3

Q. 17 Taking stability as the factor, which one of the following represents the correct relationship?

Option 1:
$TlCl_3 > TlCl$

Option 2:
$InI_3 > InI$

Option 3:
$AlCl > AlCl_3$

Option 4:
$TlI > TlI_3$

Correct Answer:
$TlI > TlI_3$

Solution:
Due to the inert pair effect, Tl^{+} is more stable than Tl^{3+}.

Hence, the answer is the option (4).

Q. 18 Given below are two statements: one labelled as Assertion A and the other labelled as Reason R:
Assertion A: Helium is used to dilute oxygen in diving apparatus.
Reason R: Helium has high solubility in O_2.

In the light of the above statements choose the correct answer from the options given below:

Option 1:
Both A and R are true and R is the correct explanation of A.

Option 2:
Both A and R are true and R is NOT the correct explanation of A.

Option 3:
A is true but R is false

Option 4:
A is false but R is true

Correct Answer:
Both A and R are true and R is NOT the correct explanation of A.

Solution:
Assertion & Reason both are correct because, helium is used to dilute oxygen for scuba diving but reason is not correct explanation, because Helium has high solubility in O_2 as gases are high miscible with each other.

Q. 19 Given below are two statements : one is labelled as Assertion A and the other is labelled a Reason R:

Assertion A : Metallic sodium dissolves in liquid ammonia giving a deep blue solution, which paramagnetic.
Reasons R : The deep blue solution is due to the formation of amide.

In the light of the above statements, choose the correct answer from the options given below:

Option 1:
Both A and R are true and R is the correct explanation of A.

Option 2:
Both A and R are true but R is NOT the correct explanation of A.

Option 3:
A is true but R is false.

Option 4:
A is true but R is false.

Correct Answer:
A is true but R is false.

Solution:
Metallic sodium dissolves in liquid ammonia giving a deep blue solution, which is paramagnetic. The deep blue solution is due to the formation of ammoniated electrons.

A is true, But R is false.

Q. 20

List -I	(Oxoaciids of Sulphur)	List - II	(Bonds of Sulphur)
A.	Peroxodisulphuric acid	I.	$\text{Two S} - \text{OH, Four S} = \text{O, One S} - \text{O} - \text{S}$
B.	Sulphuric acid	II.	$\text{Two S} - \text{OH, One S} = \text{O}$
C.	Pyrosulphuric acid	III.	$\text{Two S} - \text{OH, Four S} = \text{O, One S} - \text{O} - \text{O} - \text{S}$
D.	Sulphurous acid	IV.	$\text{Two S} - \text{OH, Two S} = \text{O}$

Choose the correct answer from the options given below:

Option 1:
$A - I, B - III, C - II, D - IV$

Option 2:
$A - III, B - IV, C - I, D - II$

Option 3:
$A - I, B - III, C - IV, D - II$

Option 4:
$A - III, B - IV, C - II, D - I$

Correct Answer:
$A - III, B - IV, C - I, D - II$

Solution:
A. Peroxodisulphuric acid :

$\text{III. Two S} - \text{OH, Four S} = \text{O, One S} - \text{O} - \text{O} - \text{S}$
B. Sulphuric acid :

$\text{IV. Two S} - \text{OH, Two S} = \text{O}$
C. Pyrosulphuric acid :

$\text{II. Two S} - \text{OH, One S} = \text{O}$
D. Sulphurous acid : $\text{I. Two S} - \text{OH, Four S} = \text{O, One S} - \text{O} - \text{S}$

$\text{(2) A-III, B-IV, C-I, D-II}$ is correct.

Q. 21 In Lassaigne's extract of an organic compound, both nitrogen and sulfur are present, which gives blood red color with Fe^{3+} due to the formation of -

Option 1:
$Fe_4[Fe(CN)_6]_3.xH_2O$

Option 2:
$NaSCN$

Option 3:
$[Fe(CN)_6NOS]^{4-}$

Option 4:
$[Fe(SCN)]^{2+}$

Correct Answer:
$[Fe(SCN)]^{2+}$

Solution:
In case, nitrogen and sulfur both are present in an organic compound, sodium thiocyanate is formed. It gives blood a red color.

$$\mathrm{Na + C + N + S \rightarrow NaSCN}$$

$$\mathrm{Fe^{3+} + SCN^- \rightarrow \underset{Blood\ Red}{[Fe(SCN)]^{2+}}}$$

Hence, the answer is the option (4).

Q. 22 The number of σ bonds, π bonds and lone pair of electrons in pyridine, respectively are:

Option 1:
11,12,0

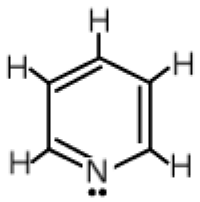
Option 2:
12,3,0

Option 3:
11,3,1

Option 4:
12,2,1

Correct Answer:
11,3,1

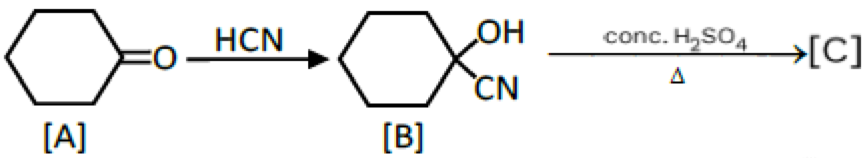
Solution:



Pyridine

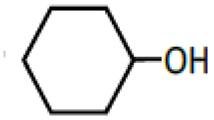
σ bond $\Rightarrow$ 11
π bond $\Rightarrow$ 3
Lone pair $\Rightarrow$ 1

Q. 23 Complete the following reaction:

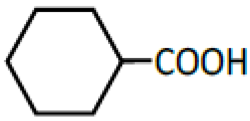


[C] is _____

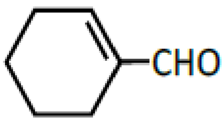
Option 1:



Option 2:

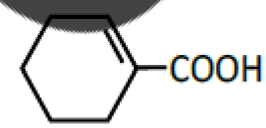


Option 3:

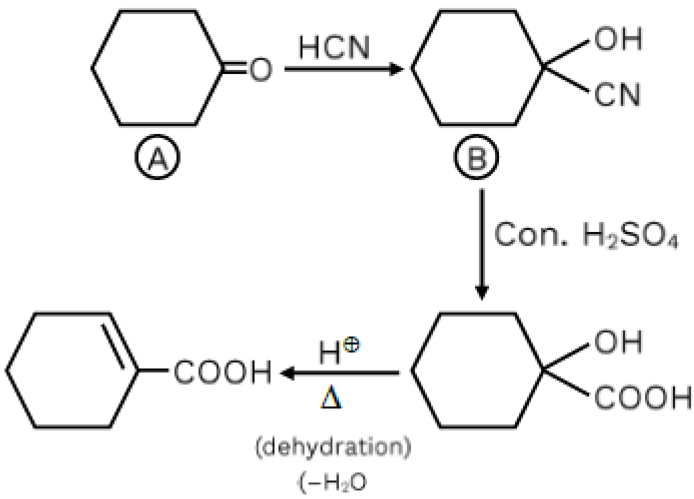


Option 4:

Correct Answer:



Solution:



Q. 24 Weight (g) of two moles of the organic compound, which is obtained by heating sodium ethanoate with sodium hydroxide in the presence of calcium oxide is:

Option 1:
16

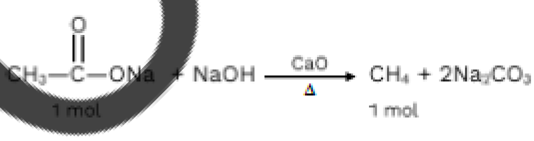
Option 2:
32

Option 3:
30

Option 4:
18

Correct Answer:
32

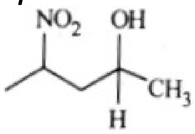
Solution:



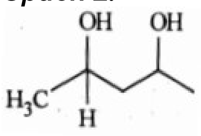
$$\text{Weight of 1 mole } CH_4 \Rightarrow 16gm$$
$$\text{So weight of 2 mole } CH_4 \Rightarrow 32gm$$

Q. 25 Which amongst the following will be most readily dehydrated under acidic conditions ?

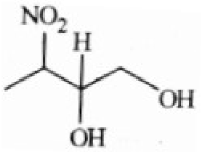
Option 1:



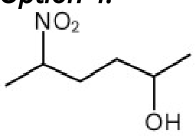
Option 2:



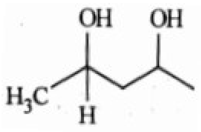
Option 3:



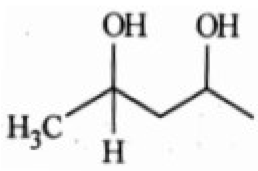
Option 4:



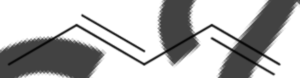
Correct Answer:



Solution:

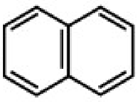
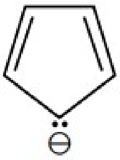
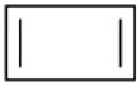
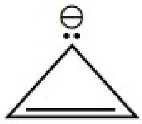


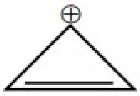
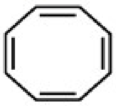
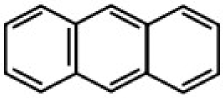
This compound will be most readily dehydrated under acidic conditions due to the presence of conjugation in the product.



Correct option 2

Q. 26 Consider the following compounds/ species:

(i)  (ii)  (iii)  (iv) 

(v)  (vi)  (vii) 

The number of compounds/species which obey Huckel's rule is ____________.

Option 1:
4

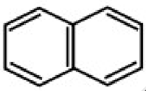
Option 2:
6

Option 3:
2

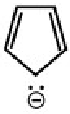
Option 4:
5

Correct Answer:
4

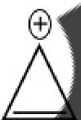
Solution:
According to Huckel's rule compound is aromatic if it is cycle, planar and have $4n+2\pi$ electrons.
According to it number of compounds species which obey huekel's rule is.

(i) 

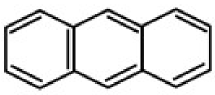
4n + 2 = 10
4n = 8
n = 2

(ii) 

4n + 2 = 6
4n = 4
n = 1

(v) 

4n + 2 = 2
4n = 0
n = 0

(ii) 

4n + 2 = 14
4n = 12
n = 3

n = 0, 1, 2, 3 .....

Q. 27 Given below are two statements:
Statement I: The nutrient deficient water bodies lead to eutrophication.
Statement II: Eutrophication leads to decrease in the level of oxygen in the water bodies.
Choose the light of the above statements choose the correct answer from the options given below:

Option 1:
Both Statement-I and Statement-II are true

Option 2:
Both Statement-I and Statement-II are false

Option 3:
Statement-I is correct but Statement-II is false

Option 4:
Statement-I is incorrect but Statement-II is true

Correct Answer:
Statement-I is incorrect but Statement-II is true

Solution:
Nutrient enriched water bodies support a dense plant population which kills animal life by depriving it of oxygen and results in subsequent loss of biodiversity is known as Eutrophication

Q. 28 A compound is formed by two elements A and B. The elements B forms cubic close packed structure and atoms of A occupy $\frac{1}{3}$ of tetrahedral voids. If the formula of the compound is $A_x B_y$, then the value of x+y is in option

Option 1:
5

Option 2:
4

Option 3:
3

Option 4:
2

Correct Answer:
5

Solution:
$B \rightarrow CCP \rightarrow 4$
$A \rightarrow \frac{1}{3}$ of THV $\rightarrow \frac{1}{3} \times 8$
$A_{8/3}B_4 \rightarrow$ formula of compound
or A_8B_{12}
or $A_2\ B_3$ (simple ratio)
So, $x = 2$ & $y = 3$
$x + y = 2 + 3 = 5$

Q. 29 What fraction of one edge centred octahedral void lies in one unit cell of fcc ?

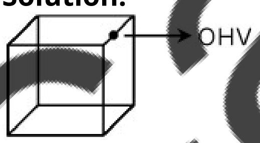
Option 1:
$\frac{1}{2}$

Option 2:
$\frac{1}{3}$

Option 3:
$\frac{1}{4}$

Option 4:
$\frac{1}{12}$

Correct Answer:
$\frac{1}{4}$

Solution:


In Fcc unit cell, OHV at Edge centre is equally shared in 4 unit cell so in one unit cell, only $\frac{1}{4}th$ part is present.

Q. 30 Given below are two statements: one is labeled as Assertion A and the other is labeled as Reason R :

Assertion A: In equation $\Delta G = -nFE_{\text{cell}}$, value of ΔG depends on n.

Reasons R : E_{cell} is an intensive property and ΔG is an extensive property.

In the light of the above statements, choose the correct answer from the options given below :

Option 1:
Both A and R are true and R is the correct explanation of A.

Option 2:
Both A and R are true and R is NOT the correct explanation of A.

Option 3:
A is true but R is false.

Option 4:
A is false but R is true.

Correct Answer:
Both A and R are true and R is the correct explanation of A.

Solution:

$\Delta G_r = -FE_{cell}$

$\Delta G_r = \text{depends upon 'n'}$

$So, \Delta_r G$ is an extensive property

Both A and R are true and R is the correct explanation of A.

Hence, the answer is the option (1).

Q. 31 The conductivity of the centimolar solution $\mathrm{KCl\ at\ 250^{\circ}C}$ is $0.0210 \mathrm{ohm}^{-1}\ \mathrm{cm}^{-1}$ and resistance of the cell containing the solution at $25^{\circ}\mathrm{C}$ is 60 ohms. The value of cell cons is

Option 1:
$1.34\ \mathrm{cm}^{-1}$

Option 2:
$3.28 \mathrm{~cm}^{-1}$

Option 3:
$1.26 \mathrm{~cm}^{-1}$

Option 4:
$3.34 \mathrm{~cm}^{-1}$

Correct Answer:
$1.26 \mathrm{~cm}^{-1}$

Solution:

$$\mathrm{K}=\frac{1}{R} \mathrm{G}^*$$

$$0.0210=\frac{1}{60} \times \mathrm{G}^*$$

$$\mathrm{G}^*=1.26 \mathrm{~cm}^{-1}$$

Hence, the answer is the option (3).

Q. 32 Given below are two statements: one is labeled as Assertion A and the other is labeled as Reason R :

Assertion A: A reaction can have zero activation energy.

Reasons R: The minimum extra amount of energy absorbed by reactant molecules so that their energy becomes equal to the threshold value, is called activation energy.

In the light of the above statements, choose the correct answer from the options given below :

Option 1:
Both A and R are true and R is the correct explanation of A.

Option 2:
Both A and R are true and R is NOT the correct explanation of A.

Option 3:
A is true but R is false.

Option 4:
A is false but R is true.

Correct Answer:
Both A and R are true and R is NOT the correct explanation of A.

Solution:
The free radical reactions can have zero activation energy. In free radical combination reaction in termination, steps have $E_a = 0$

Hence, the answer is the option (2).

Q. 33 For a certain reaction, the rate $= k[A]^2[B]$, when the initial concentration of A is tripled keeping the concentration of B constant, the initial rate would

Option 1:
decrease by a factor of nine.

Option 2:
increase by a factor of six.

Option 3:
increase by a factor of nine

Option 4:
increase by a factor of three

Correct Answer:
increase by a factor of nine

Solution:
$r = K[A]^2[B]$
$r \propto [A]^2, \text{ if } [B] \text{ remains constant}$
$r_f \propto [3A]^2$
$r_f \propto 9 \times [A]^2 \propto 9 \times r_i$

So, the rate will increase 9 times of initial rate.

Hence, the answer is the option (3).

Q. 34 Which one is an example of heterogenous catalysis?

Option 1:
Oxidation of sulphur dioxide into sulphur trioxide in the presence of oxides of nitrogen.

Option 2:
Hydroiysis of sugar catalysed by H^{+}ions.

Option 3:
Decomposition of ozone in presence of nitrogen monoxide.

Option 4:
Combination between dinitrogen and dihydrogen to form ammonia in the presence of finely divided iron.

Correct Answer:
Combination between dinitrogen and dihydrogen to form ammonia in the presence of finely divided iron.

Solution:
$N_{2(g)} + 3H_{2(g)} \longrightarrow 2NH_{3(g)}$

Since reactants and catalysts are in different phases (gas and solid respectively) so it is called as heterogeneous catalysis.

Q. 35 Pumice stone is an example of

Option 1:
sol

Option 2:
gel

Option 3:
solid sol

Option 4:
foam

Correct Answer:
solid sol

Solution:
Pumice stone is an example of solid sol with 'gas' as 'Dispersed phase' and solid as 'Dispersion medium.

Q. 36 The reaction that does NOT take place in a blast furnace between $900\ \mathrm{K}$ to $1500\ \mathrm{K}$ temperature range during extraction of iron is:

Option 1:
$\mathrm{Fe_2O_3 + CO \rightarrow 2FeO + CO_2}$

Option 2:
$\mathrm{FeO + CO \rightarrow Fe + CO_2}$

Option 3:
$\mathrm{C + CO_2 \rightarrow 2CO}$

Option 4:
$\mathrm{CaO + SiO_2 \rightarrow CaSiO_3}$

Correct Answer:
$\mathrm{Fe_2O_3 + CO \rightarrow 2FeO + CO_2}$

Solution:

At 500-800K

$\mathrm{CaCO_3 \rightarrow CaO + CO_2}$
(Limestone)

$\mathrm{3Fe_2O_3 + CO \rightarrow 2Fe_3O_4 + CO_2}$
(Iron ore)

$\mathrm{Fe_3O_4 + CO \rightarrow 3FeO + CO_2}$

At 900-1500K

$\mathrm{FeO + CO \rightarrow Fe + CO_2}$

$\mathrm{C + CO_2 \rightarrow 2CO}$

$\mathrm{CaO + SiO_2 \rightarrow \underset{(Slag)}{CaSiO_3}}$

Burning of Coke

$\mathrm{C + O_2 \rightarrow CO_2}$

$\mathrm{FeO + C \rightarrow Fe + CO}$

Correct option 1

Q. 37 The stability of Cu^{2+} is more than Cu^{+} salts in aqueous solution due to -

Option 1:
first ionisation enthalpy

Option 2:
enthalpy of atomization

Option 3:
hydration energy.

Option 4:
second ionisation enthalpy.

Correct Answer:
hydration energy.

Solution:
The stability of Cu^{2+} is more than Cu^{+} salts in an aqueous solution due to hydration energy.

Hence, the answer is the option (3).

Q. 38 Which of the following statements is incorrect?

A. All the transition metals except scandium form MO oxides which are ionic.
B. The highest oxidation number corresponding to the group number in transition metal oxides is attained in $\mathrm{Sc_2O_3}\ to\ \mathrm{Mn_2O_7}$.
C. Basic character increases from $\mathrm{V_2O_3}\ to\ \mathrm{V_2O_4} to\ \mathrm{V_2O_5}$.
D. $\mathrm{V_2O_4}$ dissolves in acids to give $\mathrm{VO_4^{3-}}$ salts.
E. CrO is basic but $\mathrm{Cr_2O_3}$ is amphoteric
Choose the correct answer from the options given below :

Option 1:
A and E only

Option 2:
B and D only

Option 3:
C and D only

Option 4:
B and C only

Correct Answer:
C and D only

Solution:
Acidic character increases with an increase in oxidation state because the tendency to gain electrons increases with the increase in oxidation state.

Basic character decreases from V_2O_3 to V_2O_4 to V_2O_5.

V_2O_4 dissolves in acids to give VO^{2+} not VO_4^{3-} salts.

C and D are incorrect.

Hence, the answer is the option (3).

Q. 39 Homoleptic complex from the following complexes is :

Option 1:
Potassium trioxalatoaluminate (III)

Option 2:
Diamminechloridonitrito-N-platinum (II)

Option 3:
Pentaamminecarbnonatocobalt (III) chloride

Option 4:
Triamminetriaquachromium (III) chloride

Correct Answer:
Potassium trioxalatoaluminate (III)

Solution:
Coordination compounds

$(1) K_3[Al(C_2O_4)_3]$

$(2) [Pt(NH_3)_2(Cl)(NO_2)]$

$(3) [Co(NH_3)_5(CO_3)]Cl$

$(4) [Cr(NH_3)_3(H_2O)_3]Cl_3$

In the complex of 2,3,4 two or more types of ligands are present hence they are heteroleptic complexes.

Hence, the answer is the option (1).

Q. 40 Which complex compound is most stable?

Option 1:
$[Co(NH_3))_4(H_2O)Br](NO_3)_2$

Option 2:
$[Co(NH_3)_3(NO_3)_3]$

Option 3:
$[CoCl_2(en)_2]NO_3$

Option 4:
$[Co(NH_3)_6]_2(SO_4)_3$

Correct Answer:
$[CoCl_2(en)_2]NO_3$

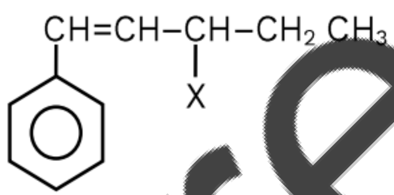
Solution:
We know,

$$\text{Stability} \propto \text{Chelation}$$

So, $[CoCl_2(en)_2]NO_3$ has an "en" ligand which provides chelation to the complex which makes it stable.

Hence, the answer is the option (3).

Q. 41 The given compound



Is an example of ________

Option 1:
Benzylic halide

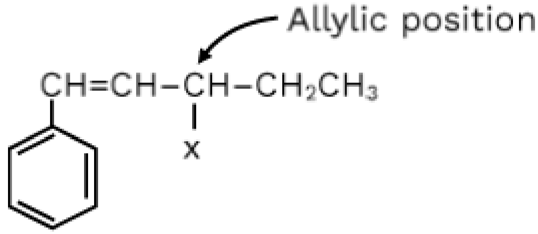
Option 2:
Aryl Halide

Option 3:
Allylic halide

Option 4:
Vinylic halide

Correct Answer:
Allylic halide

Solution:



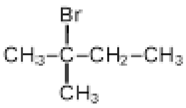
[Allylic halide]

It is an example of Allylic halide.

Q. 42 Consider the following reaction and identify the product (P).

3-Methylbutan-2-ol

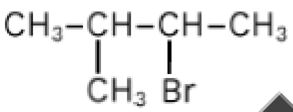
Option 1:



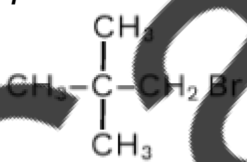
Option 2:

$CH_3CH = CH - CH_3$

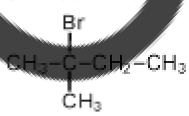
Option 3:



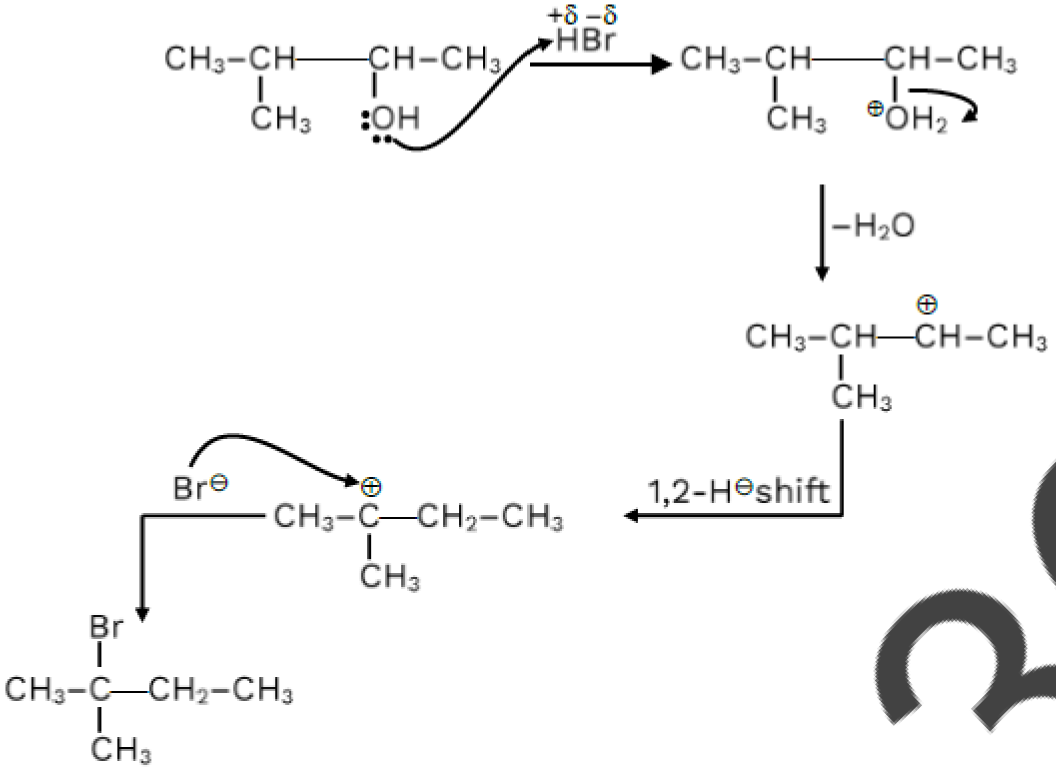
Option 4:



Correct Answer:

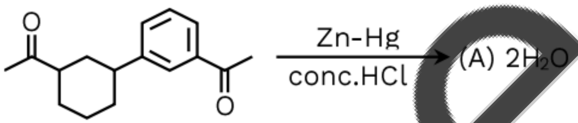


Solution:

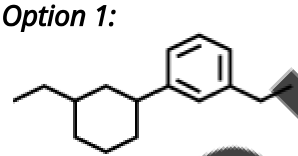


Correct option 1.

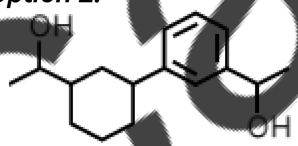
Q. 43 Identify product (A) in the following reaction.



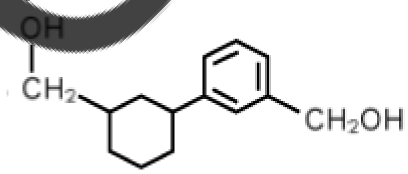
Option 1:



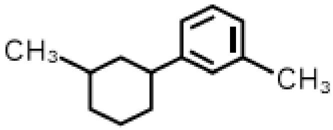
Option 2:



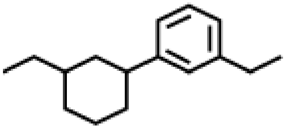
Option 3:



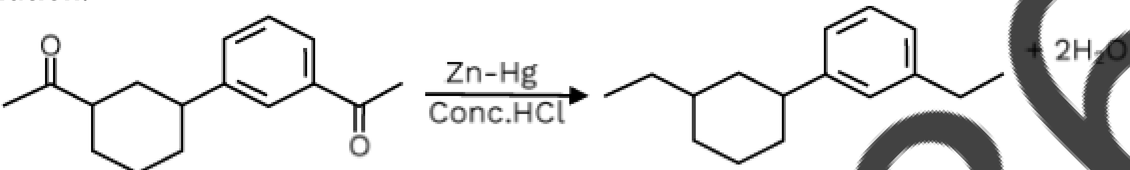
Option 4:



Correct Answer:

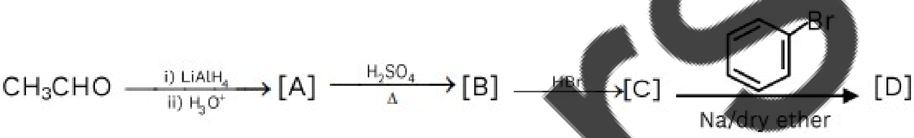


Solution:

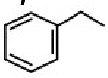


(Clemmensen reduction)

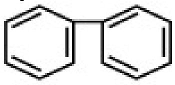
Q. 44 Identify the final product [D] obtained in the following sequence of reactions.



Option 1:



Option 2:



Option 3:

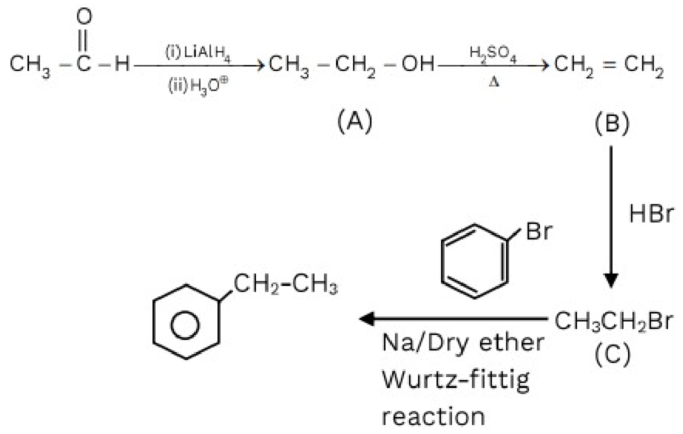
C_4H_{10}

Option 4:

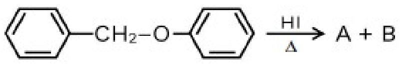
$HC \equiv C^{\ominus}Na^{+}$

Correct Answer:

Solution:



Q. 45 Consider the following reaction:



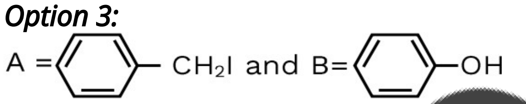
Identify products A and B

Option 1:

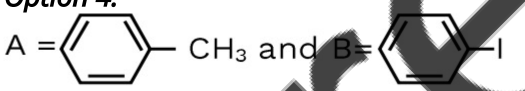
Option 2:

a

Option 3:

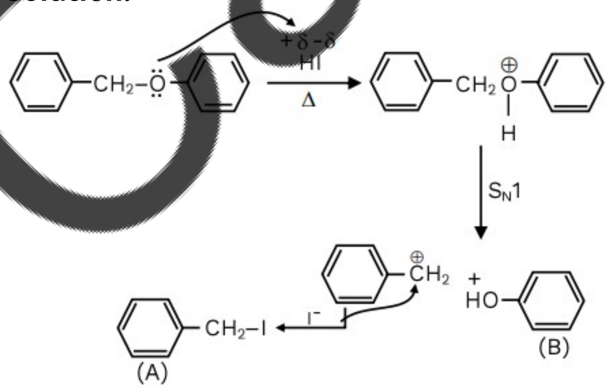


Option 4:

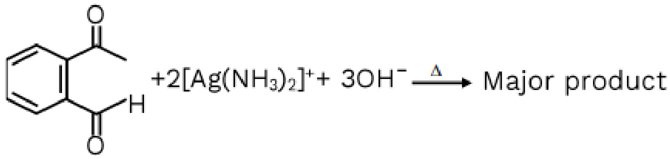


Correct Answer:

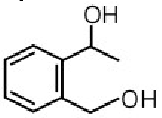
Solution:



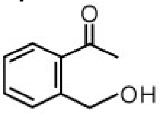
Q. 46 Identify the major product obtained in the following reaction:



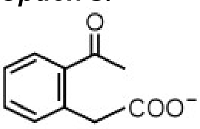
Option 1:



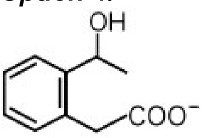
Option 2:



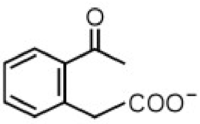
Option 3:



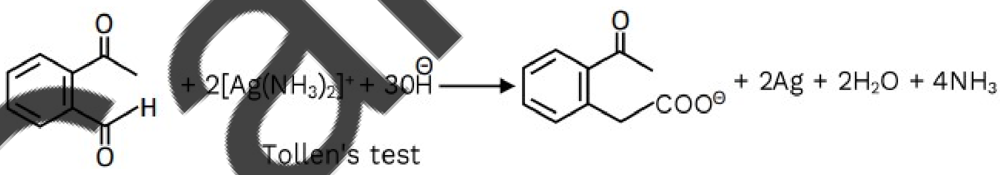
Option 4:



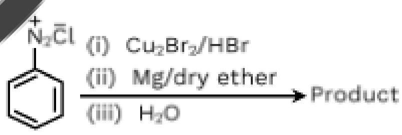
Correct Answer:



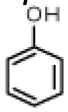
Solution:



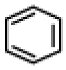
Q. 47 Identify the product in the following reaction:



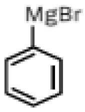
Option 1:



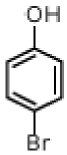
Option 2:



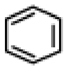
Option 3:



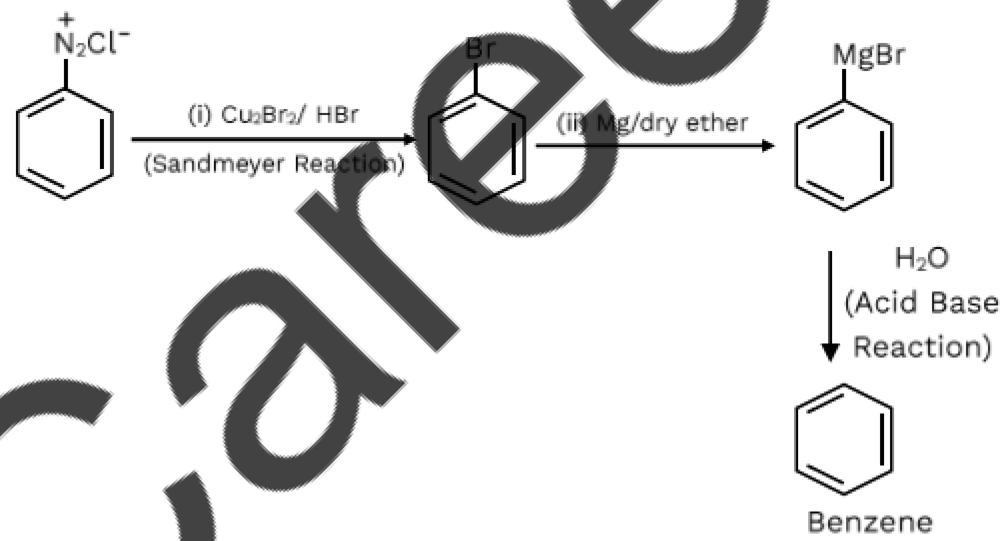
Option 4:



Correct Answer:



Solution:



Q. 48 Given below are two statements:

Statement I: a unit formed by the attachment of a base of 1' position of sugar is known as nucleoside.

Statement II: When nucleoside is linked to phosphorous acid at 5' -position of sugar moiety, we get nucleotide.

In the light of the above statements, choose the correct answer from the option given below :

Option 1:
Both Statement I and Statement II are true.

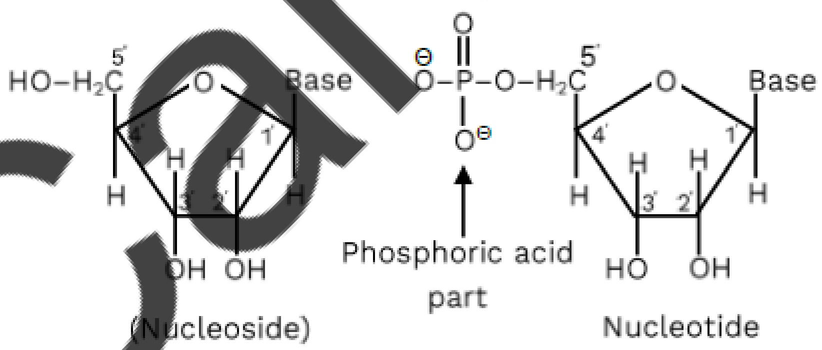
Option 2:
Both Statement I and Statement II are false.

Option 3:
Statement I is true but Statement II is false.

Option 4:
Statement I is false but Statement II is true.

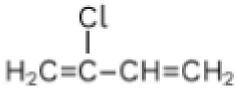
Correct Answer:
Statement I is true but Statement II is false.

Solution:

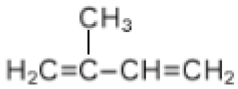


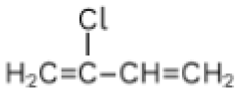
Q. 49 Which amongst the following molecules polymerization produces neoprene?

Option 1:
$H_2C = CH - CH = CH_2$

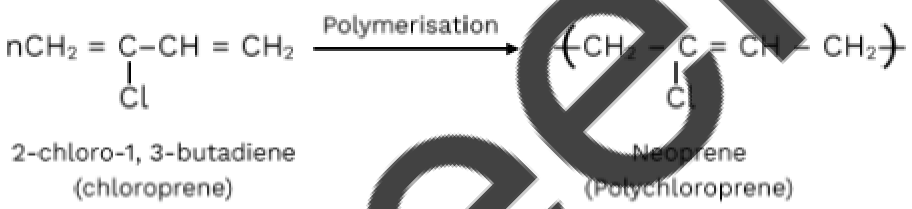
Option 2:


Option 3:
$H_2C = CH - C \equiv CH$

Option 4:


Correct Answer:


Solution:



Q. 50 Some tranquilizers are listed below which one from the following belongs barbiturates?

Option 1:
Chlordiazepoxide

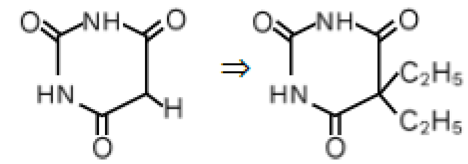
Option 2:
Meprobamate

Option 3:
Valium

Option 4:
Veronal

Correct Answer:
Veronal

Solution:



(Barbituric acid)

Veronal
(Barbiturates)
Tranquilizers

Biology

Q. 1 In recent years, DNA sequences (nucleotide sequence) of mtDNA (Mitochondrial DNA) and Y-chromosomes were considered for the study of human evolution, because

Option 1:
Their structure is known in greater detail

Option 2:
They can be studied from the samples of fossil remains

Option 3:
They are small and therefore, easy to study (elf they are uniparental in origin and do not take part in recombination

Option 4:
They are uniparental in origin and do not take part in recombination

Correct Answer:
They are uniparental in origin and do not take part in recombination

Solution:
Wilson and Sarich choose mitochondrial DNA (mtDNA) for the study of maternal line inheritance. While Y-chromosomes were considered for the study of human evolution, particularly male domain. It is possible because they are uniparental in origin and do not take part in recombination.Hence, the correct answer is option 4.

Q. 2 Given below are two statement: One is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: The first stage of gametophyte in the life cycle of moss in protonema stage.

Reason R: Protonema develops directly from spores produced in capsule. In the light of the above statements, choose the most appropriate answer from the options given below:

Option 1:
Both A and R are correct and R is the correct explanation of A.

Option 2:
Both A and R are correct but R is NOT the correct explanation of A.

Option 3:
A is true but R is false.

Option 4:
A is false but R is true.

Correct Answer:
Both A and R are correct and R is the correct explanation of A.

Solution:
The correct option is Option 1) Both A and R are correct, and R is the correct explanation of A.

Assertion A states that the first stage of the gametophyte in the life cycle of a moss is the protonema stage. This is correct. In mosses, the gametophyte generation begins with the development of a filamentous structure called a protonema. The protonema is the first stage in the life cycle and is formed from the germination of moss spores.

Reason R provides the correct explanation for Assertion A. It states that the protonema develops directly from spores produced in the capsule. This is also correct. The spores of mosses are produced within the capsule, which is part of the sporophyte generation. When the spores are released and land in a suitable environment, they germinate and give rise to the protonema.

Therefore, both Assertion A and Reason R are correct, and Reason R provides the correct explanation for Assertion A.

Q. 3 Identify the pair of heterosporous pteridophytes among the following :

Option 1:
Lycopodium and Selaginella

Option 2:
Selaginella and Salvinia

Option 3:
Psilotum and Salvinia

Option 4:
Equisetum and Salvinia

Correct Answer:
Selaginella and Salvinia

Solution:
The correct option is Option 2) Selaginella and Salvinia.

Heterospory refers to the condition in which a plant produces two different types of spores: microspores and megaspores. These spores give rise to distinct male and female gametophytes.

Among the given options, Selaginella and Salvinia are both heterosporous pteridophytes.

Selaginella is a genus of clubmosses that exhibits heterospory. It produces two types of spores: microspores, which develop into male gametophytes, and megaspores, which develop into female gametophytes. The male gametophytes produce sperm, while the female gametophytes produce eggs.

Salvinia is a genus of aquatic ferns that is also heterosporous. It produces microspores and megaspores, which give rise to male and female gametophytes, respectively.

Therefore, Option 2) Selaginella and Salvinia is the correct pair of heterosporous pteridophytes among the given options.

Q. 4 Given below are two statement:

One is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: In gymnosperms the pollen grains are released from the microsporangium and carried by air currents.

Reason R: Air currents carry the pollen grains to the mouth of the archegonia where the male gametes are discharged and pollen tube is not formed.

In the light of the above statements, choose the Correct answer from the options given below:

Option 1:
Both A and R are true and R is the correct explanation of A

Option 2:
Both A and R are true but R is NOT the correct explanation of A.

Option 3:
A is true but R is false.

Option 4:
A is false but R is true.

Correct Answer:
A is true but R is false.

Solution:
The correct answer is Option 3) A is true but R is false.

Assertion A is true. Gymnosperms are plants that produce naked seeds, and their pollen grains are typically released from the microsporangium (pollen sacs) and are dispersed by air currents. This is known as anemophily, which is a common mode of pollination in gymnosperms.

Reason R is false. Air currents do carry the pollen grains, but they are not specifically directed to the mouth of the archegonia. In gymnosperms, the male gametes are discharged from the pollen grains and are transported to the female reproductive structures called ovules. The male gametes typically travel through a pollen tube to reach the archegonia, where fertilization takes place.

Therefore, while Assertion A is true, Reason R is incorrect as it does not provide a correct explanation for Assertion A.

Q. 5 Select the correct statements with reference to chordates.

A. Presence of a mid-dorsal, solid and double nerve cord.
B. Presence of closed circulatory system.
C. Presence of paired pharyngeal gillslits.
D. Presence of dorsal heart.
E. Triploblastic pseudocoelomate animals.

Choose the **correct** answer from the option given below:

Option 1:
A, C and D only

Option 2:
B and C only

Option 3:
B, D and E only

Option 4:
C, D and E only

Correct Answer:
B and C only

Solution:
The correct answer is Option 2) B and C only.

Statement B is correct. Chordates have a closed circulatory system, meaning that the blood is contained within vessels and circulated throughout the body.

Statement C is correct. Chordates have paired pharyngeal gill slits, which are openings in the pharynx region used for respiration or filter feeding in some species.

For incorrect options,

Statement A is incorrect. Chordates have a notochord, which is a flexible rod-like structure running along the mid-dorsal region. However, they have a hollow nerve cord, not a solid and double nerve cord.

Statement D is incorrect. Chordates have a ventral heart, not a dorsal heart. The heart is located on the ventral side of the body.

Statement E is not relevant to chordates. Chordates are triploblastic, meaning they have three germ layers (ectoderm, mesoderm, and endoderm), but they are not pseudocoelomate. Pseudocoelomate animals have a body cavity called a pseudocoelom, which is not present in chordates.

Therefore, the correct answer is Option 2) B and C only.

Q. 6 The unique mammalian characteristics are:

Option 1:
Hairs, tympanic membrane and mammary glands

Option 2:
Hairs, pinna and mammary glands

Option 3:
Hairs, pinna and indirect development

Option 4:
Pinna, monocondylic skull and mammary glands

Correct Answer:
Hairs, pinna and mammary glands

Solution:
The correct answer is Option 2) Hairs, pinna, and mammary glands.

Hairs: Mammals are characterized by the presence of hairs or fur on their bodies. Hairs serve various functions such as insulation, protection, and sensory perception.

Pinna: The pinna refers to the external, visible part of the mammalian ear. It is a characteristic feature of mammals and helps in collecting sound waves and directing them into the ear canal.

Mammary glands: Mammary glands are unique to mammals and are responsible for producing milk to nourish the young. They are specialized glands that undergo development and enlargement during pregnancy and lactation.

For incorrect options,

Tympanic membrane: This is incorrect as the tympanic membrane, also known as the eardrum, is not unique to mammals. It is found in various animals, including reptiles, birds, and amphibians.

Indirect development: This is also incorrect as most mammals exhibit direct development, where the young are born in a relatively advanced state and undergo minimal or no metamorphosis.

Monocondylic skull: This is incorrect as the monocondylic skull, which refers to having a single occipital condyle, is not unique to mammals. It is found in various groups of vertebrates, including reptiles and amphibians.

Therefore, the correct answer is Option 2) Hairs, pinna, and mammary glands.

Q. 7 Radial symmetry is NOT found in adults of phylum

Option 1:
Ctenophora

Option 2:
Hemichordata

Option 3:
Coelenterata

Option 4:
Echinodermata

Correct Answer:
Hemichordata

Solution:
The correct answer is:

Option 2) Hemichordata

Radial symmetry is not found in adults of the phylum Hemichordata. Hemichordates, which include organisms like acorn worms, exhibit a bilateral symmetry as adults. Radial symmetry is commonly observed in organisms such as cnidarians (formerly referred to as Coelenterata) like jellyfish and corals, as well as in echinoderms such as starfish and sea urchins. Ctenophora, or comb jellies, also exhibit radial symmetry as adults.

Q. 8 Given below are two statements:

Statement I: RNA mutates at a faster rate.
Statement II: Viruses having RNA genome and shorter life span mutate and evolve faster.
In the light of the above statements, choose the correct answer from the options given below:

Option 1:
Both Statement I and Statement II are true.

Option 2:
Both Statement I and Statement II are false.

Option 3:
Statement I is true but Statement II is false.

Option 4:
Statement I false but Statement II is true.

Correct Answer:
Both Statement I and Statement II are true.

Solution:
Option 1) Both Statement I and Statement II are true.

Both statements are correct. RNA generally mutates at a faster rate compared to DNA due to the inherent properties of RNA molecules. Viruses with RNA genomes tend to have higher mutation rates, and their shorter life spans allow for faster evolution through the accumulation of mutations. These mutations contribute to the diversity and adaptability of RNA viruses.

Q. 9 Axile placentation is observed in

Option 1:
Mustard, Cucumber and Primrose

Option 2:
China rose, Beans and Lupin

Option 3:
Tomato, Dianthus and Pea

Option 4:
China rose, Petunia and Lemon

Correct Answer:
China rose, Petunia and Lemon

Solution:
The correct option is Option 4) China rose, Petunia, and Lemon.

Axile placentation is a type of arrangement of ovules within the ovary of a flower. In axile placentation, the ovules are attached to the central axis or the walls of a compound ovary. The ovary is divided into multiple chambers, and each chamber contains one or more ovules.

Among the given options, China rose (Hibiscus rosa-sinensis), Petunia (Petunia spp.), and Lemon (Citrus limon) exhibit axile placentation. In these plants, the ovary is composed of several chambers, and the ovules are attached to the central axis or the walls of the ovary chambers.

Mustard, Cucumber, Primrose, China rose, Beans, Lupin, Tomato, Dianthus, and Pea mentioned in the other options may exhibit different types of placentation, but not specifically axile placentation.

Therefore, Option 4) China rose, Petunia, and Lemon is the correct option for plants that exhibit axile placentation.

Q. 10 Family Fabaceae differs from Solanaceae and Liliaceae. With respect to the stamens , pick out the characteristics specific to family Fabaceae but not found in Solanaceae or Liliaceae.

Option 1:
Diadelphous and Dithecous anthers

Option 2:
Polyadelphous and epipetalous stamens

Option 3:
Monoadelphous and Monothecous anthers

Option 4:
Epiphyllous and Dithecous anthers

Correct Answer:
Diadelphous and Dithecous anthers

Solution:

The correct option is Option 1) Diadelphous and Dithecous anthers.

Family Fabaceae, also known as the legume family, is characterized by several distinguishing features, including the stamen characteristics. Diadelphous and Dithecous anthers are specific to the family Fabaceae and are not found in Solanaceae or Liliaceae.

Diadelphous anthers refer to the stamens being fused into two groups, with one group containing nine stamens and the other group containing one stamen. This arrangement is commonly observed in the flowers of the Fabaceae family.

Dithecous anthers refer to the anthers having two thecae or pollen sacs. Each theca contains pollen grains. This is another characteristic feature specific to the Fabaceae family.

In contrast, Solanaceae and Liliaceae do not typically exhibit diadelphous or dithecous anthers. Solanaceae family members often have stamens that are either free or fused together in different arrangements, and the anthers may have different structures. Liliaceae family members usually have six stamens with anthers that are usually not fused in distinct groups.

Therefore, Option 1) Diadelphous and Dithecous anthers is the correct option for the stamen characteristics specific to the family Fabaceae but not found in Solanaceae or Liliaceae.

Q. 11 Given below are two statements :

Statement I : Endarch and exarch are the terms often used for describing the position of secondary xylem in the plant body.

Statement II : Exarch condition is the most common feature of the root system. In the light of the above statements, choose the correct answer from the options given below :

Option 1:
Both Statement I and Statement II are true.

Option 2:
Both Statement I and Statement II are false..

Option 3:
Statement I is correct but Statement II is false

Option 4:
Statement I is correct but Statement II is false

Correct Answer:
Statement I is correct but Statement II is false

Solution:
The correct option is Option 4) Statement I is correct, but Statement II is false.

Statement I is correct. Endarch and exarch are terms used to describe the position of the secondary xylem in the plant body. Endarch refers to the condition where the primary xylem is located towards the center of the stem or root, and the secondary xylem is formed towards the outside. Exarch, on the other hand, refers to the condition where the primary xylem is located towards the periphery of the stem or root, and the secondary xylem is formed towards the inside.

Statement II is false. The exarch condition is not the most common feature of the root system. In fact, the exarch condition is more commonly observed in the stem rather than the root. In the root system, the most common condition is the endarch condition, where the primary xylem is located towards the center and the secondary xylem is formed towards the outside.

Therefore, Option 4) Statement I is correct, but Statement II is false is the correct answer.

Q. 12 Given below are two statements : One is labelled as Assertion A and the other is labelled as Reason R:

Assertion A : Late wood has fewer xylary elements with narrow vessels.

Reason R : Cambium is less active in winters. In the light of the above statements, choose the correct answer from the options given below :

Option 1:
Both A and R are true and R is the correct explanation of A.

Option 2:
Both A and R are true but R is NOT the correct explanation of A

Option 3:
A is true but R is false

Option 4:
A is false but R is true.

Correct Answer:
Both A and R are true and R is the correct explanation of A.

Solution:
The correct option is Option 1) Both A and R are true, and R is the correct explanation of A.

Assertion A states that late wood has fewer xylary elements with narrow vessels. This statement is true. Late wood, also known as summer wood, refers to the portion of a tree ring that is formed later in the growing season. It tends to have a higher density and a darker color compared to early wood. Late wood is characterized by having fewer xylary elements, such as tracheids or vessels, and narrower vessels compared to early wood.

Reason R states that cambium is less active in winters. This statement is also true. Cambium is a layer of cells in the stem or root of a plant that is responsible for secondary growth. During the winter season, cambium activity is reduced or even dormant due to environmental factors such as cold temperatures and shorter days. This reduced cambium activity during winter contributes to the formation of late wood with fewer xylary elements and narrow vessels.

Therefore, both Assertion A and Reason R are true, and Reason R provides a correct explanation for Assertion A. Hence, Option 1) is the correct answer.

Q. 13 Given below are two statement: One is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: A flower is defined as modified shoot wherein the shoot apical meristem changes to floral meristem.

Reason R: Internode of the shoot gets condensed to produce different floral appendages laterally at successive nodes instead of leaves. In the light of the above statements,

choose the Correct answer from the options given below:

Option 1:
Both A and R are true and R is the correct explanation of A.

Option 2:
Both A and R are true but R is NOT the correct explanation of A.

Option 3:
A is true but R is false.

Option 4:
A is false but R is true.

Correct Answer:
Both A and R are true and R is the correct explanation of A.

Solution:
Option 1) Both A and R are true and R is the correct explanation of A.

The assertion A states that a flower is defined as a modified shoot where the shoot apical meristem changes to floral meristem. This is true as flowers are indeed modified shoots.

The reason R explains that the internode of the shoot gets condensed to produce different floral appendages laterally at successive nodes instead of leaves. This is also true as floral appendages such as sepals, petals, stamens, and carpels are produced in place of leaves in the floral structure.

Therefore, both the assertion and the reason are true, and the reason correctly explains the assertion. Hence, the correct answer is Option 1.

Q. 14 Identify the correct statements :

A. Lenticels are the lens-shaped openings permitting the exchange of gases.

B. Bark formed early in the season is called hard bark.

C. Bark is a technical term that refers to all tissues exterior to vascular cambium.

D. Bark refers to periderm and secondary phloem.

E. Phellogen is single-layered in thickness.

Choose the correct answer from the options given below:

Option 1:
B, C and E only

Option 2:
A and D only

Option 3:
A, B and D only

Option 4:
B and C only

Correct Answer:
A and D only

Solution:
The correct statements are:

A. Lenticels are the lens-shaped openings permitting the exchange of gases.
D. Bark refers to periderm and secondary phloem.

Therefore, the correct answer is Option 2) A and D only.

Q. 15 Which of the following is characteristic feature of cockroach regarding sexual dimorphism?

Option 1:
Dark brown body colour and anal cerci

Option 2:
Presence of anal styles

Option 3:
Presence of sclerites

Option 4:
Presence of anal cerci

Correct Answer:
Presence of anal styles

Solution:
The correct answer is Option 2) Presence of anal styles.

Sexual dimorphism refers to the differences in physical characteristics between males and females of a species. In the case of cockroaches, one characteristic feature of sexual dimorphism is the presence of anal styles in males.

Anal styles are specialized structures located at the posterior end of the abdomen in male cockroaches. They are small, finger-like projections that are used during mating. These structures help in the transfer of sperm from the male to the female during copulation.

On the other hand, option 1) Dark brown body color and anal cerci are not specifically related to sexual dimorphism in cockroaches. Both males and females of many cockroach species can have a dark brown body color and anal cerci, which are sensory appendages located at the posterior end of the abdomen.

Option 3) Presence of sclerites refers to the hardened plates or structures that form the exoskeleton of cockroaches. While there may be slight differences in sclerites between male and female cockroaches, it is not a characteristic feature of sexual dimorphism.

Therefore, the correct answer is Option 2) Presence of anal styles.

Q. 16 Match **List I** with **List II**.

	List I		List II
A.	Mast cells	I.	Ciliated epithelium
B.	Inner surface of bronchiole	II.	Areolar connective tissue
C.	Blood	III.	Cuboidal epithelium
D.	Tubular parts of nephrons	IV.	Specialised connective tissue

Choose the correct answer from the options given below:

Option 1:
A-I, B-II, C-IV, D-III

Option 2:
A-II, B-III, C-I, D-IV

Option 3:
A-II, B-I, C-IV, D-III

Option 4:
A-III, B-IV, C-II, D-I

Correct Answer:
A-II, B-I, C-IV, D-III

Solution:
The correct matching is:

A-II (Mast cells - Connective tissue)
B-I (Inner surface of bronchiole - Ciliated epithelium)
C-IV (Blood - Specialized connective tissue)
D-III (Tubular parts of nephrons - Cuboidal epithelium)

Hence, the correct answer is Option 3) A-II, B-I, C-IV, D-III.

Q. 17 Which of the following statements are correct ?

A. Basophils are most abundant cells of the total WBCs
B. Basophils secrete histamine, serotonin and heparin
C. Basophils are involved in inflammatory response
D. Basophils have kidney shaped nucleus
N. Basophils are agranulocytes

Choose the correct answer from the options given below:

Option 1:
D and E only

Option 2:
C and E only

Option 3:
B and C only

Option 4:
A and B only

Correct Answer:
B and C only

Solution:

Statement B is correct. Basophils secrete histamine, serotonin, and heparin. These substances are involved in inflammatory and immune responses.

Statement C is correct. Basophils are involved in the inflammatory response. They release histamine and other chemicals that cause blood vessels to dilate and increase blood flow to the site of inflammation.

For incorrect options,

Statement A is incorrect. Basophils are not the most abundant cells of the total white blood cells (WBCs). Neutrophils are the most abundant WBCs.

Statement D is incorrect. Basophils have a lobed or segmented nucleus, not a kidney-shaped nucleus.

Therefore, the correct statements are B and C.

Q. 18 Given below are two statements:

Statement I: Ligament are dense irregular tissue.

Statement II: Cartilage is dense regular tissue.

In the light of the above statements, choose the **correct** answer from the options given below:

Option 1:
Both **Statement I** and **Statement II** are true.

Option 2:
Both **Statement I** and **Statement II** are false.

Option 3:
Statement I is true but **Statement II** is false.

Option 4:
Statement I is false but **Statement II** is ture.

Correct Answer:
Both **Statement I** and **Statement II** are false.

Solution:
The correct answer is Option 2) Both Statement I and Statement II are false.

Explanation:
Statement I: Ligaments are not dense irregular tissue. Ligaments are dense regular connective tissue that connects bones to other bones, providing stability and support to joints.

Statement II: Cartilage is not dense regular tissue. Cartilage is a type of connective tissue that is more flexible than bone but firmer than dense regular tissue. It provides support and cushioning between bones.

Therefore, both statements are incorrect.

Q. 19 In cockroach, excretion is brought about by.

A. Phallic gland B. Urecose gland
C. Nephrocytes D. Fat body
E. Collaterial glands

Choose the correct answer from the options given below:

Option 1:
A and E only

Option 2:
A, B and E only

Option 3:
B, C and D only

Option 4:
B and D only

Correct Answer:
B, C and D only

Solution:
he correct answer is Option 3) B, C, and D only.

B. Urecose gland: The urecose gland is involved in excretion in cockroaches. It is a specialized gland located in the posterior part of the alimentary canal, and it helps eliminate metabolic waste products, including uric acid.

C. Nephrocytes: Nephrocytes, also known as "green glands," are excretory organs found in many invertebrates. However, they are not present in cockroaches. Nephrocytes are commonly found in crustaceans, such as crabs and lobsters, and are responsible for excretion.

D. Fat body: The fat body in cockroaches is involved in various metabolic functions but is not directly responsible for excretion.

For incorrect options,

A. Phallic gland: The phallic gland in male cockroaches is not involved in excretion. It is a part of the reproductive system and is responsible for producing and transferring the spermatophore during mating.

E. Collaterial glands: Collaterial glands are not involved in excretion in cockroaches. They are associated with the reproductive system and are responsible for producing and storing spermatozoa.

Based on the information above, the correct answer is Option 3) B, C, and D only.

Q. 20 Among eukaryotes, replication of DNA takes place in–

Option 1:
$M\ phase$

Option 2:
$S\ phase$

Option 3:
$G_1\ phase$

Option 4:
$G_2\ phase$

Correct Answer:
$S\ phase$

Solution:
The correct option is Option 2) S phase.

During the cell cycle of eukaryotes, DNA replication occurs during the S (synthesis) phase. The cell cycle consists of different phases, including G1 (gap 1), S phase, G2 (gap 2), and M (mitosis) phase.

In the S phase, the cell's DNA is replicated, producing two identical copies of each chromosome. This replication process ensures that each daughter cell receives a complete set of genetic information during cell division. The S phase is characterized by active DNA synthesis, with DNA polymerases synthesizing new DNA strands based on the existing template strands.

Therefore, DNA replication specifically takes place during the S phase of the cell cycle in eukaryotes.

Q. 21 Which of the following stages of meiosis involves division of centromere?

Option 1:
Metaphase I

Option 2:
Metaphase II

Option 3:
Anaphase II

Option 4:
Telophase

Correct Answer:
Anaphase II

Solution:
The correct option is Option 3) Anaphase II.

During meiosis, which is the process of cell division that produces gametes (sex cells), the division of the centromere occurs during Anaphase II.

Anaphase II is the stage in meiosis II where the sister chromatids, which are the replicated copies of the chromosomes, are separated. The centromere, which holds the sister chromatids together, divides and allows the chromatids to move towards opposite poles of the cell.

In contrast, in Metaphase I, the homologous pairs of chromosomes align at the cell's equator, and the centromeres are still intact. Metaphase II is when the individual chromosomes line up at the equator, and the centromeres are ready to divide in Anaphase II.

Therefore, it is during Anaphase II of meiosis that the division of the centromere occurs.

Q. 22 Cellulose does not form blue colour with Iodine because

Option 1:
It is a disaccharide.

Option 2:
It is a helical molecules

Option 3:
It does not contain complex helices and hence cannot hold iodine molecules.

Option 4:
It breaks down when iodine reacts with

Correct Answer:
It does not contain complex helices and hence cannot hold iodine molecules.

Solution:
The correct option is Option 3) It does not contain complex helices and hence cannot hold iodine molecules.

Cellulose is a polysaccharide composed of glucose units linked together in a linear chain. It is a major component of plant cell walls and provides structural support to plants. When iodine reacts with certain polysaccharides, such as starch, a blue color complex is formed. However, cellulose does not form a blue color with iodine.

This is because cellulose does not contain complex helices, which are necessary for the formation of the blue color complex with iodine. Starch, on the other hand, has a helical structure that allows iodine molecules to fit into the helical spaces, forming a blue color complex.

Cellulose has a more linear and rigid structure, lacking the helical arrangement required for the interaction with iodine molecules. Therefore, when iodine reacts with cellulose, it does not form the blue color complex observed with other polysaccharides like starch.

Hence, Option 3) It does not contain complex helices and hence cannot hold iodine molecules is the correct explanation for why cellulose does not form a blue color with iodine.

Q. 23 The process of appearance of recombination nodules occurs at which sub stage of prophase I in meiosis?

Option 1:
Zygotene

Option 2:
Pachytene

Option 3:
Diplotene

Option 4:
Diakinesis

Correct Answer:
Pachytene

Solution:
The correct option is Option 2) Pachytene.

Recombination nodules, also known as recombination sites or synaptonemal complexes, are structures that facilitate the exchange of genetic material between homologous chromosomes during meiosis. These structures are important for genetic recombination and the generation of genetic diversity.

The appearance of recombination nodules occurs during the substage of prophase I called Pachytene. Prophase I is the longest and most complex phase of meiosis, consisting of several sub-stages: leptotene, zygotene, pachytene, diplotene, and diakinesis.

During Pachytene, homologous chromosomes pair up and undergo synapsis, where they become closely associated. The synaptonemal complex, which includes the recombination nodules, forms between the paired chromosomes. This complex helps to align the homologous chromosomes and promote the exchange of genetic material through a process called crossing over.

After Pachytene, the chromosomes continue to condense and undergo further changes in the subsequent sub-stages of prophase I, namely diplotene and diakinesis. In diplotene, the synaptonemal complex becomes more visible, and the homologous chromosomes begin to separate but remain connected at specific points called chiasmata. Diakinesis is the final sub-stage of prophase I, where the chromosomes fully condense and prepare for metaphase I.

Therefore, the appearance of recombination nodules occurs at the substage of prophase I called Pachytene, as stated in Option 2.

Q. 24 How many different proteins does the ribosome consist of ?

Option 1:
80

Option 2:
60

Option 3:
40

Option 4:
20

Correct Answer:
80

Solution:

The correct option is Option 1) 80.

The ribosome is a complex molecular machine involved in protein synthesis. It is composed of two subunits, the large subunit and the small subunit, which come together during protein synthesis and dissociate afterward. These subunits consist of a combination of proteins and RNA molecules called ribosomal RNA (rRNA).

In eukaryotes, the ribosome consists of 80 different proteins in total. The large subunit contains approximately 50 different proteins, while the small subunit contains around 30 different proteins. These proteins play essential roles in various aspects of protein synthesis, including mRNA binding, tRNA binding, and catalyzing the formation of peptide bonds.

Therefore, Option 1) 80 is the correct answer for the number of different proteins that the ribosome consists of.

Q. 25 Match List I with List II :

	List I		**List II**
A.	M phase	I.	Proteins are synthesized
B.	G_2 Phase	II.	Inactive phase
C.	Quiescent stage	III.	Interval between mitosis and initiation of DNA replication
D.	G_1 Phase	IV.	Equational division

Choose the correct answer from the options given below:

Option 1:
A-III, B-II, C-IV, D-I

Option 2:
A-IV, B-II, C-I, D-III

Option 3:
A-IV, B-I, C-II, D-III

Option 4:
A-II, B-IV, C-I, D-III

Correct Answer:
A-IV, B-I, C-II, D-III

Solution:
A. M phase IV. Equational division
B. G_2 Phase I. Proteins are synthesized
C. Quiescent stage II. Inactive phase
D. G_1 Phase III. Interval between mitosis and initiation of DNA replication

Therefore, the correct answer is Option 3) A-IV, B-I, C-II, D-III.

Q. 26 Melonate inhibits the growth of pathogenic bacteria by inhibiting the activity of

Option 1:
Succinic dehydrogenase

Option 2:
Amylase

Option 3:
Lipase

Option 4:
Dinitrogenase

Correct Answer:
Succinic dehydrogenase

Solution:
The correct option is Option 1) Succinic dehydrogenase.

Melonate inhibits the growth of pathogenic bacteria by specifically inhibiting the activity of succinic dehydrogenase, an enzyme involved in the citric acid cycle (also known as the Krebs cycle or tricarboxylic acid cycle). By inhibiting this enzyme, melonate disrupts the energy production process in bacteria, leading to their growth inhibition.

Q. 27 Which of the following are NOT considered as the part of endomembrane system?
A. Mitochondria B. Endoplasmic Reticulum
C.Chloroplast D. Golgi complex
E. Peroxisomes

Choose the most **appropriate** answer from the options given below:

Option 1:
A and D only

Option 2:
A, C and E only

Option 3:
A and D only

Option 4:
A, D and E only

Correct Answer:
A, C and E only

Solution:
Mitochondria, Chloroplasts, and Peroxisomes are not considered part of the endomembrane system. The endomembrane system primarily includes the endoplasmic reticulum and the Golgi complex, which are involved in the synthesis, modification, and transport of proteins and lipids within the cell.

Mitochondria are responsible for cellular respiration and energy production, and they have their own distinct membrane system separate from the endomembrane system. Chloroplasts are found in plant cells and are responsible for photosynthesis, which involves the conversion of light energy into chemical energy. Peroxisomes are involved in various metabolic processes, including the breakdown of fatty acids, detoxification of harmful substances, and synthesis of certain lipids. These organelles have their own unique functions and are not part of the endomembrane system.

Option 2 is the correct answer

Q. 28 Given below are two statements:

Statement I: Low temperature preserves the enzyme in a temporarily inactive state whereas high temperature destroyes enzymatic activity because proteins are denatured by heat.
Statement II: When the inhibitor closely resembles the substrate in its molecular structure and inhibits the activity of the enzyme, it is known as competitive inhibitor.

In the light of above statements, choose the **correct** answer from the options given below:

Option 1:
Both **statement I** and **statement II** are true.

Option 2:
Both **statement I** and **statement II** are false.

Option 3:
statement I is true but **statement II** is false.

Option 4:
Statement I is false but **statement II** is true.

Correct Answer:
Both **statement I** and **statement II** are true.

Solution:
The correct answer is Option 1) Both statement I and statement II are true.

Statement I is true. Low temperatures can slow down or preserve enzymatic activity by reducing the rate of reactions. However, high temperatures can denature proteins, including enzymes, leading to a loss of enzymatic activity.

Statement II is also true. Competitive inhibition occurs when an inhibitor molecule closely resembles the substrate and competes with the substrate for binding to the active site of the enzyme. The inhibitor effectively "competes" with the substrate for the enzyme's active site, thereby reducing the enzyme's activity.

Therefore, both statement I and statement II are true, and Option 1 is the correct answer.

Q. 29 Which of the following functions is carried out by cytoskeleton in a cell ?

Option 1:
Nuclear division

Option 2:
Protein synthesis

Option 3:
Motility

Option 4:
Transportation

Correct Answer:
Motility

Solution:
The correct answer is Option 3) Motility.

The cytoskeleton is a network of protein filaments found in the cytoplasm of cells. It plays a crucial role in providing structural support, maintaining cell shape, and facilitating various cellular processes. One of the key functions of the cytoskeleton is cell motility, which includes the movement of cells themselves and the movement of cellular components within the cell.

The cytoskeleton is responsible for the movement of cells through processes such as amoeboid movement, ciliary/flagellar movement, and muscle contraction. It provides the necessary structure and organization for these movements to occur.

While nuclear division (Option 1) is primarily regulated by the spindle apparatus during mitosis and meiosis and protein synthesis (Option 2) is carried out by ribosomes, the cytoskeleton is not directly involved in these specific processes.

Transportation (Option 4) can involve the cytoskeleton indirectly, as it provides tracks or highways for intracellular transport, but it is not the primary function of the cytoskeleton. The cytoskeleton's primary role in transportation is to help maintain the cellular architecture and provide support for the transport systems.

Therefore, the correct function carried out by the cytoskeleton in a cell is Option 3) Motility.

Q. 30 Select the correct statements.
A. Tetrad formation is seen during Leptotene.
B. During Anaphase, the centromeres split and chromatids separate.
C. Terminalization takes place during Pachytene.
D. Nucleolus, Golgi complex and ER are reformed during Telophase.
E. Crossing over takes place between sister chromatids of homologous chromosome.

Choose the **correct** answer from the option given below:

Option 1:
A and C only

Option 2:
B and D only

Option 3:
A, C and E only

Option 4:
B and E only

Correct Answer:
B and D only

Solution:
The correct answer is Option 2) B and D only.

Statement A: Tetrad formation is seen during Leptotene.
This statement is incorrect. Tetrad formation, also known as synapsis, occurs during the zygotene stage of prophase I of meiosis. During leptotene, chromosomes start to condense, but they have not yet formed tetrads.

Statement B: During Anaphase, the centromeres split and chromatids separate.
This statement is correct. During anaphase of mitosis or meiosis II, the centromeres split, and the sister chromatids separate, moving toward opposite poles of the cell.

Statement C: Terminalization takes place during Pachytene.
This statement is incorrect. Terminalization occurs during diplotene, not pachytene, which is a substage of prophase I of meiosis. Terminalization refers to the movement of chiasmata (sites of crossing over) towards the ends (terminals) of the homologous chromosomes.

Statement D: Nucleolus, Golgi complex, and ER are reformed during Telophase.
This statement is correct. During telophase of mitosis or meiosis, the nuclear envelope reforms, and the nucleolus, Golgi complex, and endoplasmic reticulum (ER) are reformed.

Statement E: Crossing over takes place between sister chromatids of homologous chromosomes.
This statement is incorrect. Crossing over, also known as recombination, occurs between non-sister chromatids of homologous chromosomes during prophase I of meiosis. It results in the exchange of genetic material between homologous chromosomes.

Therefore, the correct answer is Option 2) B and D only.

Q. 31 Given below are two statements:

Statement I: During G0 phase of cell cycle, the cell is metabolically inactive.
Statement II: The centrosome undergoes duplication during S phase of interphase.
In the light of the above statements, choose the **most appropriate** answer from the options given below:

Option 1:
Both **Statement I** and **Statement II** are correct.

Option 2:
Both **Statement I** and **Statement II** are incorrect.

Option 3:
Statement I is correct but **statement II** incorrect.

Option 4:
Statement I is incorrect but **statement II** correct.

Correct Answer:
Statement I is incorrect but **statement II** correct.

Solution:
The correct answer is Option 4) Statement I is incorrect but statement II is correct.

Statement I: During G_0 phase of the cell cycle, the cell is metabolically inactive.
This statement is incorrect. The G_0 phase of the cell cycle is a resting phase where cells exit the cell cycle and temporarily halt their division. However, cells in the G_0 phase can still be metabolically active and perform their specific functions. They may carry out normal cellular activities, such as protein synthesis and energy production, even though they are not actively dividing.

Statement II: The centrosome undergoes duplication during S phase of interphase.
This statement is correct. The centrosome, a cellular organelle involved in cell division, undergoes duplication during the S phase of interphase. The centrosome consists of two centrioles, and each centriole replicates to form a new pair of centrioles. This duplication ensures that each daughter cell produced during cell division receives a complete set of centrioles.

Therefore, the correct answer is Option 4) Statement I is incorrect but statement II is correct.

Q. 32 Given below are two statements:

Statement I: A protein is imagined as a line, the left end represented by first amino acid (C - terminal) and the right end represented by last amino acid (N-terminal)
Statement II: Adult human haemoglobin, consists of 4 subunits (two subunits of α type and two subunits of β type.
In the light of the above statements, choose the correct answer from the options given below:

Option 1:
Both Statement I and Statement II are true.

Option 2:
Both Statement I and Statement II are false.

Option 3:
Statement I is true but Statement II is false.

Option 4:
Statement I is false but Statement II is true.

Correct Answer:
Statement I is false but Statement II is true.

Solution:
Option 4) Statement I is false but Statement II is true.

Statement I is incorrect because in the conventional representation of a protein, the N-terminal (amino-terminus) is depicted on the left end and the C-terminal (carboxy-terminus) is depicted on the right end.

Statement II is true. Adult human hemoglobin is indeed composed of four subunits, two of which are α subunits and two of which are β subunits.

Q. 33 How many ATP and $NADPH_2$ are required for the synthesis of one molecule of Glucose during Calvin cycle?

Option 1:
$12\ ATP$ and $12\ NADPH_2$

Option 2:
$18\ ATP$ and $12\ NADPH_2$

Option 3:
$12\ ATP$ and $16\ NADPH_2$

Option 4:
$18\ ATP$ and $16\ NADPH_2$

Correct Answer:
$18\ ATP$ and $12\ NADPH_2$

Solution:
During the Calvin cycle, also known as the light-independent reactions of photosynthesis, glucose is synthesized from carbon dioxide (CO_2) using ATP and $NADPH_2$ as energy sources. The overall reaction for glucose synthesis can be represented as follows:

$6\ CO_2 + 18\ ATP + 12\ NADPH_2 \rightarrow C_6H_{12}O_6$ (glucose) $+ 18\ ADP + 18\ Pi + 12\ NADP^+$

For the synthesis of one molecule of glucose, 6 molecules of carbon dioxide are required. Each carbon dioxide molecule undergoes a series of reactions involving ATP and $NADPH_2$ to form one glucose molecule.

Each carbon dioxide molecule requires 3 ATP molecules for its fixation and reduction, and 2 $NADPH_2$ molecules for its reduction. Therefore, for 6 CO_2 molecules, we need 6 × 3 = 18 ATP molecules and 6 × 2 = 12 $NADPH_2$ molecules.

Hence, the correct option is Option 2) 18 ATP and 12 $NADPH_2$.

Q. 34 Given below are two statements :

Statement I: The forces generated by transpiration can lift a xylem-sized column of water over 130 meters height.

Statement II: Transpiration cools leaf surfaces sometimes 10 to 15 degrees, by evaporative cooling. In the light of the above statements, choose the most appropriate answer from the options given below :

Option 1:
Both Statement I and Statement II are correct.

Option 2:
Both Statement I and Statement II are incorrect

Option 3:
Statement I is correct but Statement II is incorrect.

Option 4:
Statement I is incorrect but Statement II is correct.

Correct Answer:
Both Statement I and Statement II are correct.

Solution:

Statement I is correct because transpiration, the process of water loss from plants through stomata, creates a pulling force that can generate enough tension to lift a column of water in the xylem to considerable heights. This phenomenon is known as the transpiration pull or cohesion-tension theory.

Statement II is also correct because transpiration, as water evaporates from the leaf surfaces, leads to evaporative cooling. This cooling effect helps regulate leaf temperature and can lower leaf surface temperatures by 10 to 15 degrees Celsius.

Option 1 is the correct answer.

Q. 35 Which micronutrient is required for splitting of water molecule during photosynthesis?

Option 1:
manganese

Option 2:
molybdenum

Option 3:
magnesium

Option 4:
copper

Correct Answer:
manganese

Solution:
Manganese is absorbed by plants in the form of Mn2+ ions. It serves various important functions, including the activation of enzymes involved in respiration, nitrogen metabolism, and photosynthesis. However, its primary role lies in the process of photosynthesis, where it plays a crucial part in the splitting of water molecules to release oxygen.

Option 1 is the correct answer.

Q. 36 The reaction centre in PS II has an absorption maxima at.

Option 1:
680 nm

Option 2:
700 nm

Option 3:
660 nm

Option 4:
780 nm

Correct Answer:
680 nm

Solution:
The reaction center in Photosystem II (PS II) has an absorption maximum at approximately 680 nm. This means that the reaction center is most efficient in capturing light energy at this specific wavelength. Light energy absorbed by PS II at this wavelength is used to drive the process of photosynthesis, particularly in the initial steps of the light-dependent reactions.

Option 1 is the correct answer.

Q. 37 Given below are two statement: One is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: ATP is used at two steps in glycolysis.

Reason R: First ATP is used in converting glucose into glucose-6 phosphate and second ATP is used in conversion of fructose-6- phosphate into fructose-1-6-diphosphate. In the light of the above statements, choose the Correct answer from the options given below:

Option 1:
Both A and R are true and R is the correct explanation of A

Option 2:
Both A and R are true but R is NOT the correct explanation of A.

Option 3:
A is true but R is false.

Option 4:
A is false but R is true.

Correct Answer:
Both A and R are true and R is the correct explanation of A

Solution:
The correct answer is Option 1) Both A and R are true and R is the correct explanation of A.

Assertion A states that ATP is used in two steps in glycolysis, which is true.

Reason R provides the correct explanation for Assertion A. It states that the first ATP is used in converting glucose into glucose-6-phosphate, and the second ATP is used in the conversion of fructose-6-phosphate into fructose-1,6-diphosphate. This is an accurate description of the steps in glycolysis where ATP is utilized.

Therefore, both Assertion A and Reason R are true, and Reason R correctly explains Assertion A.

Q. 38 Spraying of which of the following phytohormone on juvenile conifers helps in hastening the maturity period, that leads to early seed production?

Option 1:
Indole-3-butyric acid

Option 2:
Gibberellic Acid

Option 3:
Zeatin

Option 4:
Abscisic Acid

Correct Answer:
Gibberellic Acid

Solution:
The correct option is Option 2) Gibberellic Acid.

Gibberellic Acid is a phytohormone that can be used to accelerate the maturity period of plants, including conifers. When sprayed on juvenile conifers, Gibberellic Acid promotes the elongation of stems, stimulates flowering, and induces early seed production.

Gibberellic Acid is a naturally occurring plant hormone that regulates various physiological processes in plants, including growth and development. By applying Gibberellic Acid to juvenile conifers, the plant's natural growth and reproductive processes can be influenced, resulting in an expedited maturity period and earlier seed production.

Therefore, Gibberellic Acid is the phytohormone that, when sprayed on juvenile conifers, helps in hastening the maturity period and leads to early seed production.

Q. 39 Which hormone promotes internode/petiole elongation in deep water rice?

Option 1:
GA_3

Option 2:
Kinetin

Option 3:
Ethylene

Option 4:
$2,4-D$

Correct Answer:
Ethylene

Solution:
The correct option is Option 3) Ethylene.

Ethylene is a plant hormone that plays a crucial role in promoting internode and petiole elongation in deep water rice plants. Deep water rice, also known as floating rice, is a variety of rice that is adapted to grow in flooded or waterlogged conditions.

When deep water rice plants are submerged in water, the lack of oxygen and the presence of ethylene stimulate elongation growth in the internodes and petioles. Ethylene acts as a signal to trigger the elongation of these plant parts, allowing the plant to grow taller and keep its leaves above the water surface for efficient gas exchange.

Gibberellic acid (GA3) is a hormone that is involved in promoting stem elongation in many plant species. However, ethylene is specifically associated with promoting internode and petiole elongation in deep water rice under flooded conditions.

Kinetin is a type of cytokinin hormone that is involved in cell division and growth regulation but is not directly associated with internode or petiole elongation in deep water rice.

2,4-D (2,4-dichlorophenoxyacetic acid) is a synthetic auxin hormone that is commonly used as a herbicide. It is not directly involved in promoting internode or petiole elongation in deep water rice.

Therefore, Option 3) Ethylene is the correct option for the hormone that promotes internode/petiole elongation in deep water rice.

Q. 40 Movement and accumulation of ions across a membrane against their concentration gradient can be explained by

Option 1:
Osmosis

Option 2:
Facilitated Diffusion

Option 3:
Passive Transport

Option 4:
Active Transport

Correct Answer:
Active Transport

Solution:
The correct option is Option 4) Active Transport.

Active transport is the process by which ions or molecules are moved across a membrane against their concentration gradient, from an area of lower concentration to an area of higher concentration. This process requires the expenditure of energy in the form of ATP (adenosine triphosphate).

During active transport, specialized transport proteins, such as pumps or carriers, are involved in the movement of ions or molecules across the membrane. These transport proteins actively move the ions or molecules, using ATP as an energy source, to create a concentration gradient that is different from the equilibrium state.

This process is in contrast to passive transport, such as osmosis or facilitated diffusion, where the movement of ions or molecules occurs along the concentration gradient and does not require energy expenditure.

Therefore, Option 4) Active Transport is the correct explanation for the movement and accumulation of ions across a membrane against their concentration gradient.

Q. 41 Match **List I** with **List II** :

	List I		**List II**
A.	Oxidative decarboxylation	I.	Citrate synthase
B.	Glycolysis	II.	. Pyruvate dehydrogenase
C.	Oxidative phosphorylation	III.	Electron transport system
D.	Tricarboxylic acid cycle	IV.	EMP pathway

Choose the correct answer from the options given below:

Option 1:
A-III, B-IV, C-II, D-I

Option 2:
A-II, B-IV, C-I, D-III

Option 3:
A-III, B-I, C-II, D-IV

Option 4:
A-II, B-IV, C-III, D-I

Correct Answer:
A-II, B-IV, C-III, D-I

Solution:

A. Oxidative decarboxylation - II. Pyruvate dehydrogenase
B. Glycolysis - IV. EMP pathway
C. Oxidative phosphorylation - III. Electron transport system
D. Tricarboxylic acid cycle - I. Citrate synthase

Therefore, the correct matching is A-II, B-IV, C-III, D-I, which corresponds to Option 4).

Q. 42 Match List I with List II :

	List I		List II
A.	Cohesion	I.	More attraction in liquid phase
B.	Adhesion	II.	Mutual attraction among water molecules
C.	Surface tension	III.	Water loss in liquid phase
D.	Guttation	IV.	Attraction towards polar surfaces

Choose the correct answer from the options given below:

Option 1:
A-II, B-IV, C-I, D-III

Option 2:
A-IV, B-III, C-II, D-I

Option 3:
A-III, B-I, C-IV, D-II

Option 4:
A-II, B-I, C-IV, D-III

Correct Answer:
A-II, B-IV, C-I, D-III

Solution:
A. Cohesion - II. Mutual attraction among water molecules
B. Adhesion - IV. Attraction towards polar surfaces
C. Surface tension - I. More attraction in liquid phase
D. Guttation - III. Water loss in liquid phase

Therefore, the correct matching is A-II, B-IV, C-I, D-III, which corresponds to Option 1).

Q. 43 Which of the following combinations is required for chemiosmosis?

Option 1:
membrane, proton pump, proton gradient, ATP synthase

Option 2:
membrane, proton pump, proton gradient, NADP synthase

Option 3:
proton pump, electron gradient, ATP synthase

Option 4:
proton pump, electron gradient, NADP synthase

Correct Answer:
membrane, proton pump, proton gradient, ATP synthase

Solution:
The correct combination required for chemiosmosis is:

Option 1) membrane, proton pump, proton gradient, ATP synthase.

Chemiosmosis is the process by which ATP (adenosine triphosphate) is synthesized using the energy generated by the movement of protons (H+) across a membrane. In this process, a membrane is necessary to create a barrier for the proton movement. A proton pump actively transports protons across the membrane, creating a proton gradient. This proton gradient serves as a source of potential energy. ATP synthase is an enzyme located in the membrane that utilizes this proton gradient to generate ATP.

Option 2) (membrane, proton pump, proton gradient, NADP synthase) is incorrect because NADP synthase is not involved in chemiosmosis. NADP synthase is not a known enzyme involved in ATP synthesis.

Option 3) (proton pump, electron gradient, ATP synthase) is also incorrect because chemiosmosis primarily involves the movement of protons (H+) and not electrons.

Option 4) (proton pump, electron gradient, NADP synthase) is incorrect for the same reasons as Option 2 and Option 3.

Therefore, the correct combination required for chemiosmosis is Option 1.

Q. 44 Match List I with List II

	List I		List II.
A.	Iron	I.	Synthesis of auxin
B.	Zinc	II.	. Component of nitrate reductase
C.	Boron	III.	Activator of catalase
D.	Molybdenum	IV.	Cell elongation and differentiation

Choose the correct answer from the options given below:

Option 1:
A–III, B–II, C–I, D–IV

Option 2:
A–II, B–III, C–IV, D–I

Option 3:
A–III, B–I, C–IV, D–II

Option 4:
A–II, B–IV, C–I, D–III

Correct Answer:
A–III, B–I, C–IV, D–II

Solution:
A. Iron - III. Activator of catalase
B. Zinc - II. Component of nitrate reductase
C. Boron - I. Synthesis of auxin
D. Molybdenum - IV. Cell elongation and differentiation

Therefore, the correct answer is Option 3) A–III, B–I, C–IV, D–II.

Q. 45 Match **List I** with **List II** with respect to human eye.

	List I		List II
A.	Fovea	I.	Visible coloured portionof eye that regulats diameter of pupil.
B.	Iris	II.	External layer of eye formed of dense connective tissue.
C.	Blind spot	III.	Point of greatest visual acuity or resolution.
D.	Sclera	IV.	leaves the eyeball and photoreceptor cells are absent.

Choose the **correct** answer from the options given below :

Option 1:
A–III, B–I, C–IV, D–II

Option 2:
A–IV, B–III, C–II, D–I

Option 3:
A–I, B–IV, C–III, D–II

Option 4:
A–II, B–I, C–III, D–IV

Correct Answer:
A–III, B–I, C–IV, D–II

Solution:
A. Fovea - III. Point of greatest visual acuity or resolution
The fovea is a small depression in the retina of the eye that contains a high concentration of cone cells, which are responsible for detailed and color vision. It is the area of the retina with the highest visual acuity.

B. Iris - I. Visible colored portion of the eye that regulates the diameter of the pupil
The iris is the colored part of the eye surrounding the pupil. It controls the size of the pupil and regulates the amount of light entering the eye.

C. Blind spot - IV. Leaves the eyeball, and photoreceptor cells are absent
The blind spot, also known as the optic disc, is the area on the retina where the optic nerve leaves the eye. It does not contain photoreceptor cells, so there is no vision in that particular area.

D. Sclera - II. External layer of the eye formed of dense connective tissue
The sclera is the tough, white, outer layer of the eye that forms the protective covering of the eyeball. It is composed of dense connective tissue.

Therefore, the correct matching is:
A–III, B–I, C–IV, D–II, which corresponds to Option 1.

Q. 46 Which of the following statement are correct regarding skeletal muscle?

A. Muscle bundles are held together by collagenous connective tissue layer called fascicle.
B. Sarcoplasmic reticulum of muscle fibre is a store house of calcium ions.
C. Striated appearance of skeletal muscle fibre is due to distribution pattern of acting and myosin proteins.
D. M line is considered as functional unit of contraction called sarcomere.

Choose the **most appropriate** answer from the options given below:

Option 1:
A, B and C only

Option 2:
B and C only

Option 3:
A, C and D only

Option 4:
C and D only

Correct Answer:
B and C only

Solution:
The correct answer is Option 2) B and C only.

Statement B is correct. The sarcoplasmic reticulum of muscle fibers is a specialized type of endoplasmic reticulum that stores and releases calcium ions (Ca^{2+}) during muscle contraction.

Statement C is correct. The striated appearance of skeletal muscle fibers is due to the arrangement and distribution pattern of actin and myosin proteins, which form the contractile units called sarcomeres.

For incorrect options,

Statement A is incorrect. Muscle bundles in skeletal muscle are held together by a collagenous connective tissue layer called perimysium, not fascicle.

Statement D is incorrect. The M line is a structural component of the sarcomere, but the functional unit of contraction in skeletal muscle is the sarcomere itself, not specifically the M line.

Therefore, the correct answer is Option 2) B and C only.

Q. 47 Match **List I** with **List II**.

	List I (Type of Joint)		List II (Found between)
A.	Cartilaginous Joint	I.	Between flat skull bones
B.	Ball and Socket Joint	II.	Between adjacent vertebrae in vertebral column
C.	Fibrous Joint	III.	Between carpal and metacarpal of thumb
D.	Saddle Joint	IV.	Between Humerus and Pectoral girdle

Choose the correct answer from the options given below:

Option 1:
A-III, B-I, C-II, D-IV

Option 2:
A-II, B-IV, C-I, D-III

Option 3:
A-I, B-IV, C-III, D-II

Option 4:
A-II, B-IV, C-III, D-I

Correct Answer:
A-II, B-IV, C-I, D-III

Solution:
The correct answer is Option 2) A-II, B-IV, C-I, D-III.

Explanation:
A. Cartilaginous Joint is found between flat skull bones.
B. Ball and Socket Joint is found between the Humerus and Pectoral girdle.
C. Fibrous Joint is found between the carpal and metacarpal of the thumb.
D. Saddle Joint is found between adjacent vertebrae in the vertebral column.

Therefore, the correct match is:
A-II, B-IV, C-I, D-III.

Option 2) A-II, B-IV, C-I, D-III is the correct answer.

Q. 48 Once the undigested and unabsorbed substances enter the caecum, their backflow is prevented by-

Option 1:
Sphincter of Oddi

Option 2:
Ileo - caecal valve

Option 3:
Gastro - oesophageal sphincter

Option 4:
Pyloric sphincter

Correct Answer:
Ileo - caecal valve

Solution:
The correct answer is: Option 2) Ileo-caecal valve

The ileo-caecal valve is a one-way valve located between the ileum (the last part of the small intestine) and the caecum (the first part of the large intestine). It prevents the backflow of undigested and unabsorbed substances from the caecum back into the ileum, ensuring that the materials move in the correct direction through the digestive system.

Q. 49 Match List I with List II.

	List I		List II
A.	P-wave	I.	Beginning of systole
B.	Q -wave	II.	Repolarisation of ventricles
C.	QRS complex	III.	Depolarisation of atria
D.	T-wave	IV.	Depolarisation of ventricles

Choose the correct answer from the options given below:

Option 1:
A-III, B-I, C-IV, D-II

Option 2:
A-IV, B-III, C-II, D-I

Option 3:
A-II, B-IV, C-I, D-III

Option 4:
A-I, B-II, C-III, D-IV

Correct Answer:
A-III, B-I, C-IV, D-II

Solution:
A. P-wave: The P-wave represents the depolarization (contraction) of the atria. It corresponds to the beginning of systole, which is the contraction phase of the cardiac cycle. Therefore, the correct pairing is A-III.

B. Q-wave: The Q-wave is a small downward deflection that appears before the R-wave in the QRS complex. It doesn't have a specific physiological event associated with it. Therefore, the correct pairing is B-I.

C. QRS complex: The QRS complex represents the depolarization (contraction) of the ventricles. It is the largest and most prominent wave complex on the ECG. It occurs during the ventricular systole phase, which is the main pumping phase of the cardiac cycle. Therefore, the correct pairing is C-IV.

D. T-wave: The T-wave represents the repolarization (relaxation) of the ventricles. It occurs during the diastole phase, which is the relaxation phase of the cardiac cycle. Therefore, the correct pairing is D-II.

Based on the explanations above, we can conclude that Option 1) A-III, B-I, C-IV, D-II is the correct answer.

Q. 50 Which of the following are NOT under the control of thyroid hormone?

A. Maintenance of water and electrolyte balance
B. Regulation of basal metabolic rate
C. Normal rhythm of sleep-wake cycle
D. Development of immune system
E. Support the process of R.B.Cs formation Choose the correct answer from the options given below:

Option 1:
A and D only

Option 2:
B and C only

Option 3:
C and D only

Option 4:
D and E only

Correct Answer:
C and D only

Solution:
The correct answer is Option 3) C and D only.

Thyroid hormone primarily influences the regulation of metabolism and growth in the body. It affects various physiological processes, but not all processes are directly controlled by thyroid hormone. Let's go through each option:

C. Normal rhythm of sleep-wake cycle: The sleep-wake cycle is primarily regulated by the hormone melatonin, which is produced by the pineal gland. Thyroid hormone does not directly control the sleep-wake cycle.

D. Development of immune system: The development of the immune system is not directly controlled by thyroid hormone. It is regulated by various other factors and immune cells.

For incorrect options,

A. Maintenance of water and electrolyte balance: This process is not directly regulated by thyroid hormone. It is primarily controlled by hormones such as antidiuretic hormone (ADH) and aldosterone.

B. Regulation of basal metabolic rate: Thyroid hormone plays a crucial role in regulating the basal metabolic rate (BMR). It affects the rate at which the body utilizes energy.

E. Support the process of R.B.Cs formation: The production of red blood cells (RBCs) is primarily controlled by erythropoietin, a hormone produced by the kidneys. Thyroid hormone does not directly support the process of RBC formation.

Based on the explanations above, we can conclude that Option 3) C and D only is the correct answer.

Q. 51 Which of the following statements are correct?
A. An excessive loss of body fluid from the body switches off osmoreceptors.
B. ADH facilitates water reabsorption to prevent diuresis.
C. ANF causes vasodilation.
D. ADH causes increase in blood pressure. E. ADH is responsible for decrease in GFR.

Choose the correct answer from the options given below:
Choose the correct answer from the options given below: ·

Option 1:
A and B only

Option 2:
B, C and D only

Option 3:
A, B and-E only

Option 4:
C, D and E only

Correct Answer:
B, C and D only

Solution:
The correct answer is Option 2) B, C, and D only.

B. ADH facilitates water reabsorption to prevent diuresis: This statement is correct. Antidiuretic hormone (ADH), also known as vasopressin, acts on the kidneys to increase water reabsorption. It helps prevent excessive water loss and promotes water retention in the body.

C. ANF causes vasodilation: This statement is correct. Atrial natriuretic peptide (ANF) is a hormone released by the heart in response to increased blood volume and pressure. It promotes vasodilation, which leads to the relaxation of blood vessels and increased excretion of sodium and water.

D. ADH causes an increase in blood pressure: This statement is correct. ADH acts on the blood vessels, causing vasoconstriction. This constriction increases peripheral vascular resistance, which can contribute to an increase in blood pressure.

For incorrect options,

A. An excessive loss of body fluid from the body switches off osmoreceptors: This statement is incorrect. Osmoreceptors are specialized cells that detect changes in osmotic pressure and regulate the release of antidiuretic hormone (ADH). Excessive loss of body fluid would lead to increased osmotic pressure and activate osmoreceptors, not switch them off.

E. ADH is responsible for a decrease in GFR (glomerular filtration rate): This statement is incorrect. ADH does not directly influence the glomerular filtration rate. It primarily affects water reabsorption in the renal tubules.

Based on the evaluations, the correct answer is Option 2) B, C, and D only.

Q. 52 The parts of human brain that helps in regulation of sexual behaviour, expressian of excitement, pleasure, rage, fear etc are.

Option 1:
Limbic system & hypothalamus

Option 2:
Corpora quadrigemina & hippocampus

Option 3:
Brain stem & epithalamus

Option 4:
Corpus callosum and thalamus

Correct Answer:
Limbic system & hypothalamus

Solution:
The correct answer is Option 1) Limbic system and hypothalamus.

The limbic system and hypothalamus are the parts of the human brain that play a crucial role in regulating sexual behavior and expressing emotions such as excitement, pleasure, rage, fear, and more.

The limbic system is a collection of structures located in the cerebral hemispheres of the brain. It includes the hippocampus, amygdala, and parts of the thalamus and hypothalamus. The limbic system is involved in various functions, including emotion, behavior, motivation, and memory.

The hypothalamus is a small but important region located below the thalamus. It serves as a control center for many autonomic functions and behaviors, including regulation of body temperature, hunger, thirst, and sexual behavior. The hypothalamus plays a key role in the release of hormones involved in sexual arousal and reproductive functions.

Therefore, Option 1) Limbic system and hypothalamus is the correct answer.

Q. 53 Vital capacity of lung is____.

Option 1:
ERV + ERV

Option 2:
IRV + ERV + TV + RV

Option 3:
ERV + ERV + TV + RV

Option 4:
IRV + ERV + TV

Correct Answer:
IRV + ERV + TV

Solution:
The vital capacity of the lungs is the maximum amount of air that can be exhaled after a maximum inhalation. It is calculated by adding the following lung volumes:

IRV (Inspiratory Reserve Volume): the additional air that can be forcefully inhaled after a normal inhalation.
ERV (Expiratory Reserve Volume): the additional air that can be forcefully exhaled after a normal exhalation.
TV (Tidal Volume): the amount of air inhaled or exhaled during normal breathing.
The residual volume (RV), which is the amount of air that remains in the lungs after maximum exhalation, is not included in the vital capacity calculation.

Therefore, the correct option for the vital capacity of the lungs is indeed Option 4) IRV + ERV + TV.

Option 4 is the correct answer.

Q. 54 Match List I with List II.

	List I (cells)		List II (Secretion)
A.	Peptic cells	I.	Mucus
B.	Goblet cells	II.	Bile juice
C.	Oxyntic cells	III.	Proenzyme pepsinogen
D.	Hepatic cells	IV.	HCl and intrinsic factor for absorption of vitamin B12

Choose the **correct** answer from the options given below:

Option 1:
A–IV, B–III, C–II, D–I

Option 2:
A–II, B–I, C–III, D–IV

Option 3:
A–III, B–I, C–IV, D–II

Option 4:
A–II, B–IV, C–I, D–III

Correct Answer:
A–III, B–I, C–IV, D–II

Solution:
A. Peptic cells - III. Proenzyme pepsinogen
Peptic cells secrete the proenzyme pepsinogen, which is later activated to the enzyme pepsin and helps in the digestion of proteins.

B. Goblet cells - I. Mucus
Goblet cells secrete mucus, which helps in lubricating and protecting the epithelial lining of various organs.

C. Oxyntic cells - IV. HCl and intrinsic factor for absorption of vitamin B12
Oxyntic cells, also known as parietal cells, secrete hydrochloric acid (HCl) to aid in the digestion of food and also produce intrinsic factor, which is essential for the absorption of vitamin B12 in the small intestine.

D. Hepatic cells - II. Bile juice
Hepatic cells, found in the liver, secrete bile juice, which is stored in the gallbladder and aids in the digestion and absorption of fats.

Therefore, the correct matching is:
A–III, B–I, C–IV, D–II, which corresponds to Option 3.

Q. 55 Match **List I** with **List II**.

	List I		**List II**
A.	CCK	I.	Kidney
B.	GIP	II.	Heart
C.	ANF	III.	Gastric gland
D.	ADH	IV.	Pancreas

Choose the **correct** answer from the options given below:

Option 1:
A-IV, B-III, C-II, D-I

Option 2:
A-III, B-II, C-IV, D-I

Option 3:
A-II, B-IV, C-I, D-III

Option 4:
A-IV, B-II, C-III, D-I

Correct Answer:
A-IV, B-III, C-II, D-I

Solution:

A. CCK - IV. Pancreas
CCK (cholecystokinin) is a hormone that is released by the small intestine and stimulates the pancreas to release digestive enzymes.

B. GIP - III. Gastric gland
GIP (gastric inhibitory peptide) is a hormone that is released by the small intestine and inhibits gastric acid secretion.

C. ANF - II. Heart
ANF (atrial natriuretic peptide) is a hormone that is produced by the heart and is involved in the regulation of blood pressure and fluid balance.

D. ADH - I. Kidney
ADH (antidiuretic hormone), also known as vasopressin, is produced by the hypothalamus and released by the pituitary gland. It acts on the kidneys to regulate water balance and urine concentration.

Therefore, the correct matching is:
A-IV, B-III, C-II, D-I, which corresponds to Option 1.

Q. 56 Match List I with List II .

	List I		**List II**
A.	Taenia	I.	Nephridia
B.	Paramoecium	II.	Contractile vacuole
C.	Periplaneta	III.	Flame cells
D.	Pheretima	IV.	Urecose gland

Choose the correct answer from the options give below:

Option 1:
A-I, B-II, C-III, D-IV

Option 2:
A-I, B-II, C-IV, D-III

Option 3:
A-III, B-II, C-IV, D-I

Option 4:
A-II, B-I, C-IV, D-III

Correct Answer:
A-III, B-II, C-IV, D-I

Solution:
A. Taenia - III. Flame cells
B. Paramoecium - II. Contractile vacuole
C. Periplaneta - IV. Urecose gland
D. Pheretima - I. Nephridia

Therefore, the correct answer is:

Option 3) A-III, B-II, C-IV, D-I

Q. 57 **Assertion (A):** Nephrons are of two types: Cortical & Juxta medullary, based on their relative position in the cortex and medulla.

Reason (R): Juxta medullary nephrons have a short loop of Henle whereas, cortical nephrons have a longer loop of Henle.

In the light of the above statements, choose the correct answer from the options given below:

Option 1:
Both A and R are true and R is the correct explanation of A.

Option 2:
Both A and R are true but R is NOT the correct explanation of A.

Option 3:
A is true but R is false.

Option 4:
A is false but R is true.

Correct Answer:
A is true but R is false.

Solution:
Assertion A is true, stating that nephrons are of two types: cortical and juxtamedullary, based on their relative position in the cortex and medulla.

However, Reason R is false. Juxtamedullary nephrons have a longer loop of Henle that extends deep into the medulla, while cortical nephrons have a shorter loop of Henle that does not extend as far into the medulla. The statement in Reason R is the opposite of the actual arrangement of the loop of Henle in cortical and juxtamedullary nephrons.

Therefore, while Assertion A is true, Reason R is false. The correct answer is Option 3 which is A is true but R is false.

Q. 58 Large, colourful, fragrant flowers with nectar are seen in:

Option 1:
insect pollinated plants

Option 2:
bird pollinated plants

Option 3:
bat pollinated plants

Option 4:
wind pollinated plants

Correct Answer:
insect pollinated plants

Solution:
- Entomophily is the scientific term used to describe insect pollination.
- Flowers that rely on insect pollination tend to be large, vibrant in color, emit a pleasant fragrance, and contain abundant nectar.
- To enhance visibility, multiple flowers are often arranged in clusters known as inflorescences.
- These flowers possess nectar glands and emit strong fragrances to attract insects.
- Option 1 is the correct answer.

Q. 59 Which one of the following common sexually transmitted diseases is completely curable when detected early and treated properly?

Option 1:
Genital herpes

Option 2:
Gonorrhoea

Option 3:
Hepatitis-B

Option 4:
HIV Infection

Correct Answer:
Gonorrhoea

Solution:
The correct option is Option 2) Gonorrhoea.

Gonorrhoea is a common sexually transmitted disease caused by the bacterium Neisseria gonorrhoeae. When detected early and treated properly with appropriate antibiotics, gonorrhoea can be completely cured. It is important to seek medical attention and adhere to the prescribed treatment to ensure effective eradication of the infection. However, it is worth noting that the emergence of antibiotic-resistant strains of Neisseria gonorrhoeae is a growing concern, emphasizing the importance of early detection and appropriate treatment.

Q. 60 Which of the following statements are correct regarding female reproductive cycle?

A. In non-primate mammals cyclical changes during reproduction are called oestrus cycle.
B. First menstrual cycle begins at puberty and is called menopause.
C. Lack of menstruation may be indicative of pregnancy.
D. Cyclic menstruation extends between menarche and menopause.

Choose the **most appropriate** answer from the options given below:-

Option 1:
A and D only

Option 2:
A and B only

Option 3:
A, B and C only

Option 4:
A, C and D only

Correct Answer:
A, C and D only

Solution:
The correct option is Option 4) A, C, and D only.

A. In non-primate mammals, cyclical changes during reproduction are indeed called the estrus cycle, not the menstrual cycle. The estrus cycle is characterized by the presence of an estrus phase, during which the female is sexually receptive.

B. The first menstrual cycle in females begins at puberty, marking the onset of reproductive capability. Menopause, on the other hand, refers to the cessation of menstrual cycles and reproductive function in women, typically occurring around the age of 45-55.

C. Lack of menstruation can be indicative of pregnancy. During pregnancy, menstruation ceases due to the hormonal changes that sustain the pregnancy.

D. Cyclic menstruation extends between menarche (the onset of the first menstrual cycle) and menopause (the cessation of menstrual cycles).

Therefore, statements A, C, and D are correct, while statement B is incorrect.

Q. 61 Given below are two statements: one is labelled as **Assertion** A and the other is labelled as **Reason R**.

Assertion A: Amniocentesis for sex determination is one of the strategies of Reproductive and Child Health Care Programme.
Reason R: Ban ion amniocentesis checks increasing menace of female foeticide.

In the light of the above statements, choose the **correct** answer from the options given below:

Option 1:
Both **A** and R are true and **R** is the correct explanation of **A.**

Option 2:
Both **A** and **R** are true and **R** is NOT the correct explanation of **A.**

Option 3:
A is true but **R** is false.

Option 4:
A is false but **R** is true.

Correct Answer:
A is false but **R** is true.

Solution:
The correct option is Option 4) A is false but R is true.

Amniocentesis is a prenatal diagnostic procedure that involves the collection of amniotic fluid from the amniotic sac surrounding the fetus. It is primarily used for genetic testing and detecting chromosomal abnormalities, not specifically for sex determination. Therefore, Assertion A is false.

On the other hand, Reason R is true. The reason for banning amniocentesis for sex determination is to address the increasing problem of female foeticide, which is the selective abortion of female fetuses due to a preference for male children. The ban is aimed at preventing the misuse of prenatal diagnostic techniques for gender-based discrimination.

Since Assertion A is false and Reason R is true, the correct option is 4) A is false but R is true.

Q. 62 Given below are two statements:

Statement I: Vas deferens receives a duct from seminal vesicle and opens into urethra as the ejaculatory duct:
Statement II: The cavity of the cervix is called cervical canal which along with vagina' forms birth canal.
In the light of the above statements, choose the correct answer from the options given below:

Option 1:
Both Statement I and Statement II are true.

Option 2:
Both Statement I and Statement II are false.

Option 3:
Statement I is correct but Statement II is false .

Option 4:
Statement I incorrect but Statement II is true.

Correct Answer:
Both Statement I and Statement II are true.

Solution:
The correct answer is Option 1) Both Statement I and Statement II are true.

Explanation:
Statement I: The vas deferens is a tube that receives a duct from the seminal vesicle and carries sperm from the epididymis to the urethra. It is a part of the male reproductive system and is involved in the transportation of sperm during ejaculation.

Statement II: The cervix is the lower part of the uterus that connects the uterus to the vagina. It contains a cavity called the cervical canal, which is a passageway for sperm to enter the uterus and for menstrual blood to exit the uterus during menstruation. The cervix, along with the vagina, forms the birth canal through which a baby passes during childbirth.

Both statements accurately describe anatomical structures and their functions in the male and female reproductive systems, respectively. Therefore, Option 1) Both Statement I and Statement II are true.

Q. 63 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: Endometrium is necessary for implantation of blastocyst.
Reason R: In the absence of fertilization, the corpus luteum degenerates that causes disintegration of endometrium.
In the light of the above statements, choose the correct answer from the options given below:

Option 1:
Both **A** and **R** are true and **R** is the correct explanation of **A .**

Option 2:
Both **A** and **R** are true and **R** is NOT the correct explanation of **A .**

Option 3:
A is true but **R** is false.

Option 4:
A is true but **R** is true.

Correct Answer:
Both **A** and **R** are true and **R** is NOT the correct explanation of **A .**

Solution:
Option 2) Both A and R are true, but R is NOT the correct explanation of A.

The assertion (A) that the endometrium is necessary for implantation of the blastocyst is true. The endometrium provides the necessary environment for the embryo to attach and establish a connection with the maternal blood supply.

The reason (R) provided, stating that the degeneration of the corpus luteum causes the disintegration of the endometrium, is not the correct explanation for the necessity of the endometrium. The degeneration of the corpus luteum and the subsequent decrease in hormone levels contribute to the shedding of the endometrium during menstruation but do not directly explain the requirement of the endometrium for blastocyst implantation.

Therefore, option 2 is the correct answer. Both statements are true, but the reason does not explain the assertion.

Q. 64 What is the function of tassels in the corn cob?

Option 1:
To attract insects

Option 2:
To trap pollen grains

Option 3:
To disperse pollen grains

Option 4:
To protect seeds

Correct Answer:
To disperse pollen grains

Solution:
The correct option is Option 3) To disperse pollen grains

Tassels are the male reproductive structures found at the top of the corn cob in corn plants. Their primary function is to produce and release pollen grains, which are the male gametes. The tassels are composed of numerous long and slender filaments, each bearing anthers at their tips.

The anthers within the tassels contain the pollen grains. When the corn plant is mature and ready for pollination, the tassels release large amounts of pollen into the air. The purpose of releasing pollen is to ensure successful fertilization of the female reproductive structures called silk.

The silks, found in the ear of the corn cob, are the female reproductive structures. The pollen released from the tassels is carried by wind or other pollinators to the silks, where the process of fertilization occurs.

The tassels, with their filaments and anthers, effectively trap and release the pollen grains, allowing them to be dispersed and transported to the female flowers for pollination and fertilization. This is essential for successful seed production in corn plants.

Therefore, the main function of tassels in the corn cob is to trap but majorly it is to release pollen grains for the purpose of pollination and fertilization.

Q. 65 Match **List I** with **List II**.

	List I		List II
A.	Vasectomy	I.	Oral method
B.	Coitus interruptus	II.	Barrier method
C.	Cervical caps	III.	Surgical method
D.	Saheli	IV.	Natural method

Choose the **correct** answer from the options given below:-

Option 1:
A-III, B-I, C-IV, D-II

Option 2:
A-III, B-IV, C-II, D-I

Option 3:
A-II, B-III, C-I, D-IV

Option 4:
A-IV, B-II, C-I, D-III

Correct Answer:
A-III, B-IV, C-II, D-I

Solution:
The correct option is Option 2) A-III, B-IV, C-II, D-I.

A. Vasectomy - III. Surgical method
B. Coitus interruptus - IV. Natural method
C. Cervical caps - II. Barrier method
D. Saheli - I. Oral method

Q. 66 The phenomenon of pleiotropism refers to

Option 1:
presence of several alleles of a single gene controlling a single crossover.

Option 2:
presence of two alleles, each of the two genes controlling a single trait.

Option 3:
a single gene affecting multiple phenotypic expression

Option 4:
more than two genes affecting a single character.

Correct Answer:
a single gene affecting multiple phenotypic expression

Solution:
Pleiotropy refers to a phenomenon in which a single gene can manifest multiple phenotypic expressions. In most cases, this occurs due to the gene's impact on various metabolic pathways that contribute to different observable traits.

Option 3 is the correct answer

Q. 67 Frequency of recombination between gene pairs on same chromosome as a measure of the distance between genes to map their position on chromosome, was used for the first time by

Option 1:
Thomas Hunt Morgan

Option 2:
Sutton and Boveri

Option 3:
Alfred Sturtevant

Option 4:
Henking

Correct Answer:
Alfred Sturtevant

Solution:
Alfred Sturtevant was a geneticist who played a significant role in the field of genetics, particularly in the mapping of genes on chromosomes. He was the first to propose the use of recombination frequency as a measure of the distance between genes on the same chromosome. Sturtevant's work was based on the observations made by Thomas Hunt Morgan, who studied the patterns of inheritance in fruit flies. However, it was Sturtevant who developed the concept of genetic mapping using recombination

frequencies, which later became known as Sturtevant's mapping function. His groundbreaking work paved the way for the understanding of gene linkage and the construction of genetic maps.
Option 3 is the correct answer.

Q. 68 Unequivocal proof that DNA is the genetic material was first proposed by

Option 1:
Frederic Griffith

Option 2:
Alfred Hershey and Martha Chase

Option 3:
Avery, Macleoid and McCarthy

Option 4:
Wilkins and Franklin

Correct Answer:
Alfred Hershey and Martha Chase

Solution:
The correct option is Option 2) Alfred Hershey and Martha Chase.

Alfred Hershey and Martha Chase conducted the Hershey-Chase experiment in 1952, which provided unequivocal proof that DNA is the genetic material. In their experiment, they used bacteriophages, which are viruses that infect bacteria. They labeled the DNA of the bacteriophage with radioactive phosphorus-32 and the protein coat with radioactive sulfur-35.

By infecting bacteria with these labeled bacteriophages and then separating the viral DNA and proteins from the infected bacteria, Hershey and Chase showed that only the radioactive DNA, not the protein, was transferred to the next generation of phages. This demonstrated that DNA, not protein, carried the genetic information and served as the hereditary material.

Their experiment provided strong evidence supporting the idea that DNA, rather than protein, is the genetic material responsible for the transmission of inherited traits. Therefore, Hershey and Chase were the ones who first proposed the unequivocal proof that DNA is the genetic material.

Q. 69 What is the role of RNA polymerase III in the process of transcription in Eukaryotes?

Option 1:
Transcription of tRNAs $(28S, 18S\ and\ 5.8S)$

Option 2:
Transcription of tRNA, 5 srRNA and snRNA

Option 3:
Transcription of precursor of mRNA

Option 4:
Transcription of only snRNAs

Correct Answer:
Transcription of tRNA, 5 srRNA and snRNA

Solution:
The correct option is Option 2) Transcription of tRNA, 5S rRNA, and snRNA.

RNA polymerase III is an enzyme responsible for transcribing specific types of RNA molecules in eukaryotes. Its primary role is the transcription of three major types of RNA:

tRNA (transfer RNA): tRNA molecules play a crucial role in protein synthesis by carrying specific amino acids to the ribosomes. RNA polymerase III transcribes the genes encoding tRNA molecules.

5S rRNA (5S ribosomal RNA): 5S rRNA is a component of the large ribosomal subunit involved in protein synthesis. RNA polymerase III transcribes the genes encoding 5S rRNA.

snRNA (small nuclear RNA): snRNA molecules are involved in the processing of pre-mRNA and in the assembly of spliceosomes, which are responsible for removing introns from pre-mRNA. RNA polymerase III transcribes the genes encoding snRNA.

Therefore, RNA polymerase III plays a critical role in transcribing tRNA, 5S rRNA, and snRNA, contributing to important cellular functions such as protein synthesis and RNA processing.

Q. 70 Expressed Sequence Tags (ESTs) refers to

Option 1:
All genes that are expressed as RNA.

Option 2:
All genes that are expressed as proteins.

Option 3:
All genes whether expressed or unexpressed.

Option 4:
Certain important expressed genes.

Correct Answer:
All genes that are expressed as RNA.

Solution:

The correct option is Option 1) All genes that are expressed as RNA.

Expressed Sequence Tags (ESTs) are short DNA sequences that are generated from complementary DNA (cDNA) libraries. These sequences represent fragments of messenger RNA (mRNA) molecules that have been transcribed from expressed genes in a specific tissue or organism.

ESTs are obtained by converting mRNA into cDNA using the enzyme reverse transcriptase. The cDNA is then cloned into a vector and sequenced. These sequences provide information about the genes that are actively transcribed and expressed in the tissue or organism from which the mRNA was isolated.

ESTs are particularly useful in gene discovery and gene expression studies. They provide valuable insights into the identity and expression patterns of genes, allowing researchers to identify and study genes that may be involved in specific biological processes or diseases.

Therefore, ESTs refer to all genes that are expressed as RNA, as stated in Option 1.

Q. 71 Which of the following statements are correct about Klinefelter's Syndrome?

A. This disorder was first described by Langdon Down (1866).

B. Such an individual has overall masculine development. However, the feminine development is also expressed.

C. The affected individual is short statured.

D. Physical, psychomotor and mental development is retarded.

E. Such individuals are sterile.

Choose the correct answer from the options given below.

Option 1:
A and B only

Option 2:
C and D only

Option 3:
B and E only

Option 4:
A and E only

Correct Answer:
B and E only

Solution:
The correct statements about Klinefelter's Syndrome are:

B. Such an individual has overall masculine development. However, the feminine development is also expressed.
E. Such individuals are sterile.

Therefore, the correct answer is Option 3) B and E only.

Q. 72 Select the correct group/set of Australian Marsupials exhibiting adaptive radiation

Option 1:
Tasmanian wolf, Bobcat, Marsupial mole

Option 2:
Numbat, Spottted cuscus, Flying phalanger

Option 3:
Mole, Flying squirrel, Tasmanian tiger cat

Option 4:
Lemur, Anteater, Wolf

Correct Answer:
Numbat, Spottted cuscus, Flying phalanger

Solution:
The correct option is Option 2) Numbat, Spotted cuscus, Flying phalanger.

Adaptive radiation refers to the diversification of a group of organisms into various ecological niches, resulting in the development of different species with specialized adaptations. In the case of Australian marsupials, there are several examples of adaptive radiation.

Numbat, Spotted cuscus, Flying phalanger: This option includes marsupials that have undergone adaptive radiation. The numbat is a small marsupial that feeds mainly on termites and has specialized adaptations for this diet. The spotted cuscus is a tree-dwelling marsupial that has adapted to life in the rainforest. The flying phalanger, also known as the sugar glider, is a marsupial that has evolved adaptations for gliding through the air.

Option 1) is incorrect because Tasmanian wolf, Bobcat, Marsupial mole: This option includes species that are not examples of Australian marsupials. The Tasmanian wolf, also known as the thylacine, is extinct. The bobcat is not a marsupial but a small wild cat found in North America. The marsupial mole is native to Australia but is not considered an example of adaptive radiation.

Option 3) is incorrect because Mole, Flying squirrel, Tasmanian tiger cat: This option includes species that are not Australian marsupials. The mole is not a marsupial but a type of mammal found in various parts of the world. The flying squirrel is not native to Australia but is found in other regions. The Tasmanian tiger cat is not a recognized marsupial species.

Option 4) is incorrect because Lemur, Anteater, Wolf: This option includes species that are not Australian marsupials. Lemurs are primates found in Madagascar, not Australia. Anteaters are found in Central and South America, and wolves are found in various parts of the world.

Therefore, the correct option is Option 2) Numbat, Spotted cuscus, Flying phalanger, as these species are examples of Australian marsupials that have undergone adaptive radiation.

Q. 73 Given below are two statements:

Statements I: In prokaryotes, the positively charged DNA is held with some negatively charged proteins in a region called nucleoid.
Statement II: In eukaryotes, the negatively charged DNA is wrapped around the positively charged histone octamer to form nucleosome.

In the light of the above statements, choose the **correct** answer from the options given below :

Option 1:
Both Statement I and Statement II are true.

Option 2:
Both Statement I and Statement II are false.

Option 3:
Statement I is correct but statement II is false.

Option 4:
Statement I incorrect but statement II is true.

Correct Answer:
Statement I incorrect but statement II is true.

Solution:
The correct answer is Option 4) Statement I is incorrect, but statement II is true.

Statement I: In prokaryotes, the positively charged DNA is held with some negatively charged proteins in a region called nucleoid.

This statement is incorrect. In prokaryotes, the DNA is not associated with proteins like histones to form nucleosomes as seen in eukaryotes. Instead, prokaryotic DNA is typically found in a region called the nucleoid, where it is condensed and organized to some extent, but without the association of histones or other proteins.

Statement II: In eukaryotes, the negatively charged DNA is wrapped around the positively charged histone octamer to form nucleosome.

This statement is true. In eukaryotes, DNA is wrapped around histone proteins to form nucleosomes. The DNA molecule is negatively charged due to its phosphate backbone, while the histone proteins are positively charged. The interaction between the negatively charged DNA and the positively charged histones allows for the compaction and organization of DNA into a higher-order structure.

Therefore, the correct answer is Option 4) Statement I is incorrect, but statement II is true.

Q. 74 Broad plam with single palm crease is visible in a person suffering from –

Option 1:
Down's syndrome

Option 2:
Turner's syndrome

Option 3:
Klinefelter's syndrome

Option 4:
Thalassemia

Correct Answer:
Down's syndrome

Solution:
The correct answer is Option 1) Down's syndrome.

A broad palm with a single palm crease, also known as a simian crease or a single transverse palmar crease, is commonly seen in individuals with Down's syndrome. Down's syndrome is a genetic disorder caused by the presence of an extra copy of chromosome 21. This condition is characterized by various physical and intellectual disabilities, and certain physical features are often associated with it.

One of the common physical features observed in individuals with Down's syndrome is a single transverse palmar crease, where the normal two creases in the palm fuse into one. This can give the appearance of a broad palm with a single crease extending across the palm.

Therefore, the correct answer is Option 1) Down's syndrome.

Q. 75 Which one of the following is sequence on corresponding coding strand, if the sequence on mRNA formed is as follows
5'AUCGAUCGAUCGAUCGAUCGAUCG AUCG 3'?

Option 1:
5'UAGCUAGCUAGCUAGCUAGCUAGC UAGC 3'

Option 2:
3'UAGCUAGCUAGCUAGCUAGCUAGCUAGC 5'

Option 3:
5'ATCGATCGATCGATCGATCGATCGATCG 3'

Option 4:
3'ATCGATCGATCGATCGATCGATCGATCG 5'

Correct Answer:
5'ATCGATCGATCGATCGATCGATCGATCG 3'

Solution:
To find the corresponding coding strand, we replace each base according to the base pairing rules:

A (mRNA) pairs with U (coding DNA)
U (mRNA) pairs with A (coding DNA)
C (mRNA) pairs with G (coding DNA)
G (mRNA) pairs with C (coding DNA)

Applying these base pairing rules to the given mRNA sequence, we get:

5'AUCGAUCGAUCGAUCGAUCGAUCG AUCG 3'
3'TAGCTAGCTAGCTAGCTAGCTAGC TAGC 5'

Therefore, the correct corresponding coding strand sequence is 3'TAGCTAGCTAGCTAGCTAGCTAGC TAGC 5'.

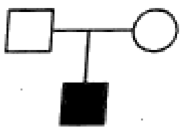
Option 3 is the correct answer.

Q. 76 Which one of the following symbols represents mating between relatives in human pedigree analysis?

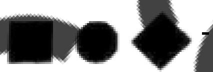
Option 1:

Option 2:

Option 3:



Option 4:



Correct Answer:

Solution:
In human pedigree analysis, mating between relatives is represented by a double horizontal line connecting a male and a female. This symbol indicates consanguineous mating, which means that the individuals involved share a common ancestor. The double horizontal line represents a marriage or mating relationship, while the vertical lines extending downward from the double line represent their offspring.

Option 2 is the correct answer.

Q. 77 Match **List I** with **List II**.

	List I		List II
A.	Gene 'a'	I.	β-galactosidase
B.	Gene 'y'	II.	Treansacetylase
C.	Gene 'I'	III.	Permease
D.	Gene 'z'	IV.	Repressor protein

Choose the **correct** answer from the options given below:

Option 1:
A-II, B-I, C-IV, D-III

Option 2:
A-II, B-III, C-IV, D-I

Option 3:
A-III, B-IV, C-I, D-II

Option 4:
A-III, B-I, C-IV, D-II

Correct Answer:
A-II, B-III, C-IV, D-I

Solution:
A. Gene 'a' - II. Transacetylase
Transacetylase is an enzyme involved in the metabolism of lactose.

B. Gene 'y' - III. Permease
Permease is a protein that facilitates the movement of molecules across the cell membrane.

C. Gene 'I' - IV. Repressor protein
Repressor protein regulates the expression of genes by binding to the operator region of DNA and preventing transcription.

D. Gene 'z' - I. β-galactosidase
β-galactosidase is an enzyme that hydrolyzes lactose into glucose and galactose.

Therefore, the correct matching is:
A-II, B-III, C-IV, D-I, which corresponds to Option 2.

Q. 78 In tissue culture experiments, leaf mesophyll cells are put in a culture medium to form callus. This phenomenon may be called as :

Option 1:
Differentiation

Option 2:
Dedifferentiation

Option 3:
Development

Option 4:
Senescence

Correct Answer:
Dedifferentiation

Solution:
The correct option is Option 2) Dedifferentiation. In tissue culture experiments, when leaf mesophyll cells are placed in a culture medium and start to divide and proliferate, they undergo dedifferentiation. Dedifferentiation refers to the process where specialized cells revert to a less specialized or undifferentiated state. In the case of tissue culture, dedifferentiation allows the cells to lose their specialized characteristics and regain their ability to divide and form callus, which is an undifferentiated mass of cells. This process provides an opportunity for researchers to manipulate and regenerate new plants from these dedifferentiated cells.

Q. 79 Which of the following is NOT an advantage of inbreeding?

Option 1:
It decreases homozygosity.

Option 2:
It exposes harmful recessive genes that are eliminated by selection.

Option 3:
Elimination of less desirable genes and accumulation of superior genes takes place due to it.

Option 4:
It decreases the productivity of inbred population, after continuous inbreeding.

Correct Answer:
It decreases the productivity of inbred population, after continuous inbreeding.

Solution:
The correct answer is Option 4) It decreases the productivity of the inbred population after continuous inbreeding.

Inbreeding refers to the mating of individuals who are closely related, such as siblings or close relatives. While inbreeding can have certain advantages in specific situations, such as in selective breeding programs, it also has several disadvantages.

Advantages of inbreeding include:

- Increased uniformity: Inbreeding can increase the uniformity of a population by concentrating desirable traits and reducing genetic variation.
- Identification of recessive traits: Inbreeding can expose harmful recessive genes that are otherwise masked in outbred populations. This allows for the identification and elimination of these harmful traits through selective breeding.
- Fixation of desired traits: Inbreeding can help fix or stabilize desirable traits in a population by repeatedly mating individuals with those traits. This is particularly useful in breeding programs aimed at developing specific breeds or lines.

Option 4 states that inbreeding decreases the productivity of the inbred population after continuous inbreeding. This is a correct statement as continuous inbreeding can lead to inbreeding depression. Inbreeding depression is a phenomenon where reduced fitness, lower fertility, and decreased productivity are observed in inbred populations due to the accumulation of deleterious recessive traits.

Therefore, Option 4 is the correct answer as it describes a disadvantage of inbreeding.

Q. 80 Match **List I** with **List II**

	List I		List II
A.	Heroin	I.	Effect on cardiovascular system
B.	Marijuana	II.	Slow down body function
C.	Cocaine	III.	Painkiller
D.	Morphine	IV.	Interfere with transport of dopamine

Choose the **correct** answer from the options given below:

Option 1:
A–II, B–I, C–IV D–III

Option 2:
A–I, B–II, C–III, D–IV

Option 3:
A–IV, B–III, C–II, D–I

Option 4:
A–III, B–IV, C–I, D–II

Correct Answer:
A–II, B–I, C–IV D–III

Solution:
The correct answer is Option 2) A–I, B–II, C–III, D–IV.

Explanation:
A. Heroin is a narcotic drug that acts as a painkiller.
B. Marijuana is known to slow down body functions.
C. Cocaine is a stimulant drug that interferes with the transport of dopamine, a neurotransmitter.
D. Morphine is a strong painkiller, often used in medical settings.

Therefore, the correct match is:
A–I (Heroin - Painkiller)
B–II (Marijuana - Slow down body function)
C–III (Cocaine - Interfere with transport of dopamine)
D–IV (Morphine - Effect on cardiovascular system)

Q. 81 In which blood corpuscles, the HIV undergoes replication and produces progeny viruses ?

Option 1:
T_H Cells

Option 2:
B–lymphocytes

Option 3:
Basophils

Option 4:
Eosinophils

Correct Answer:
T_H Cells

Solution:
The correct answer is Option 1) TH Cells (T-helper cells).

HIV (Human Immunodeficiency Virus) primarily infects and replicates in T-helper cells, which are a type of white blood cell involved in the immune response. HIV specifically targets CD4 receptors on the surface of T-helper cells. Once inside the T-helper cells, the virus replicates and produces progeny viruses, leading to the destruction of T-helper cells and the weakening of the immune system. This is why HIV infection can result in immunodeficiency and increased susceptibility to various infections and diseases.

Q. 82 Match **List I** with **List II.**

	List I		List II
A.	Ringworm	I.	Haemophilus influenzae
B.	Filariasis	II.	Trichophyton
C.	Malaria	III.	Wuchereria bancrofti
D.	Pneumonia	IV.	Plasmodium vivax

Choose the **correct** answer from the options given below:

Option 1:
A-II, B-III, C-IV, D-I

Option 2:
A-II, B-III, C-I, D-IV

Option 3:
A-III, B-II, C-I, D-IV

Option 4:
A-III, B-II, C-IV, D-I

Correct Answer:
A-II, B-III, C-IV, D-I

Solution:

A. Ringworm is caused by the fungus Trichophyton.
B. Filariasis is caused by the parasite Wuchereria bancrofti.
C. Malaria is caused by the parasite Plasmodium vivax.
D. Pneumonia is caused by the bacterium Haemophilus influenzae.

Therefore, the correct match is:
A-II, B-III, C-IV, D-I.

Option 1) A-II, B-III, C-IV, D-I is the correct answer.

Q. 83 In gene gun method used to introduce alien DNA into host cells, microparticles of _______ metal are used.

Option 1:
Copper

Option 2:
Zinc

Option 3:
Tungsten or gold

Option 4:
Silver

Correct Answer:
Tungsten or gold

Solution:
The correct option is Option 3) Tungsten or gold.

In the gene gun method, which is a technique used to introduce foreign DNA into host cells, microparticles of either tungsten or gold are commonly used.

The gene gun method, also known as biolistic particle delivery system, involves coating tiny microparticles with foreign DNA and propelling them into the target cells. The microparticles act as carriers for the foreign DNA and are propelled into the host cells using a high-pressure gas or a mechanical force.

Tungsten and gold are preferred materials for the microparticles because they are dense and can effectively penetrate the cell walls and membranes of the target cells. The microparticles are typically coated with the desired DNA and then shot into the host cells, where they can integrate into the cell's genome and express the foreign genes.

Therefore, in the gene gun method, microparticles of either tungsten or gold are commonly used as carriers for the introduction of alien DNA into host cells.

Q. 84 Upon exposure to UV radiation, DNA stained with ethidium bromide will show

Option 1:
Bright red colour

Option 2:
Bright blue colour

Option 3:
Bright yellow colour

Option 4:
Bright orange colour

Correct Answer:
Bright orange colour

Solution:
The correct option is Option 4) Bright orange colour.

Ethidium bromide is a fluorescent dye commonly used to stain DNA and visualize it under ultraviolet (UV) light. When ethidium bromide binds to DNA, it intercalates between the base pairs of the DNA molecule.

When exposed to UV radiation, ethidium bromide emits a fluorescent signal that can be visualized. Under UV light, DNA stained with ethidium bromide typically appears as bright orange bands or spots.

The orange color is a result of the interaction between ethidium bromide and the DNA molecule. The intercalation of ethidium bromide into the DNA helix causes a shift in the absorbance and emission wavelengths, resulting in the characteristic orange fluorescence when exposed to UV light.

Therefore, Option 4) Bright orange colour is the correct option for how DNA stained with ethidium bromide appears when exposed to UV radiation.

Q. 85 During the purification process for recombinant DNA technology, addition of chilled ethanol precipitates out

Option 1:
RNA

Option 2:
DNA

Option 3:
Histones

Option 4:
Polysaccharides

Correct Answer:
DNA

Solution:
The correct option is Option 2) DNA.

During the purification process in recombinant DNA technology, chilled ethanol is often used to precipitate out DNA. This technique is based on the principle that DNA is less soluble in cold ethanol than in aqueous solutions.

When chilled ethanol is added to the DNA solution, it causes the DNA molecules to come out of solution and form a visible precipitate. This is because the ethanol disrupts the hydrogen bonding and hydrophobic interactions that hold the DNA molecules together, leading to their aggregation and precipitation.

By centrifuging the mixture, the precipitated DNA can be collected as a pellet, while other contaminants remain in the supernatant. The DNA pellet can then be further purified and processed for various applications in recombinant DNA technology.

Therefore, Option 2) DNA is the correct answer for the component that precipitates out when chilled ethanol is added during the purification process for recombinant DNA technology.

Q. 86 Main steps in the formation of Recombinant DNA are given below.

Arrange these steps in a correct sequence.

A. Insertion of recombinant DNA into the host cell.

B. Cutting of DNA at specific location by restriction enzyme.

C. Isolation of desired DNA fragment.

D. Amplification of gene of interest using PCR.

Choose the correct answer from the options given below:

Option 1:
B, C, D, A

Option 2:
C, A, B, D

Option 3:
C, B, D, A

Option 4:
B, D, A, C

Correct Answer:
B, C, D, A

Solution:
The correct sequence of steps in the formation of Recombinant DNA is:

Option 1) B, C, D, A

The correct sequence is as follows:

Step 1: Cutting of DNA at specific locations by a restriction enzyme (B)
Step 2: Isolation of the desired DNA fragment (C)
Step 3: Amplification of the gene of interest using PCR (D)
Step 4: Insertion of the recombinant DNA into the host cell (A)

Therefore, the correct answer is Option 1.

Q. 87 Which one of the following techiques does not serve the purpose of early diagnosis of a disease for its early treatment ?

Option 1:
Recombinant DNA Technology

Option 2:
Serum and Urine analysis

Option 3:
Polymerase Chain Reaction (PCR) technique

Option 4:
Enzyme Linked Immuno-Sorbent Assay (ELISA) technique

Correct Answer:
Serum and Urine analysis

Solution:
The correct answer is Option 2) Serum and Urine analysis.

Serum and urine analysis are useful diagnostic techniques, but they do not specifically serve the purpose of early diagnosis of a disease for its early treatment. Serum analysis involves testing the blood serum for various markers, such as antibodies or specific substances, to detect the presence of a disease or assess its progression. Urine analysis involves examining urine for abnormalities that may indicate certain diseases or conditions.

On the other hand, the other options mentioned do serve the purpose of early diagnosis for early treatment:

Option 1) Recombinant DNA Technology: This technology involves the manipulation of DNA to produce specific molecules or substances for diagnostic purposes, such as the production of recombinant proteins or DNA probes for detecting disease markers.

Option 3) Polymerase Chain Reaction (PCR) technique: PCR is a powerful molecular biology technique used to amplify specific DNA sequences. It is widely used in disease diagnosis, including the early detection of infections or genetic disorders.

Option 4) Enzyme-Linked Immuno-Sorbent Assay (ELISA) technique: ELISA is a technique used to detect and quantify specific substances, such as antibodies or antigens, in samples. It is commonly used in disease diagnosis, including the early detection of infections or the presence of specific molecules associated with diseases.

Therefore, the correct answer is Option 2) Serum and Urine analysis.

Q. 88 Which of the following is not a cloning vector?

Option 1:
BAC

Option 2:
YAC

Option 3:
pBR322

Option 4:
Probe

Correct Answer:
Probe

Solution:

The correct answer is:

Option 4) Probe

A probe is not a cloning vector. It is a short, single-stranded DNA or RNA molecule that is used to detect a specific sequence in a sample. It is not used for the replication and cloning of DNA fragments like the other options (BAC, YAC, pBR322) which are commonly used cloning vectors.

Q. 89 Identify the correct statements :

(A) Detrivores perform fragmentation.

(B) The humus is further degraded by some microbes during mineralization.

(C) Water soluble inorganic nutrients go down into the soil and get precipitated by a process called leaching.

(D) The detritus food chain begins with living organisms.

(E) Earthworms break down detritus into smaller particles by a process called catabolism.

Option 1:
A, B, C only

Option 2:
B, C, D only

Option 3:
C, D, E only

Option 4:
D, E, A only

Correct Answer:
A, B, C only

Solution:
The correct option is Option 1) A, B, C only.

(A) Detrivores perform fragmentation. This statement is correct. Detrivores are organisms that break down organic matter into smaller particles through fragmentation, facilitating the decomposition process.

(B) The humus is further degraded by some microbes during mineralization. This statement is correct. Humus, the partially decomposed organic matter, undergoes further degradation by microbes during the process of mineralization, where organic compounds are converted into inorganic forms.

(C) Water-soluble inorganic nutrients go down into the soil and get precipitated by a process called leaching. This statement is correct. Leaching is the process by which water-soluble inorganic nutrients are washed down through the soil profile and can get precipitated or accumulate in deeper layers, affecting nutrient availability.

(D) The detritus food chain begins with living organisms. This statement is incorrect. The detritus food chain begins with dead organic matter, such as fallen leaves or dead animals. It involves the decomposition and consumption of detritus by organisms.

(E) Earthworms break down detritus into smaller particles by a process called catabolism. This statement is incorrect. Earthworms break down detritus into smaller particles through a process called physical fragmentation or grinding, not catabolism.

Therefore, Option 1) A, B, C only is the correct choice.

Q. 90 The thickness of ozone in a column of air in the atmosphere is measured in terms of :

Option 1:
Dobson units

Option 2:
Decibels

Option 3:
Decameter

Option 4:
Kilobase

Correct Answer:
Dobson units

Solution:
The correct option is Option 1) Dobson units. The thickness of ozone in a column of air in the atmosphere is commonly measured in Dobson units (DU). Dobson units are a unit of measurement used to quantify the concentration of ozone in the Earth's atmosphere. They are named after G.M.B. Dobson, a pioneer in atmospheric ozone research.

Q. 91 In the equation

GPP – R = NPP

GPP is Gross Primary Productivity

NPP is Net Primary Productivity

R here is ___________.

Option 1:
Photosynthetically active radiation

Option 2:
Respiratory quotient

Option 3:
Respiratory loss

Option 4:
Reproductive allocation

Correct Answer:
Respiratory loss

Solution:
The correct option is Option 3) Respiratory loss.

In the given equation, GPP (Gross Primary Productivity) represents the total amount of energy fixed by plants through photosynthesis. R represents the respiratory loss, which is the energy consumed by the plant through respiration. Respiration is the process by which organisms, including plants, convert stored energy (such as glucose) into usable energy (ATP) for cellular activities.

NPP (Net Primary Productivity) represents the net amount of energy available for plant growth and biomass production after accounting for the energy lost through respiration. Therefore, in the equation GPP - R = NPP, R represents the respiratory loss, indicating the energy consumed by plants during respiration.

Q. 92 Among 'The Evil Quartet', which one is considered the most important cause driving extinction of species?

Option 1:
Habitat loss and fragmentation

Option 2:
Over exploitation for economic gain

Option 3:
Alien species invasions

Option 4:
Co-extinctions

Correct Answer:
Habitat loss and fragmentation

Solution:
The correct option is Option 1) Habitat loss and fragmentation.

"The Evil Quartet" refers to four major drivers of species extinction, as identified by the renowned ecologist Paul Ehrlich. These drivers are habitat loss and fragmentation, over exploitation for economic gain, alien species invasions, and co-extinctions.

Among these, habitat loss and fragmentation are widely considered the most significant cause driving the extinction of species. As human activities continue to modify and destroy natural habitats, species are losing their homes and the resources they need to survive. Fragmentation refers to the breaking up of habitats into smaller, isolated patches, which further exacerbates the negative impacts on species by limiting their mobility, access to resources, and gene flow.

While the other factors listed in the options (over exploitation for economic gain, alien species invasions, and co-extinctions) also contribute to species loss, habitat loss and fragmentation are recognized as the primary drivers with the most far-reaching and detrimental consequences for biodiversity.

Q. 93 The historic Convention on Biological Diversity, 'The Earth Summit' was held in Rio de Janeiro in the year :

Option 1:
1985

Option 2:
1992

Option 3:
1986

Option 4:
2002

Correct Answer:
1992

Solution:
The correct option is Option 2) 1992.

The historic Convention on Biological Diversity, also known as "The Earth Summit," took place in Rio de Janeiro in the year 1992. The summit was officially called the United Nations Conference on Environment and Development (UNCED) and was held from June 3 to June 14, 1992.

During this summit, the Convention on Biological Diversity (CBD) was adopted, which is an international treaty aimed at promoting the conservation and sustainable use of biodiversity and the fair and equitable sharing of its benefits. The CBD has been ratified by numerous countries and is considered a key global framework for addressing biodiversity loss and environmental sustainability.

Therefore, the Earth Summit, where the Convention on Biological Diversity was established, occurred in Rio de Janeiro in 1992.

Q. 94 Given below are two statements :

Statement I : Gause's Competitive Exclusion Principle' states that two closely related species competing for the same resources cannot co-exist indefinitely and competitively inferior one will be eliminated eventually.

Statement II : In general, carnivores are more adversely affected by competition than herbivores. In the light of the above statements,

choose the correct answer from the options given below :

Option 1:
Both Statement I and Statement II are true.

Option 2:
Both Statement I and Statement II are false..

Option 3:
Statement I is correct but Statement II is false.

Option 4:
Statement I is incorrect but Statement II is true.

Correct Answer:
Statement I is correct but Statement II is false.

Solution:
The correct option is Option 3) Statement I is correct but Statement II is false.

Statement I, known as Gause's Competitive Exclusion Principle, is true. It states that two closely related species competing for the same resources cannot coexist indefinitely, and the competitively inferior species will be eliminated eventually.

However, Statement II is false. It is not true that carnivores are more adversely affected by competition than herbivores in general. The impact of competition on different species can vary depending on various factors such as resource availability, ecological niche, and adaptive traits.

Therefore, the correct answer is Option 3) Statement I is correct but Statement II is false.

Q. 95 Which one of the following statements is NOT correct?

Option 1:
The micro-organisms involved in biodegradation of organic matter in a sewage polluted water body consume a lot of oxygen causing the death of aquatic organisms.

Option 2:
Algal blooms caused by excess of organic matter in water improve water quality and promote fisheries

Option 3:
Water hyacinth grows abundantly in eutrophic water bodies and leads to an imbalance in the ecosystem dynamics of the water body

Option 4:
The amount of some toxic substances of industrial waste water increases in the organisms at successive trophic levels.

Correct Answer:
Algal blooms caused by excess of organic matter in water improve water quality and promote fisheries

Solution:
The correct answer is Option 2) Algal blooms caused by excess of organic matter in water improve water quality and promote fisheries.

This statement is not correct. Algal blooms caused by excess organic matter, particularly from sources like agricultural runoff or sewage pollution, can have negative impacts on water quality. The excessive growth of algae can deplete oxygen levels in the water, leading to oxygen-deprived zones harmful to aquatic organisms. It can also block sunlight from reaching other aquatic plants, affecting their growth. Additionally, algal blooms can lead to the release of toxins harmful to fish and other organisms.

Q. 96 Match List I with List II :

	List I (Interaction)		List II (Species A and B)
A.	Mutualism	I.	+(A), O(B)
B.	Commensalism	II.	–(A), O(B)
C.	Amensalism	III.	+(A), –(B)
D.	Parasitism	IV.	+(A), +(B)

Choose the correct answer from the options given below:

Option 1:
A-IV, B-II, C-I, D-III

Option 2:
A-IV, B-I, C-II, D-III

Option 3:
A-IV, B-III, C-I, D-II

Option 4:
A-III, B-I, C-IV, D-II

Correct Answer:
A-IV, B-I, C-II, D-III

Solution:
A. Mutualism - IV. +(A), +(B)
B. Commensalism - II. –(A), O(B)
C. Amensalism - I. +(A), O(B)
D. Parasitism - III. +(A), –(B)

Therefore, the correct answer is Option 2) A-IV, B-I, C-II, D-III.

Q. 97 Match **List I** with **List II**.

	List I (Interacting species)		**List II** (Name of interaction)
A.	A Leopard and a Lion in a forest/grassland	I.	Competition
B.	A Cuckoo laying egg in a Crow's nest	II.	Brood parasitism
C.	Fungi and root of a higher plant in Mycorrtizae	III.	Gastric gland
D.	A cattle egret and a Cattle in a field	IV.	Commensalism

Choose the **correct** answer from the options given below:

Option 1:
A-I, B-II, C-III, D-IV

Option 2:
A-I, B-II, C-IV, D-III

Option 3:
A-III, B-IV, C-I, D-II

Option 4:
A-II, B-III, C-I, D-IV

Correct Answer:
A-I, B-II, C-III, D-IV

Solution:
A. A Leopard and a Lion in a forest/grassland - I. Competition
Leopard and Lion compete for resources, such as prey, within their shared habitat.

B. A Cuckoo laying an egg in a Crow's nest - II. Brood parasitism
Cuckoos lay their eggs in the nests of other bird species, such as Crows, for them to raise the cuckoo offspring.

C. Fungi and root of a higher plant in Mycorrhizae - III. Mutualism
Mycorrhizae is a mutually beneficial association between fungi and the roots of higher plants, where the fungi provide nutrients to the plant, and the plant provides carbohydrates to the fungi.

D. A cattle egret and a Cattle in a field - IV. Commensalism
Cattle egrets follow cattle and feed on insects stirred up by the grazing activity of the cattle. The cattle are unaffected by the presence of the egret.

Therefore, the correct matching is:
A-I, B-II, C-III, D-IV, which corresponds to Option 1.

Q. 98 Given below are two statements:

Statement I: Electrostatic precipitator is most widely used in thermal power plant.
Statement II: Electrostatic precipitator in thermal power plant removes ionising radiations
In the light of the above statements, choose the most appropriate answer from the options given below:

Option 1:
Both Statement I and Statement II are correct.

Option 2:
Boh Statement I and Statement II are incorrect.

Option 3:
Statement I is correct but Statement II is incorrect.

Option 4:
Statement I incorrect but Statement II is correct.

Correct Answer:
Statement I is correct but Statement II is incorrect.

Solution:
The correct answer is Option 3) Statement I is correct but Statement II is incorrect.

Statement I: Electrostatic precipitators are indeed widely used in thermal power plants. They are used to control and reduce air pollution by removing particulate matter (such as dust and ash) from the flue gases emitted during the combustion of fossil fuels.

Statement II: However, electrostatic precipitators are not used to remove ionizing radiations in thermal power plants. Ionizing radiation refers to radiation with enough energy to remove tightly bound electrons from atoms, causing them to become charged ions. Electrostatic precipitators are not designed to capture or control ionizing radiation; their main function is the removal of particulate matter from the exhaust gases.

Q. 99 Which of the following statements is correct?

Option 1:
Eutrophication' refers to increase in domestic sewage and waste water in lakes.

Option 2:
Biomagnification refers to increase in concentration of the toxicant at successive trophic levels.

Option 3:
Presence of large amount of nutrients in water restricts 'Algal Bloom'

Option 4:
Algal Bloom decreases fish mortality

Correct Answer:
Biomagnification refers to increase in concentration of the toxicant at successive trophic levels.

Solution:
The correct statement is:

Option 2) Biomagnification refers to an increase in the concentration of the toxicant at successive trophic levels.

Biomagnification is the process in which the concentration of certain substances, such as pollutants or toxins, increases as they move up the food chain. This phenomenon occurs because organisms at higher trophic levels consume a larger number of organisms from lower trophic levels, leading to the accumulation of the substances in their bodies.

Eutrophication (Option 1) refers to the excessive enrichment of water bodies with nutrients, typically from sources such as agricultural runoff or excessive sewage, which can lead to increased algal growth and subsequent ecosystem imbalances.

Option 3 states that the presence of large amounts of nutrients in water restricts algal bloom, which is incorrect. Algal blooms are often associated with an abundance of nutrients in the water, particularly phosphorus and nitrogen.

Option 4 suggests that algal blooms decrease fish mortality, which is also incorrect. Algal blooms can have detrimental effects on fish and other aquatic organisms due to factors such as oxygen depletion and the release of harmful algal toxins.

Q. 100 Match **List I** with **List II.**

	List I		List II
A.	Logistic growth	I.	Unlimited resource availability condition
B.	Exponential growth	II.	The limited resource availability condition
C.	Expanding age pyramid	III.	The percent individuals of pre-reproductive age is the largest followed by reproductive and post-reproductive age groups
D.	Stable age pyramid	IV.	The percent individuals of pre-reproductives and reproductive age groups are same

Choose the **correct** from the option given below:

Option 1:
A-II, B-I, C-III, D-IV

Option 2:
A-II, B-III, C-I, D-IV

Option 3:
A-II, B-IV, C-I, D-III

Option 4:
A-II, B-IV, C-III, D-I

Correct Answer:
A-II, B-I, C-III, D-IV

Solution:
A. Logistic growth corresponds to II. Limited resource availability condition. Logistic growth occurs when a population's growth rate slows down as it approaches the carrying capacity of its environment, where resources become limited.

B. Exponential growth corresponds to I. Unlimited resource availability condition. Exponential growth occurs when a population's growth rate is not limited by resources and continues to increase exponentially over time.

C. Expanding age pyramid corresponds to III. The percent individuals of pre-reproductive age is largest followed by reproductive and post-reproductive age groups. An expanding age pyramid indicates a population with a high proportion of individuals in the younger age groups, suggesting a high birth rate and potential for future population growth.

D. Stable age pyramid corresponds to IV. The percent of individuals of pre-reproductives and reproductive age groups are the same. A stable age pyramid indicates a population with a relatively balanced distribution across different age groups, indicating a stable population size over time.

Therefore, the correct matching is A-II, B-I, C-III, D-IV.Hence, the correct answer is option 1.